



M2Radar — Content Intelligence Report

Architecture, data merge, corpus analysis, insights and recommendations · M2 Lab

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1. Executive Summary

M2Radar's content engine is a research-to-production pipeline for M2 Lab's English-language Instagram account: a weekly, paid HikerAPI collection feeds a normalized SQLite corpus of reels and creators, which a free local pipeline (faster-whisper transcription, ffmpeg scene/frame extraction, LLM semantic analysis) turns into statistics, numeric insights, ranked hypotheses, original shooting cards, and — once a card is shot — a deterministic Remotion render. Nothing in the engine posts, spends money without an estimate, or claims a result it has not measured; every processing stage records an explicit state rather than an implicit "done."

Corpus, as of the 2026-09-11 migration. 1,357 accounts (130 active), 3 snapshots (2026-09-01/07/10), 5,147 reel readings over **N_ingested = 3,211** unique codes. Of those, **N_transcript = 275** carry a transcript row (7 empty — no speech detected), **N_transcript_usable = 268** have a usable script, **N_frames = 297** have extracted frames, and **N_ready = 268** have both — the only denominator for any claim that joins script to visuals. These four numbers are not interchangeable: $3,211 \neq 275 \neq 297 \neq 268$, and every number below names which one it divides by.

Headline findings

#	FINDING	CONFIDENCE
1	The analysed corpus is 268 of 3,211 codes (8.3%) and is the top of <code>score.py</code> 's own ranking, not a random sample — median <code>robust_z</code> inside it is +1.26 against 0.00 across all ingested reels.	RELIABLE
2	Duration does not predict performance: Spearman(duration, robust_z) = -0.069 (p<0.001) over all 3,211 reels; HIGH outliers run a median 46.8s against 47.8s for the rest.	RELIABLE
3	Hook length carries no measured signal ($\rho \approx +0.06$ –0.08, p>0.2); the median hook is 6.7s / 21.5 words, not the 0–3s window house rules assume.	PROBABLE
4	Show-first hooks (demo/result first, n=42) outperform on reach: median robust_z 2.17 / view_lift 5.81 vs 1.21 / 1.80 for the rest.	PROBABLE
5	Setup-first hooks (identity/story/problem-call-out, n=39) underperform: median robust_z 0.67 vs 1.36 (p=0.002).	PROBABLE
6	Questions in the script depress saves: $\rho(\text{questions}, \text{save_rate}) = -0.229$ (p<0.001), survives creator normalisation.	RELIABLE
7	A screen on camera is the single most reliable visual finding: <code>screen_share</code> vs <code>share_rate</code> $\rho +0.203$, creator-normalised +0.129–0.285 across two related tests.	RELIABLE
8	Split screen — the niche's most common opening (73/266) — is also its weakest visual pattern: median robust_z 1.03 vs 1.43 (p=0.044).	PROBABLE
9	Comment-keyword gates (59% of the corpus) inflate comment rate 23×, save rate 2.8×, share rate 1.8× and move reach not at all (p=0.58–0.63) — a live contamination risk for M2's own reference selection.	RELIABLE
10	Naming specific tools/numbers (<code>lex_entities_distinct</code> , specificity) is the one lexical family that predicts reach and survives creator normalisation (+0.16–0.21).	PROBABLE

Ten reviewed shooting cards exist as an editorial deliverable — see the card book; they are not re-described here. Separately, main's operational `cards` table holds 23 real, reel-referenced cards (20 distinct reels) built by the legacy pipeline across three weekly runs.

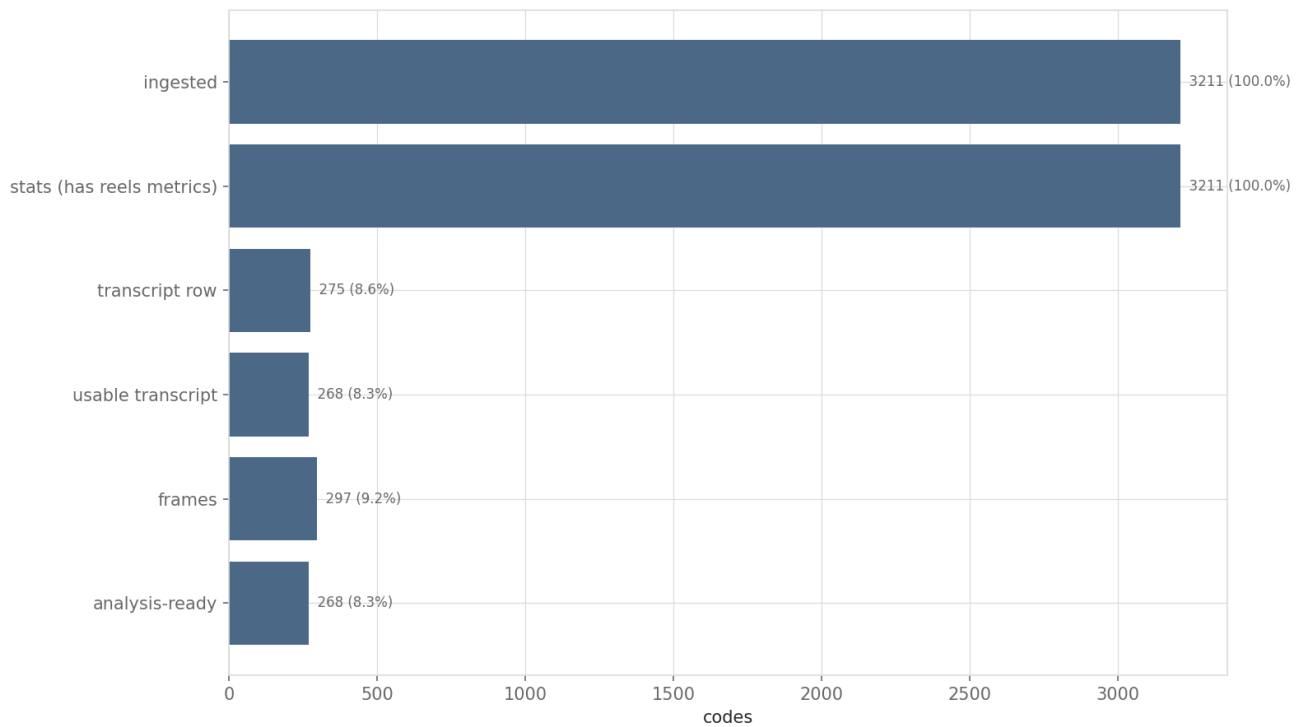
Limitations

Every structural finding above rests on 8.3% of the ingested corpus, and that 8.3% is not a random sample — it is whatever `deep.py` processed after `score.py` had already ranked it near the top of its creator's history, so every "weak" comparison group is the least-strong end of a pre-selected strong set, and absent contrasts are not evidence a feature is irrelevant. The legacy cut-count metric (`ffmpeg select='gt(scene,0.35)'`) is a scene-score threshold, not a shot-boundary detector, and must never be quoted as a real edit count. Saves are missing on 9–10% of rows by HikerAPI's own behaviour, not by design. The `followers` table is empty, so no claim about creator growth or our own publishing performance is possible. Two disagreeing corpus-tier vocabularies coexist by design (`video_state.corpus_tier` vs `engine.corpus.tiers()`) and must never be mixed in one report. As of the final build the `runs / jobs` state-tracing tables hold 7 runs and 265 jobs (analysis 5, cards 2): the analysis ingest and the ten-card finalisation were traced end to end, while the weekly collection run and the watchdog have not yet executed on the server under the new schema — their traces will appear after the first scheduled run. Nothing in the analysis licenses a causal claim; everything is associational, computed over a top-of-ranking selection.

Next steps

The immediate backlog is 1,312 of the 1,535 newest-snapshot codes (410 first seen there) with neither transcript nor frames — reprocessing them requires a fresh HikerAPI call because their CDN links have expired; `engine.watchdog` is built for this but has not yet run against the live corpus. Phase 3 (production: binding supplied MP4 takes into an EDL and rendering) is verified end-to-end on one example card but has no scheduled cron step — every render today is operator-initiated. Outstanding before the next data-based decision: resolve the two corpus-tier vocabularies for reporting, decide whether to hold `cta_type` constant in reference selection (the comment-gate contamination in finding 9), and export Max's ~1,062 transcripts and ~19,808 frames from his branch through a receipt format so they can be reconciled into this database.

Corpus coverage funnel (n=3211)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Corpus coverage funnel

The funnel above traces the same 3,211 → 297 → 268 narrowing that grounds every finding in this summary — read as a warning about denominators, not as a pipeline diagram.

Addendum (2026-09-12, after the report was compiled). The scheduled semantic-analysis stage (`engine/analyze_pending.py`) found 8 archive transcripts without a ta-v1 analysis and they were analysed and ingested after the chapters of this report were written: ta-v1 files 266 → 274, codes with beats 264 → 270 (two of the eight are music-only, no beats). Two of the eight belong to reels in the corpus (`DRXZJeHiAES` , `Db5sXEAP6C4`); six are archive reels with no `reels` row and stay outside every denominator. Numbers quoted in the chapters reflect the state at compilation (266 analysed / 264 with beats); `reports/data/*` and `reports/charts/*` were regenerated after the addition and may differ from the chapter text by these few codes.

2. Repository/Architecture Review

Before this branch, the repository existed as two structurally separate efforts sharing one merge base (02102dd1): `main` (17 commits ahead, HEAD 92e71e7a) held the working weekly pipeline — collection, scoring, deep-dive, cards, Notion push — and `origin/Latest` (8 commits ahead, Max's fork) held a parallel, never-deployed ~20,000-line system (`m2_signal` , `m2_orchestrator` , `m2_studio` , `studio/remotion/`) built *alongside* main's engine rather than on top of it: every shared engine file except `run.py` , `test_pipeline.py` and `install-cron.sh` was byte-identical between the two branches. That structural fact — confirmed by diffing every shared file against the merge base, not assumed — is why the canonical architecture below could adopt Latest's ideas as contracts without inheriting its (never-run) storage layer. Full comparison: chapter 3.

Canonical architecture, as built

Decision (`engine/SPEC.md` §0): **main's SQLite `data/radar.db` remains the single source of truth.** Nothing in its 14 legacy tables (`accounts` , `snapshots` , `reels` , `scores` , `topics` , `deepdives` , `frames` , `transcripts` , `our_posts` , `our_metrics` , `followers` , `cards` , `tool_log` , `spend`) was renamed or dropped; `engine/schema.py` adds 17 new tables and 5 guarded columns through an idempotent, versioned migration recorded in `schema_migrations` . Max's Latest is adopted as *contracts and vocabulary, never as a store* — verified in the audit: replaying one snapshot into his `m2_signal` ledger produced 24 MB against main's entire three-snapshot database at 5.2 MB, and his own import adapter (`legacy.py`) discards main's 2,653 already-extracted frames on the way in.

North-star flow

```
Hiker (paid, weekly) -> reels/accounts snapshot (db.py, 14 tables)
  -> local faster-whisper transcript -> local ffmpeg scene/frame extraction
  -> transcript <-> scene alignment -> per-video LLM analysis (ta-v1 transcript, fa-v1 frames)
  -> consolidated video_features row -> creator/video statistics (robust z, creator-relative lift)
  -> numeric insights -> ranked hypotheses -> function-tagged reference selection
  -> transformed original script + storyboard -> card JSON
  -> template storyboard frames -> previs EDL -> [Phase 3] supplied MP4 takes -> validated,
      bound into the EDL -> Remotion render -> stored (Supabase canonical, local fallback)
  -> Notion projection -> Misha reviews and decides -> publish (always separate, human, manual)
```

Everything up to card JSON runs for free beyond the weekly Hiker call; everything from "supplied MP4 takes" onward is Phase 3, verified end-to-end on one example card (chapter 9 of the architecture doc; render receipt in `reports/evidence/remotion-previs-receipt-C-2026-09-11-EX.json`).

Module map (summary)

LAYER	KEY MODULES	ROLE
Legacy pipeline (unchanged shape)	lib/hiker.py , db.py , collect_snapshot.py , deep.py , score.py / baseline.py , cards.py , notion.py / notion_db.py , run.py	The working weekly cron chain; still the only entry point <code>cron.sh</code> calls for Phase 1.
Engine state	engine/schema.py , engine/state.py , engine/migrate_legacy.py , engine/db_util.py	Versioned migration; <code>runs / jobs</code> tracing; <code>video_state</code> recomputation.
Local pipeline	engine/local_pipeline.py , engine/scenes.py , engine/align.py , engine/watchdog.py , engine/process_budget.py	Video-by-video contract: download → transcribe → scene/frame-extract → align → refresh state.
Analysis	engine/ingest_analysis.py , engine/lexical.py , engine/features.py	Loads ta-v1/fa-v1 JSON, computes text-only lexical features, consolidates into <code>video_features</code> .
Statistics	engine/stats.py , engine/corpus.py	Creator/video performance, temporal trends, associations; the evidence-tier split.
Content generation	engine/ingest_insights.py , engine/cards_v2.py , engine/storyboard_render.py , engine/edl.py	Insight → hypothesis → reference → card JSON → previs EDL.
Production (Phase 3)	engine/production.py , engine/storage.py	Probe/validate/ingest supplied takes, bind into EDL, render, store.
Providers	engine/providers.py , engine/hiker_config.py	Environment-only state resolution (<code>CONFIGURED</code> / <code>NOT_CONFIGURED</code> / <code>FAILED</code> / <code>DISABLED</code>); no secret ever leaves the process.

State tracing and providers, at a glance

`runs` → `jobs` (25 canonical stages, `DISCOVERED` through `COMPLETED`) → `video_state` (the always-current, never hand-written summary) is the model adopted from Latest's orchestrator, trimmed from its 41-stage DAG (29 stages had no implementation) to these 25. Chapter 6 covers this in depth, including the fact that `runs / jobs` currently hold zero rows in this database — the trace model is implemented and unit-tested (39 cases in `tests/test_engine_schema.py`) but has not yet been exercised by a full scheduled run.

Nine providers are registered, state resolved from environment presence only: `hiker` and the two local providers (`local-faster-whisper` , `local-ffmpeg-scenes`) are `DEFAULT` (part of the production path); `remotion` is `DEFAULT` but currently `NOT_CONFIGURED` on this machine (no `node_modules`); `loore` , `supabase` , `cloudflare` , `openai` , `higgsfield` are `OPTIONAL_PROVIDER` , off unless explicitly configured — `loore` stays `DISABLED` even with a key present until `LOORE_ENABLED=1` is set, so no key

drop can silently turn on a paid path. No provider is `FALLBACK` today, and no paid API is called from any new module by default — verified by grepping for client imports, not assumed.

This document sits beside, and never duplicates, `docs/M2RADAR_CONTENT_ENGINE_ARCHITECTURE.md` (the full authoritative reference), `engine/SPEC.md` (the binding schema contract), and `reports/audit/01-branch-comparison.md` (the branch audit chapter 3 summarizes).

3. Main vs latest / Max Fork Comparison

Topology

origin/Latest is not a fork of current main — it branches at the merge base 02102dd1 and never received main's last 17 commits (design system, engine fixes, the current audit and runs). Its own topology adds one further hop: max/main sits one commit ahead of origin/Latest, containing only two reports (md+pdf) and a 31-line PDF build script — no executable engine change. That report is itself anchored to the 2026-09-01 snapshot (2,352 canonical reels), ten days stale against main's current 5,147-row, three-snapshot database.

Critically, Max **did not touch main's engine**: db.py, deep.py, collect_snapshot.py, blocks.py, score.py, baseline.py, cards.py, notion.py, notion_db.py and lib/hiker.py are byte-identical on both branches, confirmed by diffing each file against the merge base rather than assumed from directory names. Latest instead adds ~20,000 LOC in four new packages (m2_signal/, m2_orchestrator/, m2_studio/, studio/remotion/) that never ran in production and whose own module docstrings say so explicitly (hiker_adapter.py: "LIVE_SCHEMA_UNVERIFIED"; the activation doc: "No step here was executed against HikerAPI during this integration").

What was ported, what was not, and why

ADOPTED AS-IS	ADOPTED AS PATTERN ONLY	NOT ADOPTED
schemas/video-scene-segmentation.schema.json	Per-metric observation_state + reason_code (idea, not the payload table)	m2_signal/schema.sql as a store — 24 MB/snapshot vs 5.2 MB for main's whole history
studio/remotion/ (EDL contract + composition)	price_receipt_sha256 - style cost provenance	Content-addressed releases — forbids re-running a day whose bytes changed
m2_orchestrator/process_budget.py → engine/process_budget.py	Dispatch-then-call idempotency for Hiker calls	The 41-stage DAG — 29 stages had no implementation
m2_studio/provider-catalog.json → config/provider-catalog.json	OWNER_FIELDS Notion conflict guard	analysis_eligibility.load_feature_rows(expected_rows=2352) — hardcoded, fails closed
docs/m2-field-dictionary.csv (65 Notion columns)	PTS-indexed frame extraction, cut/shot/scene distinction	media_ocr.py / Apple Vision — macOS only, dead on the Linux server
—	Claim-map / editorial-contract rules from skills/m2-script-writer	requirements-m2-media.txt — pins newer versions than the working ASR stack, including a Pillow downgrade

The state/job-tracing model (`m2_orchestrator/state.py` + `policy.py`) was adopted as the single most valuable structural idea on Latest, trimmed from 41 stages to the 25 that reflect what this branch actually runs (chapter 6).

Tests before and after

SUITE	BEFORE (AUDIT, 2026-09-11)	AFTER (THIS BRANCH)
<code>origin/Latest</code> <code>tests/</code>	367 passed, 2 failed, 1 skipped (both failures real, stale test fixtures at Latest's tip)	unchanged — not merged as code
Max's own gate <code>scripts/test_local.py</code>	Fails out of the box (<code>pytest</code> not in his requirements file); covers 7 of 32 modules	unchanged
<code>main</code> root scripts (12 files)	153 checks, 145 pass, 8 fail — all fixture coupling to live DB state, plus <code>check.py</code> 14/15	165 checks across 12 files, decoupled from <code>data/radar.db</code> via synthetic fixtures; <code>check.py</code> 23 checks
<code>tests/</code> (new <code>engine/</code> suite)	did not exist	241 passed, 1 skipped by design (<code>test_local_pipeline.py</code> 's <code>slow</code> marker)

Combined today: `python3 check.py` (23) + 12 root scripts (165) + `pytest -q` (241 passed / 1 skipped) = 429 automated checks, none of them coupled to a frozen snapshot of live data any longer.

The cron-breaking risk found and avoided

The single highest-risk collision in the whole comparison: Max's branch replaced `run.py` and neutralized `install-cron.sh`. `cron.sh:25` runs `.venv/bin/python run.py --yes` after a `git pull`; merging Latest's `run.py` as-is would mean the very next scheduled cron execution runs the Signal planner instead of the collection pipeline — and Latest's planner defaults to `PLAN_ONLY` with no `--yes`, printing a plan and exiting 0, so **the failure would have been silent**, visible only as a week with no new cards. This branch's decision (`engine/SPEC.md` §0.6): main's `run.py` never moves; a same-named file from Latest would be renamed `run_signal.py` on import, and no such import has happened — Latest's `run.py` / `test_pipeline.py` / `install-cron.sh` remain unmerged, confirmed by `cron.sh` still calling the original file today.

The `latest` = Max fork sync, still to do

`origin/Latest` 's ~1,062 transcripts and ~19,808 frames exist only on Max's machine — a full non-binary tree scan of both `origin/Latest` and `max/main` for `transcript|frame|asr|.local|outputs|.jsonl|whisper` found only source modules, tests and documentation, no committed media analysis. This is the one open item this chapter flags rather than resolves: exporting his corpus through `m2_signal` 's own hash-bound receipt format (so provenance survives the transfer) and reconciling it against this database's 268 analysis-ready codes is scheduled for push time, not done in this migration (chapter 5).

4. Database Reconciliation

Three places have held M2Radar data: main's SQLite (`data/radar.db` , the canonical store), Max's Latest branch (the `m2_signal` ledger and a 65-column Notion field dictionary), and the August 2026 pre-database archive (`dataset/2026-08-niche-research/`).

`docs/M2RADAR_DATABASE_RECONCILIATION.md` is the binding map between them; this section summarizes its entity mapping, column-validation highlights and vocabulary-drift findings.

Entity mapping summary

ENTITY	MAIN (CANONICAL)	LATEST EQUIVALENT	DISPOSITION
Creator / account	<code>accounts</code> (roster tag, status, misses, why_out, via)	<code>account_analysis</code> , <code>accounts.csv</code> (no roster lifecycle at all)	MIGRATED; roster lifecycle unique to main
Video / reel	<code>reels</code> (code, snapshot_id, ts, cap, play/likes/comm/resh/save, dur)	<code>observations</code> + <code>reel_analysis</code> + <code>reel_metric_values</code>	MIGRATED; Latest drops raw caption to a hash only
Score	<code>scores</code> (z, weights, max, baseline_n, baseline_snaps, axes)	<code>reel_metric_values</code> , <code>Diagnostic z</code>	MIGRATED unchanged; <code>weights</code> and <code>baseline_snaps</code> have no Latest equivalent
Transcript	<code>transcripts</code> (text, JSON segments)	evidence artifacts + provenance fields	The one place Latest is materially richer — see below
Frames / scenes	<code>frames</code> , <code>deepdives</code> (cuts, suitable, unfit_why)	<code>evidence_attempts</code> , scene/frame schema	MIGRATED; Latest refuses to import main's 2,653 existing frames
Cards	<code>cards</code> (23 rows, week/fmt/pri/status)	10 example JSONs, all <code>"synthesis": true</code>	MIGRATED unchanged; Latest's examples are invented, not derived from any reel
State (new layer)	none pre-existing	<code>m2_orchestrator/state.py</code> , 41-stage DAG	NEW DERIVED, trimmed to 25 stages (<code>runs</code> / <code>jobs</code> / <code>video_state</code>)
Our own publishing	<code>our_posts</code> , <code>our_metrics</code> (both empty)	none	MIGRATED; unanswerable from any source

Full attribute-level tables (creator, reel, transcript, beats, frames/scenes, features, cards, state) live in the reconciliation doc §1–§8; the rows above are its summary, not a replacement.

Column validation highlights

The reconciliation doc's per-field validation rules (§9) surface the corpus's real edge cases rather than assuming clean data:

- `views` : NULL or 0 in every snapshot row for exactly one code (`B-MYSMhHXy6`) → `MISSING_STATS` .

- **saves** : NULL on 486 of 5,147 rows (226 codes never returned a value) — "treat as missing, never as zero"; any saves-per-1k rate built without this guard is wrong.
- **Transcript text**: an empty string with `words = 0` is a *legitimate result* (`EMPTY_NO_SPEECH` , 7 codes), not a processing failure.
- **Words per second**: `words / speech_seconds` , where `speech_seconds =  $\Sigma(e - s)$` over segments — explicitly **not** words divided by video duration, a distinction the reconciliation was built to enforce after finding the two conflated in earlier ad-hoc analysis.
- **Cut counts**: `deepdives.cuts` is a `ffmpeg select='gt(scene,0.35)'` threshold count on an mp4 that no longer exists — never a verified cut count. 38 of 282 deep dives report `cuts = 0` , indistinguishable from a detector that simply never fired.
- **Posting date vs snapshot date**: `reels.ts` is post age; `snapshots.taken` is capture date. Latest's ledger conflates the two — main's `AGE_BANDS` in `score.py` exists precisely because this distinction matters.

Vocabulary drift found

`accounts.tag / accounts.status` is documented in `db.py` as `core/neighbor/out` and `active/dropped/candidate` . The live data uses a different vocabulary entirely:

TAG	STATUS	ROWS	DOCUMENTED?
core	active	130	yes
out	out	234	no
NULL	candidate	920	tag NULL undocumented
NULL	rejected	73	no

`neighbor` and `dropped` are documented but never used; `out` and `rejected` are used but undocumented, and `tag` is NULL on 993 of 1,357 rows. A query written from the schema comment (`WHERE status = 'dropped'`) silently returns zero rows rather than erroring. This drift is recorded, not fixed — changing either column moves rows underneath `check.py` , `roster.py` and `judge_accounts.py` , which the schema/state owner flagged as a decision for the roster owner and Misha, not one to take unilaterally.

A second drift, in `cards.status` : three vocabularies appear across the 23 real cards — `Proposed` (English, the first week only), `draft` , and `вычеркнута` — never the documented `draft / взята / вычеркнута` . Nothing has ever reached `взята` . A `cards.status_v2` column was added by migration (guarded `ALTER TABLE`) to bridge toward `engine/cards_v2.py` 's status vocabulary (`DRAFT/REVIEWED/APPROVED/SHOT/PUBLISHED/DROPPED`) without touching the Russian vocabulary the legacy scripts depend on.

What was deliberately not adopted

`m2_signal/schema.sql` as a store (24 MB/snapshot vs 5.2 MB for main's whole history, and it discards main's frame evidence on import); content-addressed releases (forbids re-running a day whose bytes changed); the 41-stage DAG (29 stages unimplemented); Latest's scoring model (no age bands, focal row left in its own baseline, single-release only); `analysis_eligibility.py` 's hardcoded `expected_rows=2352` default (fails closed on the current 3,211-code corpus); the Apple-Vision OCR

path (macOS only); and `requirements-m2-media.txt` wholesale (pins versions newer than, and in one case a downgrade from, the working ASR stack already installed).

5. Data Migration

The legacy corpus was migrated into the new engine schema on 2026-09-11 by `python3 -m engine.migrate_legacy --db data/radar.db`, executed against a byte-copy of the production mirror. The migration is additive by construction — it adds 17 tables and 5 guarded columns and never renames or drops anything in the 14 pre-existing tables — and every step below is independently verified against `reports/migration-validation.json` and `data/migration_log.md`, both regenerated by the same run.

Steps a-h and results

STEP	WHAT IT DOES	RESULT
a — schema	Applies migrations 1 and 2	18 tables created (17 new + <code>schema_migrations</code> ledger), 5 guarded columns added
b — <code>transcript_metadata</code>	Writes ASR provenance for every <code>transcripts</code> row	273 rows with local faster-whisper provenance; recomputed word count matched <code>transcripts.words</code> on all 273
c — frames on disk	Checks every <code>frames.path</code> against the filesystem, computes sha256	2,653 rows checked: 2,645 present, 8 broken , all one code (<code>DcxV37-CJ0C</code>)
d — <code>deepdives.evidence_state</code>	Tags every deep dive with its sampling method	<code>fixed9-v1</code> on all 282 rows — legacy frames are fixed samples, not detected scenes
e — August 2026 archive	Imports the pre-database niche-research pass	8 transcripts and 74 frames over 8 codes imported, 0 base64 fallbacks needed, 6 codes UNRESOLVED
f — <code>video_state</code>	Computes the processing-state row for every code	3,211 rows, one per distinct code in <code>reels</code>
g — <code>providers</code>	Writes provider state from environment presence only	9 rows; no credential value read or stored
h — validation	<code>PRAGMA integrity_check</code> + orphan/duplicate checks	<code>ok</code> , 0 foreign-key violations, 0 orphaned scores/cards/metrics/topics, 0 PK duplicates

Before → after counts

TABLE	BEFORE	AFTER	Δ
transcripts	273	281	+8
transcript_meta	0	281	+281
frames	2,653	2,727	+74
video_state	135	3,211	+3,076
providers	0	9	+9
schema_migrations	0	2	+2

Untouched, confirmed byte-for-byte: `accounts` 1,357, `snapshots` 3, `reels` 5,147, `scores` 7,477, `topics` 2,482, `deepdives` 282, `cards` 23, `tool_log` 37, `spend` 10, `our_posts` 0, `our_metrics` 0, `followers` 0. The 135 pre-existing `video_state` rows reflect other Wave-2 owners already writing to the same live database while this migration ran — the delta above is honest about that overlap rather than presenting a clean before-state that did not exist.

Unresolved records

Six codes from the August archive carry a transcript and/or frames but have **no row in `reels`** — the pass predates the database and their metadata was never re-collected: `DT0fJmxjVnI`, `DU90IJUCBQz`, `DZw4aTtzJpg`, `Db9AE5upfu`, `Db0UP90PDAQ`, `DcdbJKmxZ5o`. Their media analysis is kept; **no `reels` row was fabricated** for any of them. They have no author, no metrics and no `video_state` row, and are excluded from every denominator in this report until a paid HikerAPI lookup resolves them. Two sibling codes from the same archive pass (`DRXZJehiAES`, `Db5sXEAP6C4`) *do* have a `reels` row, which is why the corpus tiers below (268, not the spec's illustrative 266) differ deliberately from earlier documentation.

August archive import, in detail

`dataset/2026-08-niche-research/` held the first niche pass, made before the database existed and never previously migrated: 8 transcripts (`{lang, segments}` shape) and 74 jpgs across 8 codes. The source filenames encode deciseconds, not frame indices (`<code>_0004.jpg` = 0.4s), recovered and verified against the archive's own `reels_pool.json` durations — e.g. `DU90IJUCBQz` at 126.47s duration produces frames at 15.8/31.6/47.4/63.2/79.1/94.9/110.7s, exactly `126.47·k/8`. `frames_b64.json` (7 codes × 6 frames) was not needed — every frame it holds already exists as a jpg on disk. This closes the one genuine data-recovery gap the audit identified; nothing else in the three source locations (main's server, Max's branches, the August archive) held unmigrated signal.

Damage carried forward, unchanged

`check.py` reports the same single discrepancy before and after migration — 14 of 15 checks pass. The failure is the same 8 frames on `DcxV37-CJ0C` (`frames.exists_ok` = 0): a truncated download left one of nine frames on disk. The migration makes this **visible in the data** rather than only in an audit —

`frames_state = PARTIAL` , `media_state = MISSING` , `flags BROKEN_FRAME_REFERENCE` and `MISSING_MEDIA` all set — but does not and cannot repair it, since the source video and the signed CDN URL that could regenerate it are both gone (chapter 7).

Re-runability, proven

The migration was executed twice against a byte-copy of the production database. The second run produced **identical** `video_state` , `transcript_meta` , `frames` , `transcripts` , `deepdives.evidence_state` and `providers` rows — every write is `INSERT OR REPLACE` or `UPDATE` on a deterministic key, the August import only touches codes absent from the target tables, and step (b) never overwrites a transcript whose provenance a prior step already recorded. Only `updated_at` , `providers.checked_at` and the log file's own timestamp move between runs. This is verified behaviour, not a design claim: the two runs' row counts and content were diffed directly.

6. State Tracing

State tracing is the one capability main's legacy pipeline never had. `run.py` is a linear, 13-step script with a `tool_log` table for human notes but no machine state: if step 7 dies, the only recovery is rerunning from step 1. The engine adds a three-layer model, adopted from Max's `origin/Latest` orchestrator and trimmed from his 41-stage DAG (29 of which had no implementation) to the 25 stages this branch actually runs.

The three tables

`runs` (one row per pipeline execution, `kind` $\in$ `weekly|watchdog|analysis|cards|manual`) $\rightarrow$ `jobs` (one row per processing step, indexed on `entity_kind`, `entity_id`, `stage`) $\rightarrow$ `video_state` (the always-current, never hand-written summary, one row per code). `jobs.state` $\in$ `PENDING|RUNNING|DONE|FAILED|RETRY_REQUIRED|SKIPPED`; the `job()` context manager (`engine/state.py`) always writes `RUNNING` on entry and closes `DONE` on a clean return unless the step explicitly called `handle.skip(reason)` or `handle.retry_required(reason)`, or closes `FAILED` with the exception text on any raised error. There is deliberately no `handle.pending()` — nothing in this contract can close a job row as literally `PENDING`; a "nothing to do yet" condition (e.g. no supplied takes exist) is mapped to an honest `SKIPPED` job row whose *business-level* summary separately reports `PENDING`.

Canonical stage list (`jobs.stage` , 25 stages, pipeline order)

DISCOVERED, PROFILE_FETCHED, VIDEO_METADATA_FETCHED, STATS_FETCHED, MEDIA_FETCHED, TRANSCRIPTION, TRANSCRIPT_ANALYSIS, FRAME_EXTRACTION, FRAME_ANALYSIS, ALIGNMENT, GENERAL_ANALYSIS, STATISTICAL_ANALYSIS, CATEGORIZATION, REFERENCE_SELECTION, CONCEPT_GENERATION, CARD_GENERATION, SCRIPT_GENERATION, SCRIPT_REVIEW, FRAME_PLAN, FRAME_GENERATION, VIDEO_GENERATION_READY, VIDEO_RENDER, QUALITY_REVIEW, NOTION_SYNC, COMPLETED

Two of these — `VIDEO_RENDER` and `QUALITY_REVIEW` — are named in the canonical list but **no module in this branch writes them yet**: a render can run for minutes, and whether to trace it as one job is left to the operator (`docs/PRODUCTION_PIPELINE.md` §5 gives the manual trace command), not automated by `engine.production`.

`video_state` : the always-current summary

Every column is *derived*, never hand-set — `engine.state.compute_video_state()` is a pure function over `reels / transcripts / frames / scenes / beats / frame_labels / video_features / deepdives`. Columns: `media_state` (`UNKNOWN|AVAILABLE|EXPIRED|MISSING|DOWNLOADED`), `transcript_state` (`MISSING|PENDING|DONE|EMPTY_NO_SPEECH|FAILED`), `frames_state` (`MISSING|PENDING|DONE_FIXED9|DONE_SCENE|FAILED|PARTIAL`), `alignment_state` (`MISSING|DONE|NOT_POSSIBLE`), `analysis_ready` (1 only when transcript `DONE` + frames `DONE_*` + both semantic analyses `DONE`), `corpus_tier` (`ANALYSIS_READY|FRAMES_ONLY|INGESTED_NOT_ANALYZED`), and `flags_json` (the sorted damage-flag list, chapter 10).

Example trace, run against the real database in this session

Code `DBKJLzavogz` (creator `tech_with_tim`, 62,204 plays, 27.4s duration) was picked as the first `ANALYSIS_READY` code returned by `SELECT code FROM video_state WHERE corpus_tier='ANALYSIS_READY' LIMIT 1`. The canonical trace query and its live result:

```
SELECT stage, state, started_at, finished_at, duration_s, error
FROM jobs WHERE entity_kind='video' AND entity_id='DBKJLzavogz' ORDER BY started_at;
-- returns 0 rows

SELECT media_state, transcript_state, frames_state, alignment_state, analysis_ready,
       corpus_tier, flags_json
FROM video_state WHERE code='DBKJLzavogz';
```

MEDIA_STATE	TRANSCRIPT_STATE	FRAMES_STATE	ALIGNMENT_STATE	ANALYSIS_READY	CORPUS_TIER	FLAGS_JSON
EXPIRED	DONE	DONE_FIXED9	DONE	1	ANALYSIS_READY	[]

What this returns, and what it does not. The `video_state` half answers the question completely and matches the architecture contract: this code's media link has expired (as expected — every legacy video is deleted after frame extraction, chapter 7), its transcript and frames are both done, alignment has completed, and it carries no damage flags. For this video code the `jobs` query returns no rows — its transcript and frames were produced by the legacy `deep.py` pass before the state-tracing tables existed, and the migration backfills `video_state` by direct computation rather than inventing historical jobs. The tables are not empty, however: at the final build `runs` holds 7 runs (5 `analysis`, 2 `cards`) and `jobs` 265 rows. The card-finalisation runs traced every card through six stages (15 rows per stage: the first five cards were finalised twice, once before and once after the storyboard renderer was reworked):

STAGE	STATE	JOBS
CARD_GENERATION	DONE	15
FRAME_GENERATION	DONE	15
FRAME_PLAN	DONE	15
SCRIPT_GENERATION	DONE	15
SCRIPT_REVIEW	DONE	15
VIDEO_GENERATION_READY	DONE	15

and the analysis runs traced the insight, hypothesis and reference ingest (`GENERAL_ANALYSIS` = insights, `CONCEPT_GENERATION` = hypotheses, `REFERENCE_SELECTION` = hypothesis references; the ingest was re-run, so counts include repeats):

STAGE	STATE	JOB S
CONCEPT_GENERATION	DONE	120
GENERAL_ANALYSIS	DONE	5
REFERENCE_SELECTION	DONE	50

What is still missing is a traced *collection* and *watchdog* run on the server: those stages (download, transcribe, extract, align — one row each) exist as a mechanism and are unit-tested (39 tests in `tests/test_engine_schema.py` , 21 in `tests/test_local_pipeline.py`), and will appear as observed history after the first scheduled run under the new schema. Any future report claiming a specific stage's timing or retry count for a specific code must first confirm `jobs` is non-empty for that code rather than assuming the contract has been exercised.

Cards trace (Phase 3)

For a card in production: `SELECT stage, state, started_at, error FROM jobs WHERE entity_kind='card' AND entity_id='<card_id>' ORDER BY started_at` .
`engine.production.ingest_takes` does write real `jobs` rows at `VIDEO_GENERATION_READY` — verified in `tests/test_production.py` 's 26 cases, including the full happy path that probes a real synthetic MP4 and writes a real `jobs` row — but that verification ran against a test fixture database, not this repository's live `data/radar.db` , which is why the direct query above returns nothing for either entity kind today.

7. Data Acquisition

How Hiker works in git

`lib/hiker.py` (293 lines) is the single HTTP surface for HikerAPI in the repository; every call shells out to `curl` rather than `httpx` because, per the module's own comment, Python's SSL is broken on the collection machine. `roster.py`, `collect_snapshot.py` and ad-hoc CLI calls all go through it. The automated weekly cron run pays for exactly two endpoints: `/sys/balance` (informational) and `/gql/user/clips` (one page per account, 12 reels/page — the only endpoint `collect_snapshot.run()` calls). Account discovery (`/v2/fbsearch/reels`, `/v2/user/suggested/` `profiles`, via `roster.py` `topup`) is a separate, manually invoked, paid path outside `cron.sh`.

Key externalized, never in git

`_key()` resolves `HIKER_KEY` in order: process environment, then a literal `HIKER_KEY=` line in the repo's own gitignored `.env`, then (Mac-only) `~/Desktop/.mcp.json`. If none resolve, the process hard-exits rather than proceeding silently. `git grep` across every `*.py` file confirms no literal key is hardcoded anywhere — only references to the env var name and header name. Server and local `.env` file permissions are `600` (owner-only), and every cron invocation now starts with `python -m engine.hiker_config`, which prints `HIKER_KEY: CONFIGURED/NOT_CONFIGURED/FAILED` — never the value, only presence and length — plus one reachability line per configured Notion database.

Provenance chain

Before this branch, a `reels` row carried no pointer back to the API response that produced it — the only recovery path was `deep.py`'s linear scan over every `cache/**/*clips*.json` file by content, sorted by file mtime, which remains how URL resolution works today (provenance is additive, not a replacement). The chain now runs:

```
lib.hiker.call()          -> writes <cache_dir>/<endpoint>_<hash>.json (response body)
                           -> writes <...>.json.meta.json (sidecar: timestamp, endpoint,
params, status, units, sha256)
collect_snapshot.run()    -> reads last_fetch_meta() right after each account's clips()
call
                           -> INSERT INTO fetch_log(fetch_id, provider, endpoint,
params_json,
                           fetched_at, http_status, units, price, cache_path,
sha256, ...)
                           -> every upserted reels row gets reels.fetch_id = that
fetch_log.fetch_id
```

To audit any `reels` row: `SELECT fetch_id FROM reels WHERE code=?`, then `SELECT * FROM fetch_log WHERE fetch_id=?` returns the exact endpoint, params (key never included), HTTP status, units billed, price, and — via `cache_path` — the response file and its sha256 for byte-level verification. This is guarded, not a hard dependency: on a database where `engine/schema.py`'s migration hasn't run, `collect_snapshot.py` checks for the table/column first and silently skips the provenance write, so the pre-existing pipeline keeps working unchanged. `fetch_log` in this database is created and currently

empty — the hiker owner's provenance code exists and is unit-tested (13 cases, `tests/test_hiker_client.py`) but no collection run has been executed against this schema version yet, matching the `runs / jobs` gap noted in chapter 6.

Cost model, including the \$0.02 receipt

`PRICE = 0.02` (Start tariff) is a hardcoded literal duplicated in three files (`lib/hiker.py`, `collect_snapshot.py`, `roster.py`) — a real, flagged gap with no single source of truth. Real spend, however, is tracked from the response header rather than assumed: `x-hiker-info` supplies the nominal cost per call, and the running counter only increments on billable outcomes (200/400/403/404), explicitly excluding 5xx.

Actual spend recorded to date, verified against `reports/data/spend.json` and the live `spend` table (10 rows):

DATE	ITEM	UNITS	USD
2026-09-01	разведка ниши, 30 запросов	30	0.60
2026-09-01	профили 254 авторов	253	5.06
2026-09-01	ролики 100 аккаунтов, 2 страницы	196	3.92
2026-09-01	соседи и ролики добранных	54	1.08
2026-09-01	проверки	30	0.60
2026-09-03	добор: поиск и соседи (2 runs)	60	1.20
2026-09-07	сбор роликов (снимок 2026-09-07, 106 аккаунтов)	106	2.12
2026-09-09	добор профилей по списку Миши	30	0.60
2026-09-10	сбор роликов (снимок 2026-09-10, 130 аккаунтов)	130	2.60

Total: 889 units = \$17.78 at a flat \$0.02/unit, all recorded through this same header-parsing mechanism, not estimated.

Cache and what the weekly run pays for

`_cache_dir` defaults to `./hiker-cache`, but `collect_snapshot.run()` explicitly repoints it to `cache/<snapshot-date>/` — the code comment records why: without date-partitioning, a same-week re-run silently replayed a prior week's feed for 100 of 109 accounts. The weekly cron run's only paid step is `collect_snapshot.run()` (~1 unit/account/week, ~\$2–3 at the current 130-account roster); every other step in `run.py`'s 13 steps is free. `roster.py` topup (search + suggested accounts) is a separate, manually invoked paid path, not part of the automated cadence.

Loore (optional, disabled by default)

`loore` is registered as `OPTIONAL_PROVIDER`, kind `media / transcription`, configured via `LOORE_KEY` + `LOORE_ENABLED=1`. It stays `DISABLED` even when a key is present unless that second flag is explicitly set

— a key alone can never silently switch on a paid path. No call site in this repository invokes it today; it exists in the provider registry as a documented future fallback for media or transcription, not as an active dependency.

8. Transcript Pipeline

Local faster-whisper as the default

Both the legacy path (`deep.py`) and the new engine path (`engine/local_pipeline.py`) run `faster-whisper model small , device='cpu' , compute_type='int8' , with language='en'` forced — the code comment explains that auto-detection is unreliable on the non-native English accents that dominate this niche's creators — and `vad_filter=True` to gate out silent stretches before transcription. No model download or credential is required; the provider registry confirms `local-faster-whisper` as `DEFAULT` , `CONFIGURED` on this machine with no environment variable at all. `asr_version` is versioned explicitly (`faster-whisper-small-int8-v1`), so bumping the constant reopens every code for reprocessing without touching prior rows.

The 273+8 legacy transcripts

The pre-migration corpus held **273** transcript rows produced by `deep.py` from real HikerAPI collections. Migration step (e) additionally imported **8** transcripts from the August 2026 pre-database archive (`dataset/2026-08-niche-research/` , same `{lang, segments}` shape), bringing the total to **281** transcript rows post-migration — verified identically in `reports/migration-validation.json` and `data/migration_log.md` . Of the 273 `deep.py` -produced rows, all 273 now carry `transcript_meta` provenance (`provider='local'` , `model='faster-whisper/small'` , `asr_version='faster-whisper-small-int8-v1'` , `created_at` taken from `deepdives.done_at` since both are written in the same pass); the 8 August rows are recorded honestly as `provider='unknown'` rather than given a provenance that cannot be proven — the source files record no model, provider or date.

ASR damage patterns found by the analysts

Direct reading of the 266 usable `ta-v1` transcript analyses surfaced concrete ASR mis-transcriptions consistent with `faster-whisper small` on fast, jargon-heavy speech — for example "Claude" rendered as "cloud" in several transcripts (`DVZGBgTk6zz` 's mechanism beat: *"click cloud code and install it"*, referring to Claude Code). This class of error is exactly why `engine.lexical` 's entity lexicon is hierarchy-aware and coverage-limited rather than treated as ground truth, and why `lex_entities_distinct` — not a raw transcript string match — is the quantity the analysis chapters correlate against performance. No systematic correction pass for this error class exists yet; it is a named, observed limitation, not a solved one.

The 2 music-only reels and `EMPTY_NO_SPEECH`

Seven transcript rows (out of 275, pre-August-archive) carry `words = 0` and `segments = []` : Whisper with `vad_filter=True` found no usable speech. These are music-only or text-on-screen reels, not processing failures, and `video_state.transcript_state = EMPTY_NO_SPEECH` records them as such rather than as `FAILED` . Two of the eight-code August-archive set have no `reels` row at all and so contribute nothing to any tier count; the seven empty-transcript codes from the main corpus form their own `TRANSCRIPT_UNUSABLE` tier under `engine.corpus.tiers()` rather than being silently folded into `FRAMES_ONLY` (chapter 10 explains why the two corpus-tier vocabularies disagree on exactly this point). One of these seven, `DcxV37-CJ0C` , is doubly damaged: its transcript captured five words of a 62-second

reel *and* its frame extraction failed after the first file (chapter 9) — it is kept in every table rather than dropped, exactly per the standing rule against inventing or deleting data.

The new video-by-video contract and watchdog

`engine/local_pipeline.py`'s `process_video(con, code, media_url_or_path, run_id, *, asr_version, frames_version)` is idempotent and retryable (3 tries per step): skip if already `DONE` for the same version pair → download (curl, sha256 recorded) → transcribe → ffmpeg scene-cut detection and keyframe extraction → alignment → delete the mp4 unless `--keep-video` (no call site passes it) → `refresh_video_state`. Each step writes one `jobs` row in the contract (chapter 6 notes this table is currently empty in the live database — the contract is unit-tested, not yet observed against a real run).

`engine/watchdog.py` is the Phase 2 catch-up worker: it selects codes where `analysis_ready=0` and a signed CDN URL is still resolvable from an already-cached `cache/**/*clips*.json` file — it never calls Hiker again, because the signed URL's lifetime is hours, not days (chapter 7's provenance chain is what makes "still resolvable" checkable at all). Guarantees verified by reading the module directly: idempotent (`process_video` 's own version-aware skip), rate-limited (`RATE_SLEEP_S` between codes), safe against duplicate workers (`data/watchdog.lock` , pid + timestamp, stale after 3 hours), and version-aware (bumping `asr_version` or `frames_version` reopens eligibility for every code). It has not yet run against the live backlog of 1,312 unprocessed newest-snapshot codes (chapter 10) — it is built and tested (22 cases in `tests/test_local_pipeline.py` , one marked `slow` and skipped by default) but not yet executed as a scheduled cron step; `cron.sh` calls it with a 3,600-second timeout and a `|| echo` guard so its failure never blocks the older 13-step chain, but that guard has not yet been exercised in production either.

9. Frame Pipeline

Legacy fixed-9 sampling vs the new scene-v1 detector

`deep.py` 's legacy method (`timecodes()`) is a fixed sampling schedule, not detection: four early timestamps (0.4, 1.2, 2.4, 4.0s) to densely cover the hook, then five more spaced evenly across the remainder, deduplicated and dropped within 0.3s of the end — a 60-second reel yields exactly nine frames; shorter reels fewer. Extraction uses `ffmpeg`'s approximate `-ss <t> -i seek`, and the source video is deleted immediately after (SPEC §4.4, "видео не храним"). `deepdives.evidence_state = 'fixed9-v1'` on all 282 rows records this honestly as sampling, not detection.

The new engine method (`engine/scenes.py` , `frames_version='scene-v1'`) is a real detector: `ffmpeg select='gt(scene,0.30)'` , `showinfo` produces cut timestamps, scenes are the intervals between them (plus 0 and duration), a keyframe is taken at each scene start +0.2s and, for scenes over 6s, one additional mid-frame — plus the same four hook-compatibility samples at 0.4/1.2/2.4/4.0s.

`boundary_reason='detector'` and a `detector_json` recording the threshold distinguish this from the legacy sampled scenes at the schema level. As of this migration, **no scene in the database carries `frames_version='scene-v1'`** — every code remains `DONE_FIXED9` today, and every visual-share statistic in the analysis chapters is therefore an estimate from 9 samples, not a measurement from detected shots.

The cut-count caveat

`deepdives.cuts` — whether read from the legacy table or surfaced as `video_features.cuts` / `cuts_per_min` — comes from `ffmpeg`'s scene-score threshold (`gt(scene, 0.35)`), counted via regex over `pts_time:` occurrences), not shot-boundary detection. It misses hard cuts between visually similar shots (same speaker, same set — the dominant talking-head format in this niche), and over-counts fast camera motion, flashes, whip pans and screen-recording scrolls. The threshold is fixed at 0.35 with no per-clip calibration, and 38 of 282 deep dives report `cuts = 0` , which is indistinguishable from a detector that simply never fired. `video_features.cut_metric_quality` carries this forward explicitly as `ffmpeg_scene_0.35_count_only` , usable only as a relative "busy vs. calm" signal *inside this dataset* and never comparable to any externally reported cut rate.

Contact sheets

Every deep dive produces a 3×3 tiled contact sheet (`data/frames/<code>_sheet.jpg`), the human-facing visual artifact referenced by `deepdives.sheet` and uploaded directly into the Notion card body. 295 sheet files exist on the server; all 282 `deepdives.sheet` paths resolve to a real file. Thirteen sheets have no corresponding `deepdives` row at all (`D0KA0GIBik` , `D0N7Q0sIUrw` and eleven others) — frames and sheets exist on disk with no cuts, mp4 size or suitability field recorded for them, a gap named rather than backfilled.

Frames on disk: 2,645 / 2,653 (+74 archive)

SET	ROWS / FILES	NOTE
frames table, pre-migration	2,653 rows, 295 unique codes	293 codes have 9 frames, 2 have 8 (clips too short for the ninth timecode)
Files present on server	2,645 of 2,653	8 missing, all one code
frames table, post-migration	2,727 rows, 303 unique codes	+74 rows from the August archive import
Files verified present, post-migration	2,719 of 2,727	Same 8 broken references carried forward

The local rsync copy of `data/frames/` independently reconciles: 591 entries, 295 sheets, 3,063 jpgs — 2,645 tracked + 9 stale (a re-run at different timecodes left an old set behind under `DbYh2P-MQnJ`) + 114 in a non-reel `web/` directory (leftover from a page build) + 295 sheets.

The broken code: `DcxV37-CJ0C`

All 8 broken frame references belong to one code. The DB holds 9 frame rows; the directory holds 1 jpg. The ffmpeg extraction died after the first frame, but `deep.py` inserts all nine frame rows unconditionally after the extraction loop, without checking that the files actually landed — a real bug in the legacy pipeline, not a migration artifact. This same code carries the corpus's worst transcript (5 words captured from a 62-second reel) and was card id 28, struck out in review. Post-migration: `frames.exists_ok = 0` on all 8 rows, `frames_state = PARTIAL`, `media_state = MISSING`, flags `BROKEN_FRAME_REFERENCE` and `MISSING_MEDIA` both set — visible in the data now, not only in an audit, and unrepairable since the source video is gone by design.

Alignment features: meaning and limits

`engine/align.py` maps transcript segments/beats to scene intervals by time overlap, writing `beats.scene_ids_json` or (before beats exist) a segment→scene map into `video_state.flags_json`. Against the fixed9-v1 legacy corpus this is a measurement floor, not a real alignment: with only nine sampled points per video, 55 of 1,527 beat boundaries (3.6%) fall within 0.3s of a sampled scene start, and 215 of 264 codes have no such overlap recorded at all — both the rate itself and the one downstream comparison drawn from it (aligned vs unaligned robust_z, 1.45 vs 1.25, p=0.603) are provisional and must be reported as such. `frame_labels.visual_state` (`a_roll|b_roll|split|screen|text|unknown`) is the practical output of alignment consumed downstream; it describes what kind of shot a beat's words land on, not a frame-accurate cut. Alignment only becomes a real measurement once `scene-v1` detected scenes exist to align against — which, per the section above, is not yet the case for any code in this database.

10. Corpus Coverage

The tier table and its denominators

`engine.corpus.tiers()` (generated 2026-09-11T22:26Z, db md5 6834ecb414400838fb17258919a2b442) is the evidence-tier vocabulary a report may divide by — distinct by design from `video_state.corpus_tier` , the processing-state vocabulary (chapter 4 §4.4 of the architecture doc explains why the two disagree and must never be mixed in one number):

TIER (<code>ENGINE.CORPUS.TIERS()</code>)	CODE S	MEANING
ANALYSIS_READY	268	usable transcript (words>0, timed segments) and frames
FRAMES_ONLY	22	frames extracted, no <code>transcripts</code> row at all
TRANSCRIPT_UNUSABLE	7	a transcript row exists but is empty — no speech detected
INGESTED_NOT_ANALYZED	2,914	Hiker metadata only
Total unique codes	3,211	

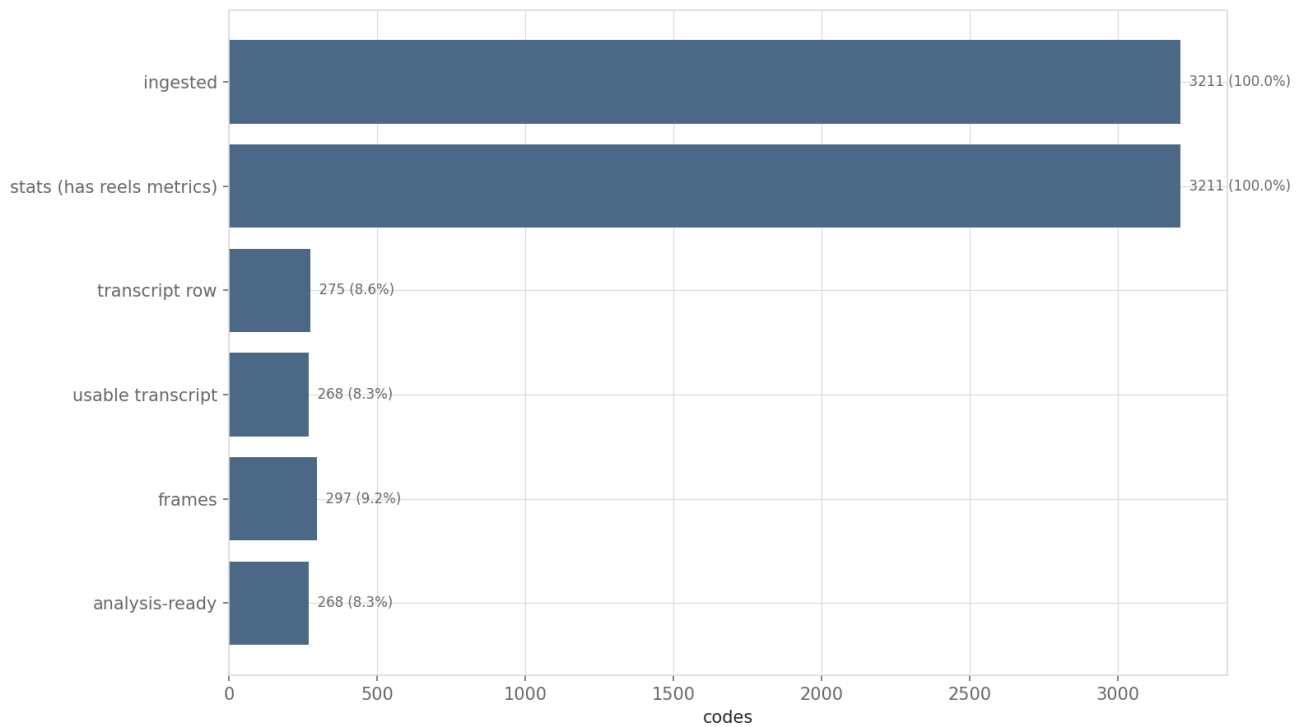
The second vocabulary, `video_state.corpus_tier` , reports `ANALYSIS_READY` 268 / `FRAMES_ONLY` 29 / `INGESTED_NOT_ANALYZED` 2,914 — it folds the 7 empty transcripts into `FRAMES_ONLY` rather than giving them their own tier. Both are correct for what they measure; neither substitutes for the other.

Denominators, named explicitly per number as the method requires:

DENOMINATOR	N	THE ONLY QUESTIONS IT MAY ANSWER
N_ingested	3,211	Metric-only statements: performance, creator norms, temporal trends
N_transcript	275	"How many transcript rows exist" — nothing else (7 of them empty)
N_transcript_usable	268	Every statement about language, script or wording
N_frames	297	Every statement about what is on screen
N_ready	268	Every statement joining script and visuals

266/3,211 ≈ 8.3% is the honest coverage figure for any structural claim about the niche; it is not a random 8.3% (chapter 8/9 and the analysis chapters explain the selection bias this implies).

Corpus coverage funnel (n=3211)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Corpus coverage funnel

The funnel traces the same narrowing — 3,211 ingested down to 268 fully analysed — as a single picture; read it alongside the denominator table above, never in place of it.

The newest-snapshot backlog, queued for the watchdog

Snapshot 3 (2026-09-10, id 3) holds 1,535 codes. **1,312** have neither transcript nor frames, **410** of those seen for the first time in that snapshot (the pre-migration audit figure was 1,313/411; the one-code shift is the August-archive resolution). Their signed CDN media links have expired, so reprocessing any of them costs a fresh HikerAPI call, not a free pass. `engine.watchdog` selects and processes exactly this set from already-cached URLs without paying Hiker again, but as chapter 8 records, it has not yet run against this backlog.

Missingness flags (reports/data/flags.json)

FLAG	CODES	DEFINITION
MISSING_TRANSCRIPT	2,936	No transcripts row
MISSING_FRAMES	2,914	No frames rows
STALE_METRICS	1,676	Not present in the newest completed snapshot
DUPLICATE_VIDEO	56	Same author, byte-identical caption over 30 chars, different code (25 groups)
UNALIGNED_TRANSCRIPT	7	transcripts.segments empty or unparseable
MISSING_STATS	1	play NULL or 0 in every snapshot row (B-MYSMhHXy6)
MISSING_MEDIA	1	Deep dive ran but no usable frame files landed (DcxV37-CJ0C)
BROKEN_FRAME_REFERENCE	1 (8 rows)	frames.path points at a non-existent file
UNRESOLVED_CREATOR_ID	1	More than one (pk_user, username) pair across snapshots (DcFm9i5s5q8)

3,006 of 3,211 codes (94%) carry at least one flag — mostly MISSING_TRANSCRIPT / MISSING_FRAMES , i.e. simply unprocessed, not damaged. Real damage is concentrated in a handful of codes named above by their exact code, not by count alone.

Data-loss audit, per stage

STAGE	EXPECTED	OBSERVED	MISSING	FAILED	DUPLICATES	ORPHANS	FIX APPLIED	UNRESOLVED
Ingestion (Hiker → reels)	2,363 reels declared, snapshot 1	2,352 rows	11	0	1,936 rows (by design, one reading/snapshot)	0 from unknown account	none	11 reels lost in JSON→SQLite migration, snapshot 1 only
Transcription	3,211 candidate codes	275 rows (281 post-migration)	2,936	7 empty (EMPTY_NO_SPEECH)	0	6 (transcripts_without_reel)	August archive import, +8	6 orphan codes, no paid re-lookup done
Frame extraction	297 codes sampled	2,653 rows / 2,645 files (2,727/2,719 post-migration)	2,914 codes with none	8 rows, 1 code (Dcx-V37-CJOC)	9 stale jpgs (DbYh2P-MQnj) + 114 non-reel files (web/)	6 (frame_codes_without_reel)	August archive import, +74	8 broken references, source video gone, unrecoverable
Scoring	3,211 codes	3,205 scored (99.8%), 2,961 eligible	6 (no usable baseline)	0	n/a	0 (scores_without_reel)	none	6 codes (1 zero-play, 5 by one author)
Topic tagging	3,211 codes	1,454 tagged (45.3%)	1,757 untagged	0	n/a	0 (topics_without_reel)	ongoing manual/sample tagging	1,757 backlog + 80 captions in unmatched.md
Cards (legacy)	n/a, candidate-driven	23 rows, 20 distinct codes	ids 15–23 absent (9 rows)	0	3 codes recur across weeks (by design)	0 (cards_without_reel)	none	9 deleted/rolled-back ids, not investigated further
Identity resolution	3,211 codes, 1 creator each	3,210 clean	n/a	n/a	n/a	n/a	none	1 code (DcFm9i5s5q8), two (pk_user, username) pairs
August archive import	8 legacy codes	8 transcripts + 74 frames imported	n/a	n/a	0 base64 fallbacks needed	6 (orphans_not_in_reels)	full import, this migration	6 codes with no reels row, excluded from every denominator

foreign_key_violations: [] and integrity_check: ok database-wide (reports/migration-validation.json) — the loss above sits at the data-completeness layer (unprocessed backlog, one truncated download, one pre-schema archive), not at referential integrity. No fix in this audit silently discarded a record; every unresolved item is named by exact code and kept in whichever table already held it.

11. Creator Analysis

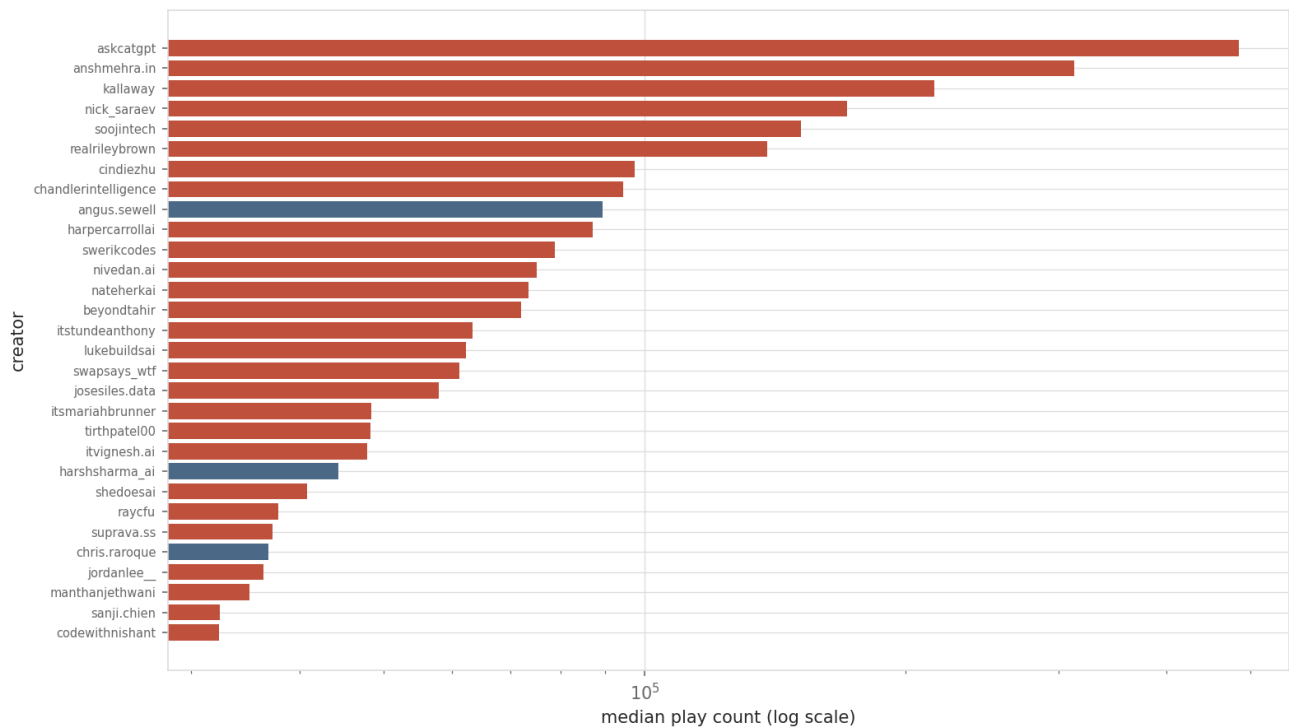
Denominators. 132 creators carry a `creator_stats` row at snapshot 3 (2026-09-10). 102 of them contribute at least one reel to the analysis-ready corpus of 268, at a median of **2 reels each** (p75 4, max 6). Creator-level statements below use n=132; reel-level statements inside the analysed set use n=268 and are strong-vs-strong by construction [I-01]. The `followers` table is empty, so **creator growth cannot be answered at all** — no statement about audience gain appears anywhere in this report.

The landscape

	VALUE
creators with a <code>creator_stats</code> row	132
CONSISTENT (CV < 1 and n ≥ 8)	17 (13 %)
HIGH_VARIANCE	115 (87 %)
SMALL_SAMPLE (n < 5)	0
median videos in baseline	26
median creator median play	14 936 (p10 3 099, p90 71 198)
median consistency score, $1/(1+CV)$	0.352
median MAD-to-median ratio	0.47
Spearman(median_play, mad_play)	+0.967
max CV	4.77 (<code>adamstewartmarketing</code>)
posts above 5× their creator's rolling median	269 of 3 078 (8.7 %), 101 creators

The table describes a lottery, not a craft. Eighty-seven per cent of tracked creators cannot repeat their own results, the median account's typical reel lands within roughly half a median of its own median, and dispersion rises almost perfectly with level (+0.967) — which is exactly why both `score.py` and `engine/stats.py` compute the z on $\ln(1 + \text{play})$ rather than on raw views. The practical consequence is that **a creator's median is the only fair yardstick for their own reels and is useless for comparing one creator with another**: a 2× at a 483 982-play median and a 2× at a 459-play median are not the same event.

Creator median views, top 30 (n=30)

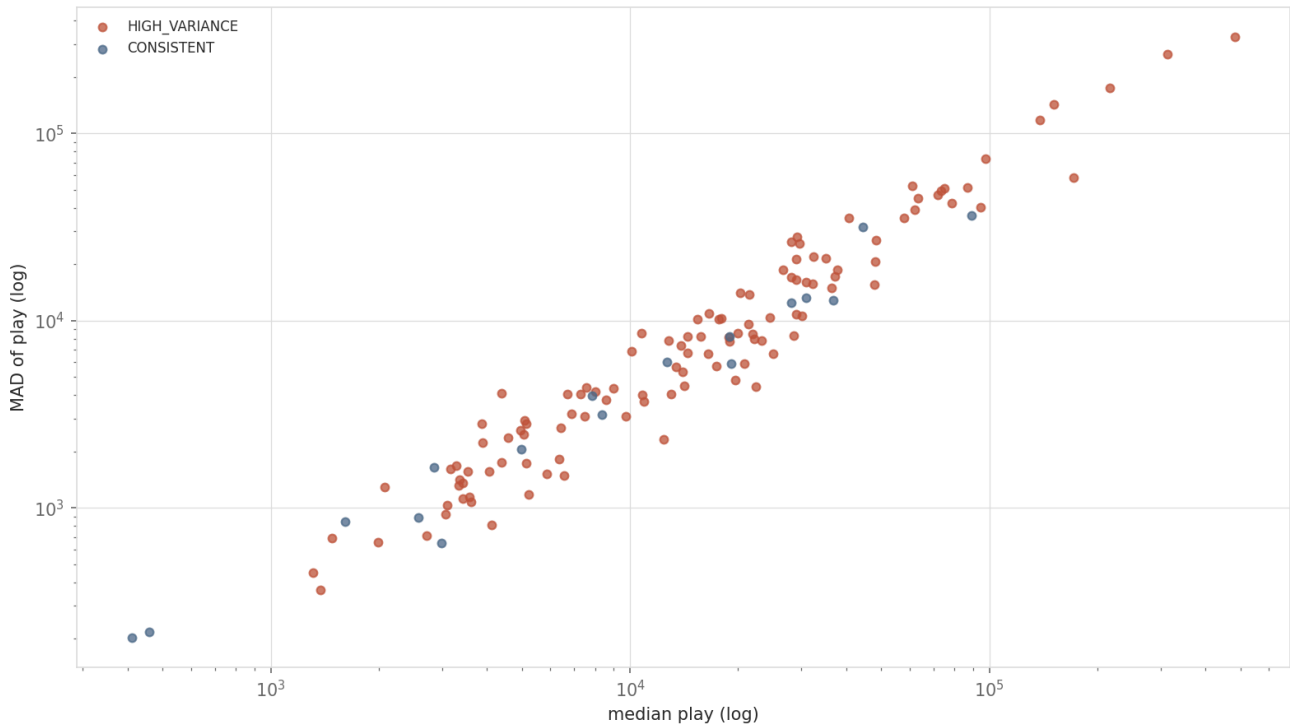


M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Creator median views, top 30 – the roster spans more than an order of magnitude

The top 30 accounts run from `askcatgpt` at a 483 982-play median down to `codewithnishant` at 32 292, against a roster-wide median of 14 936. Only 3 of those 30 carry the CONSISTENT label; the other 27 are HIGH_VARIANCE, so most of the bars represent a median that any single post can swing. Reading this chart as a ranking of "who is good" would be a mistake — it is a ranking of who has an audience, and the two are separate questions answered by the next figure.

Creator median vs MAD (dispersion) (n=132)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Creator median vs MAD – dispersion rises with level

Every creator sits close to the diagonal: bigger accounts are not steadier accounts, they are bigger and equally unsteady. The 17 CONSISTENT creators cluster below the line and the 115 HIGH_VARIANCE creators above it. Because MAD and median are both plotted on the raw play scale, a creator with one enormous outlier can still show a low MAD if the rest of their output is flat — a second reason the log-space z is the number that travels into every downstream table.

The seventeen consistent creators

USERNAME	N	MEDIAN PLAY	CV	POSTS/WEEK	SHARE/1K	SAVE/1K
angus.sewell	12	89 439	0.98	4.39	9.6	23.1
harshsharma_ai	12	44 377	0.96	2.59	13.9	—
chris.raroque	12	36 820	0.80	5.60	6.8	23.3
decodingai.vs	29	30 932	0.73	4.09	12.6	24.5
ajsahni.ai	12	28 067	0.95	1.64	4.4	—
justyn.ai	30	19 173	0.91	0.72	5.8	17.9
divyannshisharma	27	18 948	0.78	1.47	22.0	36.1
damini.knows	14	12 678	0.64	3.06	19.9	31.4
seb.ai	42	8 344	0.74	14.55	13.5	32.0
umangratani	23	7 825	0.51	0.13	1.3	0.9
nicholas.puru	32	4 988	0.72	5.14	2.9	11.6
omarmeski	24	2 979	0.37	0.81	4.0	8.6
seymourofdevin	12	2 842	0.69	7.71	4.8	20.7
brycenwood.ai	24	2 572	0.42	1.39	5.3	7.1
pritam.nagrle	29	1 613	0.61	3.49	2.3	6.2
automationatlas.co	48	459	0.90	32.13	0.0	0.4
hcmetellus	25	410	0.68	0.47	0.0	0.0

Consistency and distribution are separate properties, and only the top seven rows have both. `automationatlas.co` posts 32 times a week to a 459-play median and `hcmetellus` is steady at 410 — a steady nothing is still nothing. The two rows worth copying are `divyannshisharma` and `damini.knows` : consistent **and** above the analysed-set medians on both hi-intent rates (corpus medians 0.0152 share, 0.0278 save, n=268 and 245). Cadence itself explains nothing — the roster spans 32.1 posts a week (`automationatlas.co`) to 0.13 (`umangratani`) with no relation to performance, and `posts_per_week` is a lower bound throughout because the corpus holds only what was visible on a profile page at collection time.

The measurement trap this creates

The 28 analysed reels from CONSISTENT creators carry a median `robust_z` of **0.60** and `view_lift` **0.65**. The 240 from HIGH_VARIANCE creators carry **1.37** and **2.38**. Their scripts are the same object: median hook 6.7 s on both sides, spoken length 53.8 s against 53.5 s, 3.4 against 3.5 words per second. The gap is arithmetic, not editorial — both metrics divide by the creator's own spread, so a steady creator produces small lifts from genuinely good reels and a lottery account produces enormous lifts from

ordinary ones [I-23]. Two real differences in shape survive: consistent creators spend less on proof (11.2 s vs 18.6 s) and longer on the closing line (6.3 s vs 4.8 s).

Transcript reading. `damini.knows` is the corpus's cleanest example of what consistency looks like in a script. Her `DcoE5ZsK4I7`(<https://www.instagram.com/reel/DcoE5ZsK4I7/>) opens "With one prompt, you will create some crazy animated websites. First, understand the whole process." (hook, 0.0–4.9 s) and then does exactly that in 49 seconds. The reel is 1.4× her own median — a number that would be discarded by any lift-ranked filter — while carrying 36.1 shares and 62.4 saves per 1 000, both far above the corpus median. This is what a reliable creator's win looks like: unremarkable reach, exceptional intent.

Visual reading. The same reel runs `A_ROLL_MEDIUM > UI_DEMO > SPLIT_SCREEN > UI_DEMO` — host, then the generated site, then the extension, then the site again. `fa-v1` records the proof visuals as two finished website mockups and the `Woblo` browser extension itself. Nothing is asserted that is not shown, which is the visual equivalent of a low variance.

Strategic implication

Three rules follow for M2. First, **every reference must carry its creator's `reliability`, CV and baseline `n` on the card**, because a 300× from a 1 374-view median and a 1.4× from a steady 12 677-view median are not comparable claims. Second, **prefer references from CONSISTENT creators or from creators with more than one entry in the pool** — the reference pool in `06-creators-and-references.md` contains exactly one CONSISTENT row out of 40, which is a property of the selection rule, not of the niche. Third, when M2's own account becomes consistent, **expect its multipliers to shrink**, and do not read that shrinkage as decline; it is the same arithmetic operating on M2.

Confidence: RELIABLE for the landscape description (n=132, complete roster, direct measurement). **PROBABLE** for "who is worth studying", which rests on eight rows and on a share/save comparison that does not hold `cta_type` constant. **INSUFFICIENT** for anything about follower growth: the table is empty and the question cannot be asked of this dataset.

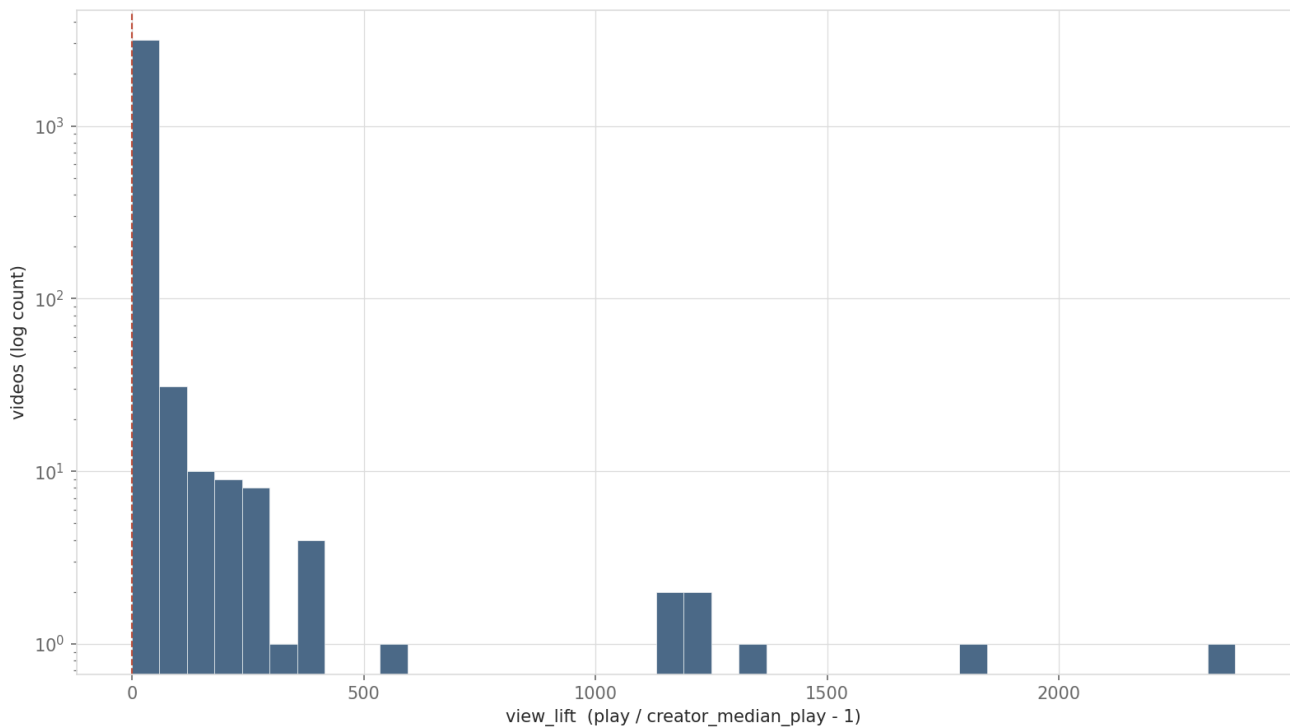
12. Video Analysis

Denominators. This chapter works at the level of the individual reel. Metric-only statements use **N_ingested = 3 211** (every unique code with a `reels` row; 3 209 have computable rates after the 100-play denominator guard). Statements that need a script or a frame read use **n=268** (analysis-ready) and are strong-vs-strong [I-01]. 56 codes carry a `DUPLICATE_VIDEO` flag and 1 676 a `STALE_METRICS` flag; 3 006 of 3 211 codes carry at least one flag of some kind, which is a statement about pipeline hygiene, not about content quality.

How a reel performs relative to its own author

	VALUE
median <code>view_lift</code> across 3 211	0.00 — the corpus is centred exactly on the author's own median
reels below their author's median	49.4 %
reels at or above 2× their author's median	25.1 %
p95 <code>view_lift</code>	+20.96
maximum	+2 381 (<code>Cq3Lkw1JaBv</code> , loganwelbaum, 7 550 238 plays on a 3 170 median)
<code>outlier_status</code> HIGH (<code>robust_z</code> ≥ +2)	375
<code>outlier_status</code> LOW (≤ -2)	49
NORMAL	2 787
rows with <code>robust_z_floored</code> = 1	0 — no magnitude in this wave is an upper bound for that reason

Video performance ratio distribution (view_lift) (n=3211)



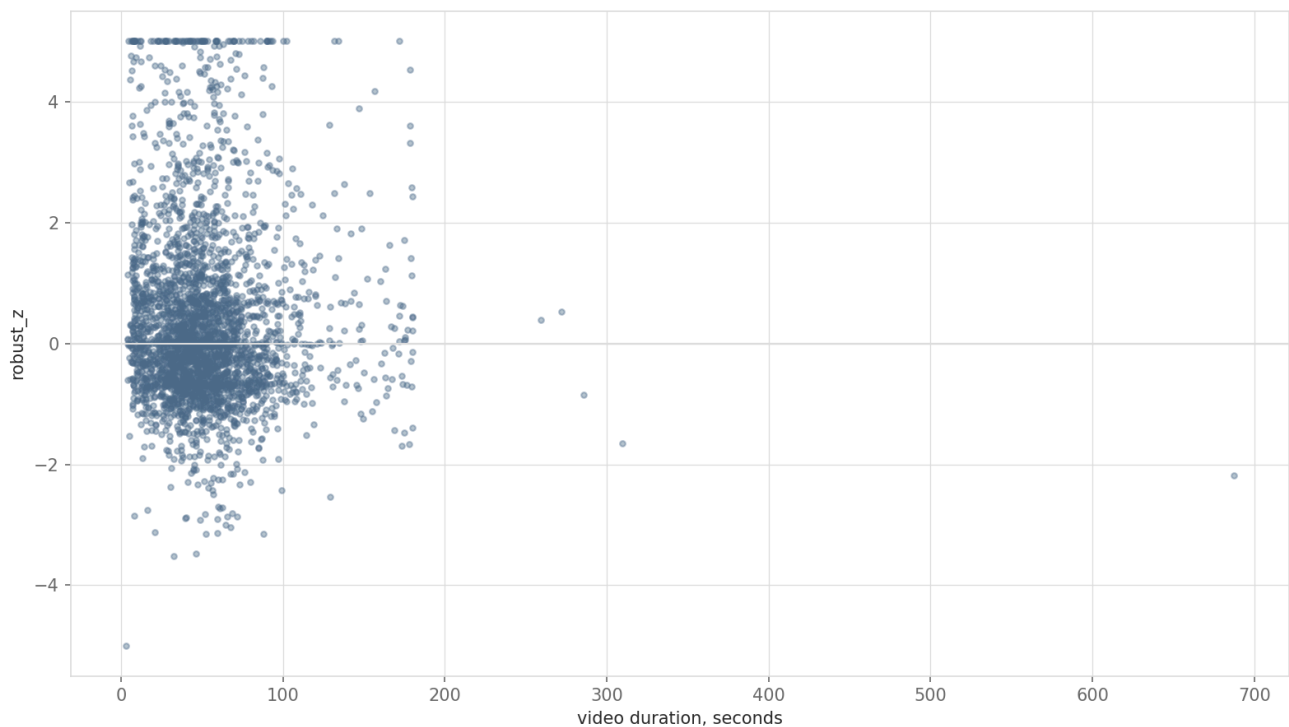
M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Video performance ratio distribution – view_lift across 3 211 reels, log count axis

The log y-axis is not cosmetic. The bulk of the mass sits between -1 and $+2$ and everything interesting about this niche lives in a tail that a linear axis would erase. Two readings follow. First, a "winner" in this corpus is a rare event: 375 of 3 211 reels (11.7 %) reach `robust_z` ≥ 2 , and 150 (4.7 %) exceed ten times their creator's rolling median. Second, the tail is so long that the top of it is uninterpretable on its own — a $+2$ 381 is a statement about a 3 170-view baseline, and RULES.md §6's label for that case ("резонировало у своих" — resonated with its own audience) is the honest one. Every multiplier in this report should be read next to its denominator, never alone.

Duration is not a lever

Video duration vs performance (n=3211)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

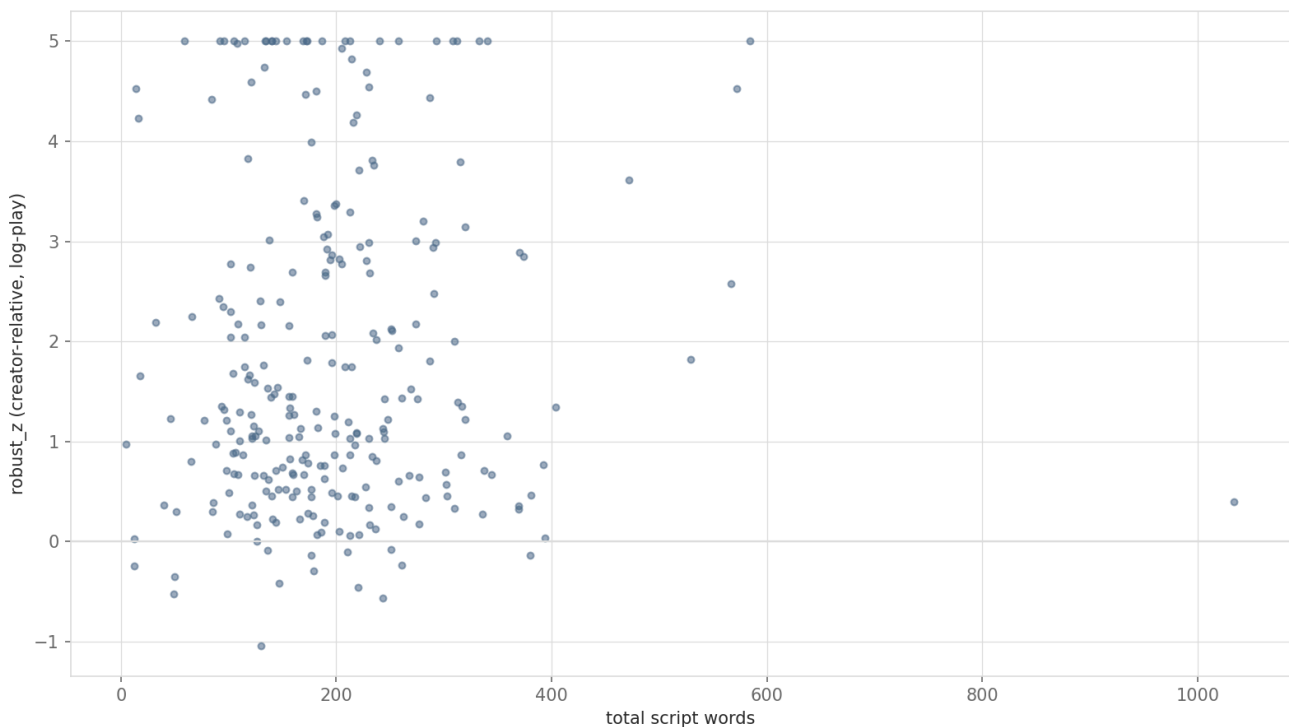
Video duration vs performance across all 3 211 ingested reels

Across the full ingested corpus the Spearman correlation between duration and `robust_z` is **-0.069** ($p = 9.5e-05$ — significant only because n is 3 211) and against `view_lift` **-0.0004**. The band comparison is flat: under 20 s median `robust_z` **0.25**, 20–45 s **0.00**, 45–70 s **-0.01**, 70–90 s **-0.19**, over 90 s **+0.05**. The 375 HIGH outliers have a median duration of **46.8 s** against **47.8 s** for the other 2 836, and **14 %** of them are under 20 s against **12 %** of the rest.

Two numbers previously published inside M2 do not reproduce here and are corrected in chapter 33: PRODUCTION.md's "the winners' median is 55 seconds against 47 for the rest" and "under twenty seconds is 5 % of winners and 13 % of the rest". Both came from a 100-reel top slice of an earlier 2 352-reel export; on 3 211 creator-normalised reels they become 46.8 s vs 47.8 s and 14 % vs 12 %. The direction of the second claim is reversed, not merely attenuated [I-02].

Script length is not a lever either

Script length vs performance (n=268)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Script length vs performance, 268 analysed reels

Across the 268 codes with both values, total spoken word count correlates with `robust_z` at **+0.048** ($p = 0.44$), and the strong `robust_z` quartile carries a median 198 words against 179 for the weak quartile (Mann-Whitney $p = 0.33$). The cloud is as wide at 100 words as at 400. Median spoken length in the analysed set is **53.6 s / 182 words / 3.52 words per second**; median video duration is 54.2 s against 47.6 s across all 3 211, which is a selection effect of the analysed tier, not a finding.

Transcript reading — both extremes work. The corpus's shortest script is [DZgBxjnBzMe](https://www.instagram.com/reel/DZgBxjnBzMe/): **14 spoken words in 27 seconds**, at **207×** its author's median and 1 570 719 plays. Its entire hook beat is "Welcome back." (5.3–7.3 s) and its entire proof beat is "Unlock my phone. Open Geohot Star in my phone. Call my sister." (7.3–23.0 s). Its opposite is [DLzPQzGNEmY](https://www.instagram.com/reel/DLzPQzGNEmY/) — 340 words in 100 seconds, 72.5× on a 22 121 median. A rule expressed in words or seconds would have to condemn one of them.

Visual reading. The two reels are also visual opposites and both are legible. [DZgBxjnBzMe](https://www.instagram.com/reel/DZgBxjnBzMe/) holds a single `B_ROLL_PROCESS` state across all nine samples — fa-v1 records a camera filming a laptop running a Jarvis Python script, a "J.A.R.V.I.S V6.5" HUD, and a phone changing state beside it, under one white title pinned for the whole reel. [DLzPQzGNEmY](https://www.instagram.com/reel/DLzPQzGNEmY/) is a held `SCREEN_RECORDING`. What both share is not a length or a cut rate: it is that **the thing being claimed is visible while the claim is made**.

What this chapter licenses

Nothing about form. Of the video-level properties measurable here — duration, spoken length, words per second, cut rate, scene count, caption density — not one separates the top from the bottom `robust_z` quartile at $p < 0.01$, and the negative result was obtained inside a sample pre-selected for strength, which

should have made a difference *easier* to see [I-22]. The video-level variable that does carry signal is not a shape at all: it is how many distinct named things the script contains (`lex_entities_distinct` → `view_lift` +0.161 pooled, **+0.198 creator-normalised**, the only reach correlation in 460 pairs that survives normalisation).

Strategic implication

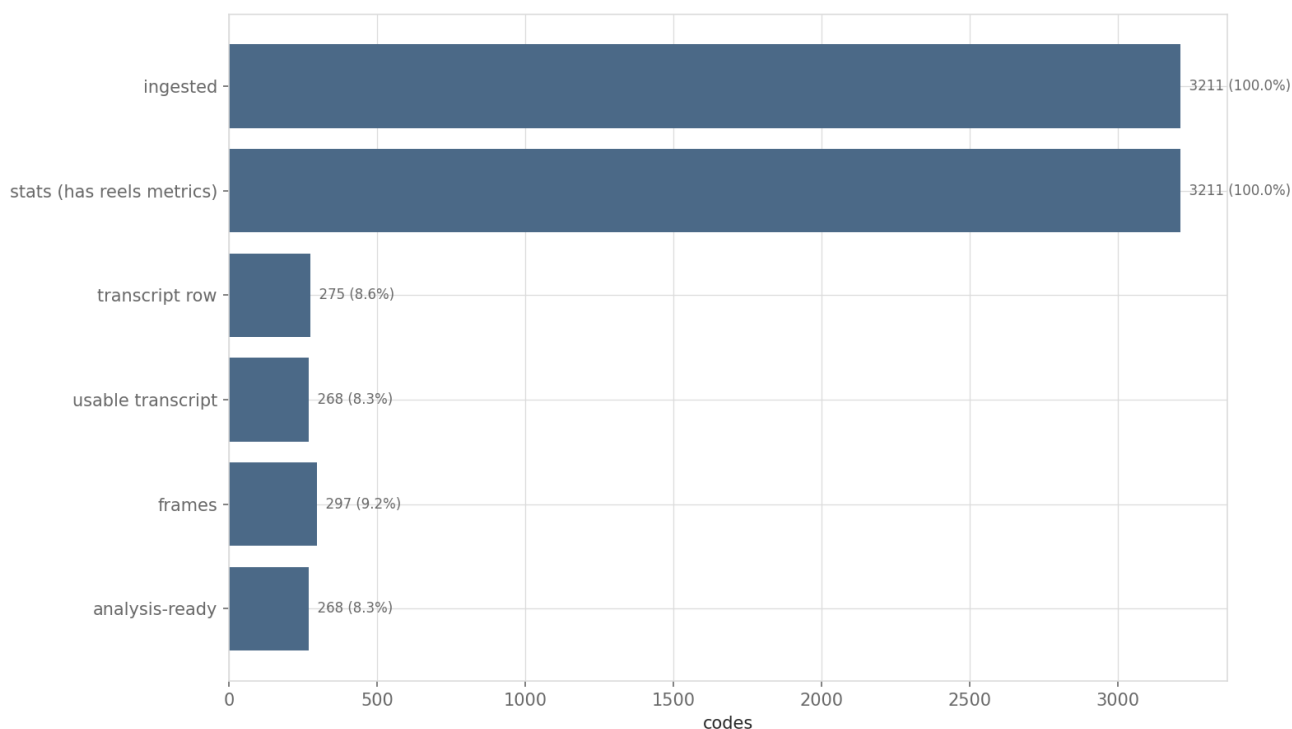
Keep M2's 50–70 second band as a **shooting convenience** — it fits one filming day and one take — and stop citing it as an evidence-backed reach lever. Never cut a good idea short to hit the band and never pad a thin one to reach it. Judge an individual M2 reel against M2's own accumulated median, carry the baseline `n` with every multiplier, and expect the first six months of M2 numbers to be uninterpretable in isolation: with fewer than 5 baseline videos the engine itself returns `SMALL_SAMPLE` rather than a score.

Confidence: RELIABLE for the duration null and the lift distribution (n=3 211, complete corpus, direct measurement). **RELIABLE** for the script-length null within the analysed tier (n=268). **PROBABLE** for the two-extremes reading, which rests on two named reels chosen because they are extremes. No causal claim is licensed by any number above.

13. Transcript Categorization

Denominators. 275 codes have a `transcripts` row; **7 of those rows are empty** (Whisper with `vad_filter` heard no speech — music-only or text-on-screen reels) and get their own `TRANSCRIPT_UNUSABLE` tier rather than being folded into frames-only. **268** transcripts are usable (words > 0 **and** timed segments); **266** carry a ta-v1 semantic read; **264** produced a beat parse, yielding **1 527 beats**. Every language, script and wording statement in this report divides by 266 or 264 and says which. The four codes with a transcript and frames but no beat-derived script block (`features_json.script.part_source != 'beats'`) are excluded from all `*_s` columns by design — the clock-based timing fallback (method §5.1) never writes those columns, because a five-second hook by the clock is not a hook by meaning.

Corpus coverage funnel (n=3211)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Corpus coverage funnel — 268 of 3 211 codes carry both a script and a frame read

The funnel is a denominator comparison, not a drop-off sequence: frames and transcripts are extracted independently, so the stages are not strictly nested. What it establishes is the number that governs this entire report — **268 / 3 211 = 8.3 %**, and not a random 8.3 %. `deep.py` only ever processed reels `score.py` had already ranked near the top of their creator's output, so the median `robust_z` inside the analysed tier is **+1.264** against **0.000** across all 3 211 [I-01]. Read every categorical table below as a description of the already-successful part of the niche.

What ta-v1 produces per reel

Each of the 266 semantic reads returns three layers. **Closed vocabularies** — `topic`, `subtopic`, `pain`, `solution_type`, `proof_type`, `hook_type`, `cta_type`, `narrative`, `positioning_type`, `funnel_role` — are normalised against a fixed list and the raw label is always kept alongside. **Free prose** — `subject`, `thesis`, `audience`, `desire`, `mechanism`, `objection`, `payoff`, `rhetorical_devices`,

`visual_to_script_sync` — is a judgement, not a measurement. **Counts** — `open_loops` , `pattern_interrupts` , `numbers_used` , `tools_mentioned` , `models_mentioned` — are the model's own tallies over the transcript. A fourth block, `quality` (specificity, novelty, clarity, info_density, grounded), and a self-assessed `confidence` travel with every row.

TA-V1 SELF-ASSESSED CONFIDENCE	CODES
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RELIABLE	161
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PROBABLE	94
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INSUFFICIENT	11
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The 11 INSUFFICIENT rows are kept, not dropped, and this is the right decision: a suppressed row is a fact about the data. The clearest case is [DcxV37-CJOC](https://www.instagram.com/reel/DcxV37-CJOC/)(<https://www.instagram.com/reel/DcxV37-CJOC/>), where ASR captured five words — **"Did you know that the-"** (hook beat, 0.0–0.8 s) — of a 62-second reel that reached **559 515 plays** on an 83 168 median and carries the corpus's third-highest save rate at **72.2 per 1 000**. Nothing in this report's script findings can see that reel. It is the standing reminder of what transcript-only analysis cannot reach.

The frames can see it, though. fa-v1's read of the same code records a host holding up a small printed card with a pink dot pattern under a cut-out caption reading "How _ Advanced Design Effects", one held `A_ROLL_MEDIUM` state across all nine samples and no proof visual at all. That is a complete description of a reel the transcript pipeline registered as four broken words — and it is the argument for keeping both analyses joined at the code rather than choosing one.

Beat roles and how they fold

The beat vocabulary has twenty roles. Raw frequencies over 1 527 beats: hook 259 · mechanism 250 · cta 205 · example 157 · payoff 145 · proof 104 · explanation 96 · solution 84 · context 40 · setup 31 · tension 31 · problem 29 · closing 23 · objection 19 · rehook 15 · pain 12 · transformation 10 · other 5 · audience 3 · hook_extension 2. Those fold into the eight script-part columns of method §5 (`hook_s` ← hook + hook_extension; `problem_s` ← problem + pain + tension; `explanation_s` ← explanation + mechanism + context; `proof_s` ← proof + example; `payoff_s` ← payoff + transformation; `cta_s` ← cta + closing). `objection` , `rehook` and `other` deliberately stay outside the eight, because they are moves that can appear anywhere and folding them in would make the shares meaningless; their seconds are kept as `unbucketed_s` , which is non-zero in only **35 of 264** reels.

Two transcription defects that change how counts must be read

Punctuation. Whisper punctuates **255** of the 266 usable transcripts; the other **11** have none at all, and each timed segment becomes one pseudo-sentence at Whisper's own VAD phrase boundary. `sentence_source` records which happened on every row, so per-sentence statistics from the two groups are never silently pooled.

Product names. The keyphrase bigrams contain **"claude code" 11** and **"cloud code" 7** — Whisper mishears the product name consistently. The `tools_mentioned` vocabulary therefore carries both spellings as separate rows (Claude code n=16 at median view_lift 7.81; Claude Code n=13 at 0.87; cloud code n=8 at 3.02), and they are reported as they are rather than silently merged. The practical consequence:

`lex_entities_n` counts **mentions of anything the lexicon knows**, not distinct products — read `lex_entities_distinct` for that.

How much of the grid can actually carry a claim

Of **159** reported categorical cells across the eight vocabularies, **18 are SUFFICIENT_SAMPLE** ($n \geq 30$ and ≥ 8 creators), **96 are PROBABLE** and **45 are INSUFFICIENT** ($n < 5$ or < 3 creators). All **20** `subtopic` cells are INSUFFICIENT — every one is $n \leq 2$. Only one `hook_type` (`bold_claim`), two `pain` values, three `solution_type` values, three `proof_type` values and two `cta_type` values clear the size threshold, and **no topic cell does** [I-30]. Coverage inside the analysed set is otherwise near-complete: 266 of 268 rows carry a topic, a pain and a hook type; 256 carry a CTA type.

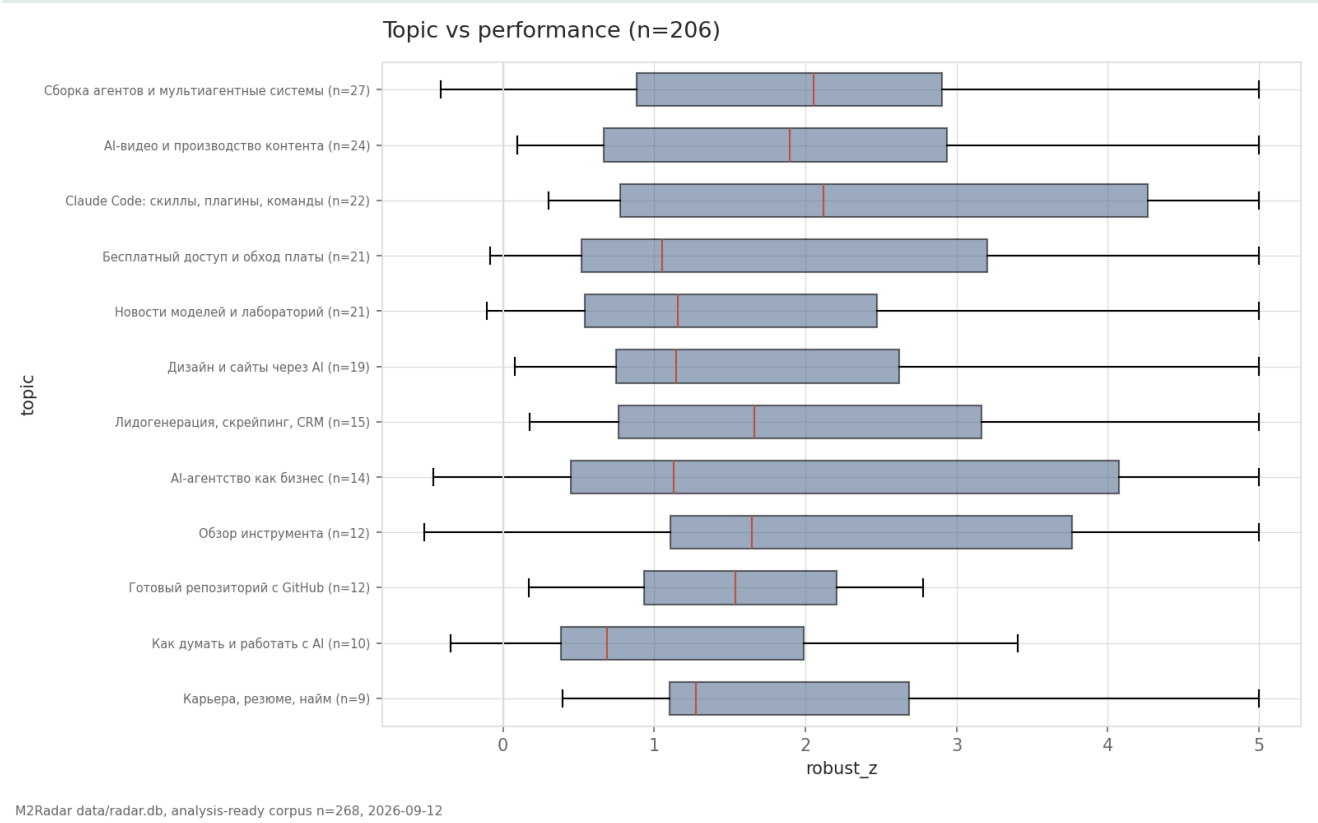
Strategic implication

Two operating rules. First, **no M2 hypothesis or card may cite a category cell below $n=5$ or 3 creators**, and every cell it does cite must carry its `n`, `n_creators` and confidence label onto the card — the thinness of the grid is itself a finding, and hiding it would convert a suppressed cell into a false claim. Second, **the next processing wave must buy breadth, not depth**: more codes per cell, not more features per code. Adding a twelfth lexical family to 266 reels changes nothing about the 20 subtopic cells that cannot be spoken about at all.

Confidence: RELIABLE for the coverage counts and the confidence distribution (direct counts over the pipeline's own outputs). **PROBABLE** for anything resting on the closed vocabularies themselves: they are LLM judgements normalised to a fixed list, not measurements, and `analysis_confidence` travels with every row for that reason.

14. Topic Analysis

Denominators. 266 of the 268 analysis-ready reels carry a ta-v1 `topic` label, spread across 27 distinct values. The chart below draws the twelve largest cells, which together hold **205** reels; groups under 5 reels are dropped from the plot but kept in the table. **No topic cell reaches the SUFFICIENT_SAMPLE threshold** of $n \geq 30$ and ≥ 8 creators, so every row is PROBABLE at best and 9 of the 27 are INSUFFICIENT [I-30]. Baselines for comparison, all inside the analysed set: median `view_lift` **2.03**, median `robust_z` **1.26**, median share rate **0.0152** (n=268), median save rate **0.0278** (n=245).



Topic vs performance – twelve largest cells, n=205

The full topic table

TOPIC	N	CREAT ORS	MEDIAN VIEW_LIFT	MEDIAN ROBUST_Z	SHARE /1K	SAVE/1K
Сборка агентов и мультиагентные системы (agent building)	27	19	6.63	2.06	17.4	31.7
AI-видео и производство контента	23	19	2.12	1.75	18.0	34.3
Claude Code: скиллы, плагины, команды	22	17	4.36	2.12	15.3	37.5
Бесплатный доступ и обход платы (free access)	21	16	1.17	1.05	22.0	37.3
Новости моделей и лабораторий (model news)	21	17	2.30	1.16	18.3	9.7
Дизайн и сайты через AI	19	16	2.70	1.14	13.5	23.7
Лидогенерация, скрейпинг, CRM	15	10	2.27	1.66	16.5	30.2
AI-агентство как бизнес	14	9	0.68	1.13	12.8	21.3
Обзор инструмента (tool review)	12	11	5.67	1.65	17.8	21.6
Готовый репозиторий с GitHub	12	8	4.03	1.54	17.4	48.7
Как думать и работать с AI	10	8	0.44	0.69	9.4	20.8
Карьера, резюме, найм	9	8	2.72	1.27	12.8	22.9
Стартапы и венчур	9	9	2.49	0.71	14.4	31.0
Программирование и код	9	9	1.63	1.04	10.4	22.7
Обучение и навыки	8	8	0.57	0.38	10.6	26.2
AI в конкретном бизнес-процессе	8	8	0.65	0.85	4.0	13.3
Личное, мотивация, влог	6	6	1.34	0.89	6.2	7.8
Токены, стоимость, лимиты	5	5	0.89	0.57	5.7	14.8
Голосовые агенты и телефония	4	4	4.55	3.11	15.2	25.8 (INSUFFICIENT)

TOPIC	N	CREATORS	MEDIAN VIEW_LIFT	MEDIAN ROBUST_Z	SHARE /1K	SAVE /1K
Развлечение и конспирология	3	3	18.32	1.48	6.5	13.6 (INSUFFICIENT)
remaining 7 cells (n ≤ 3 each)	≤ 3	≤ 3	—	—	—	INSUFFICIENT

Read down the `view_lift` column and the niche's shape is unambiguous: **reach concentrates where the subject is a thing a builder can assemble** — agents (6.63×), Claude Code skills (4.36×), a ready-made GitHub repo (4.03×), a tool review (5.67×) — and collapses where the subject is a way of thinking: "how to think and work with AI" (0.44×), "learning and skills" (0.57×). The save column tells a different and equally consistent story: **the topic people keep is the topic that hands over a file**. Ready-made GitHub repos carry 48.7 saves per 1 000 against a corpus median of 27.8, while model news — which reaches perfectly adequately at 2.30× — is saved at **9.7 per 1 000**, roughly a third of the median. Novelty gets attention and is not kept.

The cell that matters most to M2 is the weakest one

AI в конкретном бизнес-процессе — AI inside a concrete business process, which is M2's declared territory — is **n=8 across 8 creators, median view_lift 0.65, robust_z 0.85, share rate 0.0040 and save rate 0.0133**. The share rate is roughly a quarter of the analysed-set median. It is simultaneously one of the smallest cells that is not suppressed and the weakest on three of four metrics. The neighbouring cell M2 is positioned to refuse, agent building, is three times larger and ten times higher on lift [I-25].

The honest reading is that both facts are real and neither is a verdict. The corpus contains eight reels on M2's subject; eight reels across eight creators cannot establish that the subject fails, and the selection bias runs the other way — these eight are already among the top of their authors' output. What the cell does establish is that **nobody in the measured niche is producing this content at volume**, and PRODUCTION.md's earlier framing of the same gap ("the topic is open for whoever treats it as content rather than a pitch") is confirmed in direction while its supporting numbers, drawn from a different and smaller export, are not reproduced here.

Transcript reading. The gap is audible in the language. The agent-building topic's own top reel, [DbYh2P-MQnj](https://www.instagram.com/reel/DbYh2P-MQnj/) (https://www.instagram.com/reel/DbYh2P-MQnj/) (2 771 182 plays, 94.8×), opens "This is what it looks like when 137 AI agents run an entire company. Seven departments and it's not just a crazy visual." (hook, 0.0–10.8 s) and never once names a person responsible for anything. Its closest competitor for shape, [DcbnTAtxb8C](https://www.instagram.com/reel/DcbnTAtxb8C/) (https://www.instagram.com/reel/DcbnTAtxb8C/), describes four departments and 33 agents in three grammatically identical sentences. The energy is real; the process owner is absent. That absence is precisely POSITIONING.md §8's "AI-agent demos without a real process owner" and precisely the space M2 says it will occupy.

Visual reading. The topic's visual language is already invented and it is not the niche's default. DbYh2P-MQnj is a held `B_ROLL_PROCESS` : a camera filming a curved monitor showing a live network diagram with a hand pointing at labelled clusters. fa-v1 records the proof visual as an openable agent card naming what each agent replaces. That is a process artefact on camera — the exact visual grammar a Teardown needs — being used to describe an architecture rather than a workflow.

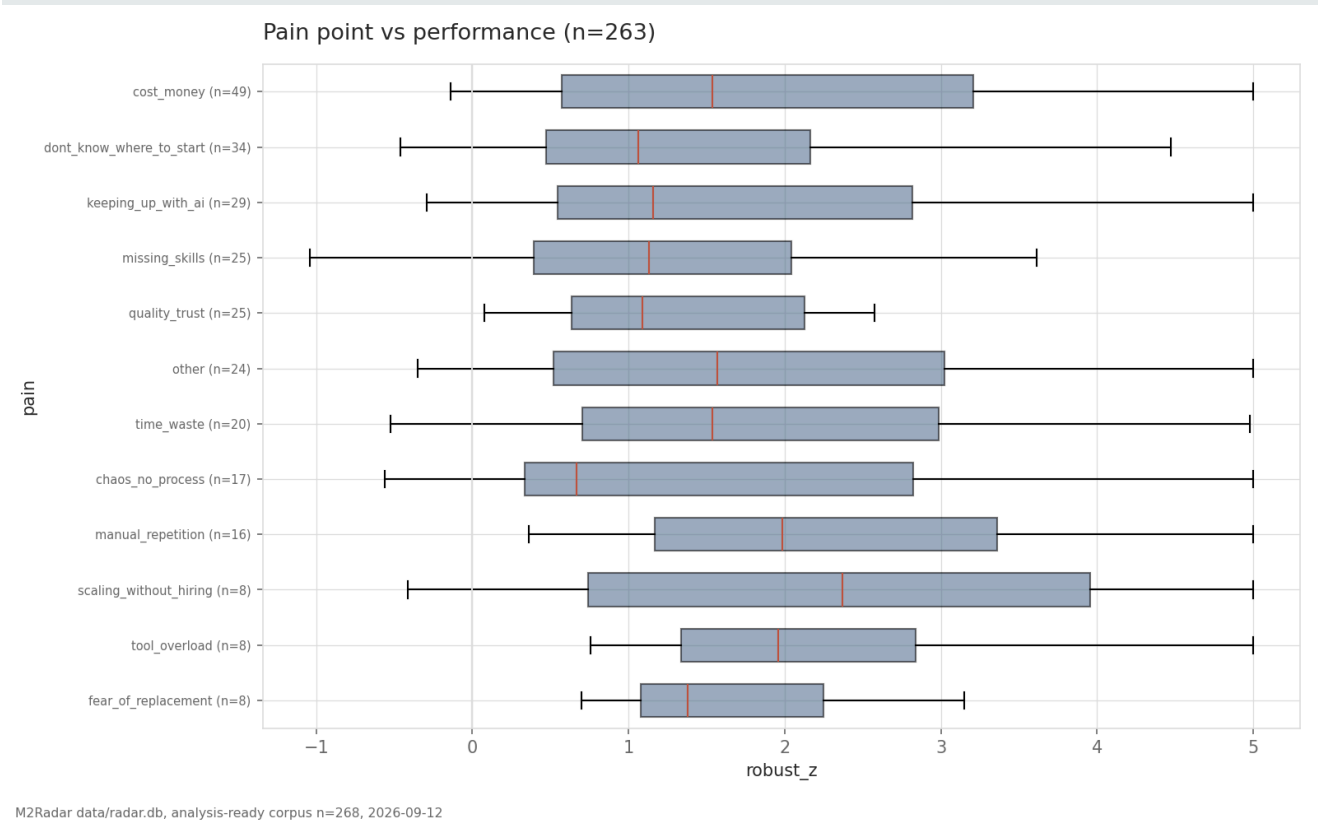
Strategic implication

Take the agent-building topic's **energy and its visual grammar** and refuse its **shape**. Build the same visible artefact, film it the same way, and attach it to one named repeated process with a named owner and an explicit AI boundary — which is the M2 angle formula in RULES.md §4 and which nobody in this corpus does. Two corollaries follow from the table. Do not build a pillar on model news: it reaches and is not kept (9.7 saves/1 000), and POSITIONING.md §8 excludes it anyway. Do not expect the business-process topic to reach like agent building in its first months; select and judge M2 Teardowns on shares and saves per 1 000, not on view lift, because the topic's own measured weakness is concentrated in reach.

Confidence: PROBABLE for every individual topic cell (no cell reaches $n \geq 30$ with ≥ 8 creators; the largest is 27 reels across 19 creators). **INSUFFICIENT** for the nine cells at $n \leq 4$, which are printed with their `suppressed_reason` rather than deleted. **RELIABLE** only for the meta-finding that the topic grid is too thin to answer topic questions [I-30]. Nothing here is causal: these are medians of a pre-selected sample, and a topic's median says nothing about what would happen if a different creator made the same reel.

15. Pain Analysis

Denominators. All 266 analysis-ready reels with a ta-v1 read carry a canonical `pain` label from a closed list of 13 values (12 named plus `other`). The chart draws the twelve largest cells, **n=261**. Two cells clear the SUFFICIENT_SAMPLE size threshold ($n \geq 30$ and ≥ 8 creators); the rest are PROBABLE. No creator-normalised test applies to a categorical median, so **no pain cell can be RELIABLE by the method's own rule** (§8.5). Baselines: median `view_lift` 2.03, median `robust_z` 1.26, share 0.0152 (n=268), save 0.0278 (n=245).



Pain point vs performance – twelve largest cells, n=261

PAIN	N	CREATORS	MEDIAN VIEW_LIFT	MEDIAN ROBUST_Z	SHARE /1K	SAVE /1K	CONFIDENCE
cost_money	48	33	1.96	1.38	16.4	30.3	SUFFICIENT_SAMPLE
dont_know_where_to_start	34	30	1.34	1.06	17.5	34.6	SUFFICIENT_SAMPLE
keeping_up_with_a_i	29	24	1.73	1.16	16.1	18.1	PROBABLE
missing_skills	25	20	1.34	1.13	9.7	24.6	PROBABLE
quality_trust	25	20	1.63	1.09	13.3	27.5	PROBABLE
other	24	21	5.30	1.57	12.5	21.2	PROBABLE
time_waste	20	17	3.54	1.54	19.9	36.4	PROBABLE
chaos_no_process	17	14	0.87	0.67	13.8	26.4	PROBABLE
manual_repetition	15	13	6.94	1.81	16.1	39.0	PROBABLE
scaling_without_hiring	8	8	4.83	2.37	20.9	35.9	PROBABLE
tool_overload	8	8	2.98	1.96	14.4	21.7	PROBABLE
fear_of_replacement	8	6	2.11	1.38	15.3	12.6	PROBABLE
slow_response_to_leads	5	3	0.47	0.35	0.1	0.1	PROBABLE

The finding: an act beats a condition

Order the table by lift and a single distinction explains most of the spread. The two strongest pains name **something the viewer physically does**: `manual_repetition` (6.94× lift, 39.0 saves per 1 000 — the highest save rate of any pain cell) and `time_waste` (3.54×, 36.4 saves). The two weakest name **a property of the viewer's business**: `chaos_no_process` (0.87×) and `slow_response_to_leads` (0.47×, and a share rate of 0.1 per 1 000, which is the lowest number in this report). `scaling_without_hiring` sits with the strong group at 4.83× on eight reels, and it too is phrased as an act the owner is avoiding.

This is uncomfortable for M2 because **the two weakest cells are the two closest to M2's own vocabulary**. "Your process is chaotic" and "you respond to leads too slowly" are the language of a process consultancy, and this corpus does not reward them. "You retype the same numbers out of a PDF every Monday" is the same underlying pain named as an act, and it sits in the 6.94× bucket. The

translation is free: nothing in POSITIONING.md requires the condition phrasing, and RULES.md §4 already asks for "процесс по шагам" — the process step by step — which is the act phrasing by another name.

Two cautions before this becomes a rule. First, `manual_repetition` is 15 reels across 13 creators and `slow_response_to_leads` is 5 reels across 3 — the second is INSUFFICIENT-adjacent and its 0.1 share rate is almost certainly one or two accounts with no distribution rather than a property of the pain. Second, the whole table is computed inside a pre-selected strong sample, so a low cell means "weak among strong reels", not "fails in the niche" [I-01].

What the pain sounds like when it works

[DYuivWzSj5k](https://www.instagram.com/reel/DYuivWzSj5k/)(https://www.instagram.com/reel/DYuivWzSj5k/) (aiforbusinesses247, 113 697 plays, 32.2×, 50.8 shares per 1 000) is 59 spoken words and names no tool, no step and no number. Its whole pain statement is the hook: **"Think, when you wake up in the morning, the clients are already interested in your service."** (0.0–3.7 s). The pain is not described — it is inverted into a morning the viewer can picture. Its sibling [DZLDk9ySi-7](https://www.instagram.com/reel/DZLDk9ySi-7/)(https://www.instagram.com/reel/DZLDk9ySi-7/) from the same account (179 093 plays, 51.4×) does the same move and then lists the friction *after* the payoff: **"No manual data scraping. No cold outreach headache. No manually demo websites. No hours waste to find leads."** (pain beat, 25.8–32.6 s). Placing the pain list third — after the outcome and the mechanism — makes the pain read as a reward rather than an accusation.

Contrast the weakest neighbourhood. [DcmK70aO5VP](https://www.instagram.com/reel/DcmK70aO5VP/)(https://www.instagram.com/reel/DcmK70aO5VP/) reaches only 1.55× its author's accumulated median and yet carries 40.3 shares and 62.1 saves per 1 000, because its three examples — law firm intake, résumé filtering, insurance document collection — are each a named repeated act. Its defect is the frame around them: **"three AI automations that can literally help you make eight thousand dollars in just a month"** (hook, 0.0–5.6 s) prices the pain in income, which costs reach (chapter 26).

Visual reading. The pain-as-act reels show the act. DZLDk9ySi-7 opens on a dark neon animated flowchart titled "AI Agent Workflow — Fully Automated, Zero Manual Work" revealing numbered steps one at a time, and its proof visuals are live pipeline stat cards counting leads, sites, outreach, replies and meetings. DcmK70aO5VP alternates presenter and `DATA_VISUAL`, showing an n8n-style diagram ("Incoming webhook → Claude classifier → Parse classification → Route switch"). In both, the friction is a *thing on screen with a count on it* — not an adjective.

Where the corpus states the pain at all

Only **64 of 264** parsed reels (24 %) contain a problem, pain or tension beat, and where the beat exists it lands early: median start at **21 % of spoken length** (p25 0.15, p75 0.39), with **56 % inside the first quarter**, and a median length of **8.95 s / 32 words** [I-17]. Three quarters of this niche never states a problem out loud. M2's mandatory editorial filter requires one, so M2 will structurally differ from the niche here and must budget the seconds for it.

Strategic implication

Three concrete rules. **(1) Name the act, never the condition.** Every M2 Teardown pain line must be a sentence with a verb the viewer performs and a frequency — "every Monday you copy fourteen rows out of a PDF" — not "your intake process is chaotic". **(2) Budget 8–10 seconds and put it in the first quarter,**

matching the niche's own minority practice rather than inventing a placement. **(3) Show the friction as an artefact:** the spreadsheet, the inbox, the paper form, on camera with a number on it. Do not price the pain in money — that is chapter 26's finding and it costs reach.

Confidence: PROBABLE for the act-versus-condition contrast (two SUFFICIENT_SAMPLE cells, but the key comparison rests on cells of 15 and 17 reels and no creator-normalised test exists for a categorical median). **INSUFFICIENT** for `slow_response_to_leads` as evidence about anything — 5 reels across 3 creators. **PROBABLE** for the placement statistics (n=64). No causal claim: this is an association between how a pain was labelled and how the reel performed for its own author.

16. Solution Analysis

Denominators. 266 analysis-ready reels carry a ta-v1 `solution_type` from a closed list of ten values. Three cells clear the SUFFICIENT_SAMPLE threshold ($n \geq 30, \geq 8$ creators): `tool_walkthrough`, `workflow_recipe` and `resource_handoff`. Save rates divide by 245 (HikerAPI drops the field on ~9–10 % of rows); everything else by 266. Baselines: median `view_lift` 2.03, `robust_z` 1.26, share 0.0152, save 0.0278.

SOLUTION_TYPE	N	CREATORS	MEDIAN VIEW_LIFT	MEDIAN ROBUST_Z	SHARE /1K	SAVE /1K	CONFIDENCE
tool_walkthrough	60	40	1.96	1.09	16.6	30.3	SUFFICIENT_SAMPLE
workflow_recipe	51	35	1.60	1.11	14.7	34.7	SUFFICIENT_SAMPLE
resource_handoff	38	28	2.05	1.29	17.2	42.9	SUFFICIENT_SAMPLE
framework_mental_model	25	16	1.49	1.08	9.2	13.5	PROBABLE
agent_build	23	17	6.33	2.18	17.4	33.5	PROBABLE
other	23	17	2.36	1.22	15.9	8.3	PROBABLE
case_story	18	15	3.74	2.35	12.6	18.9	PROBABLE
comparison	12	10	7.85	1.93	7.0	13.4	PROBABLE
prompt_technique	10	10	0.71	0.67	14.0	37.2	PROBABLE
warning_dont	6	6	0.88	0.87	8.2	19.5	PROBABLE

Two different questions, two different answers

The table answers two questions at once and they do not have the same answer. **What reaches** is `agent_build` (6.33× on 23 reels) and `comparison` (7.85× on 12) — both PROBABLE, both small, both driven by a handful of very large multipliers from high-variance accounts. **What is kept** is `resource_handoff` : **42.9 saves per 1 000**, the highest of any solution type and 1.5× the corpus median, on the largest cell that is not a walkthrough. Its opposite is `framework_mental_model` — the "here is how to think about this" answer — at **13.5 saves per 1 000**, less than a third of `resource_handoff` and the lowest of any cell above n=20.

Stated plainly: **the niche's audience does not save a way of thinking**. It saves a thing it intends to use. This is the same finding the desire classification produces independently — reels whose stated desire is a

ready-made artefact (repo, file, template, map, prompt pack) carry median save rate **0.0346 against 0.0270** for the rest ($n=27$, $p=0.052$), and reels whose stated desire is *certainty* that a thing actually works carry **0.0140 against 0.0282** ($n=19$, $p=0.017$, negative) [I-29]. Read together with chapter 26's lexical results — named platforms are the strongest save correlate in the whole run at $\rho +0.331$ — the picture is consistent: a save is a bet that the viewer will do the thing, and it requires a destination and an object.

The mechanism: deliberate incompleteness

The cleanest example of the artefact demand is also the clearest thing M2 must invert.

[DVtgbY8Av6A](https://www.instagram.com/reel/DVtgbY8Av6A/) (<https://www.instagram.com/reel/DVtgbY8Av6A/>) is **84 spoken words** carrying **57.3 saves per 1 000**. Its hook is one sentence — **"My 15 favorite Claude skills in 60 seconds."** (0.0–2.3 s) — followed by a solution beat that is a bare list of skill names ("My cost reducer skill. My internet skill. My scalability skill...", 2.3–15.3 s), a payoff naming the best one, and then the withholding turn: **"If you want all of my skills completely for free just come in skills and I'll send them to you."** (cta, 21.3–25.4 s). A skill name is useless without the file. The reel is a pure demonstration that the artefact, not the explanation, is the object of desire — and the gate is simply the mechanism by which the artefact is withheld.

The same move without the gate is [Dc_TX1CttIc](https://www.instagram.com/reel/Dc_TX1CttIc/) (https://www.instagram.com/reel/Dc_TX1CttIc/) (562 094 plays, 24.1×, 31.7 shares per 1 000, ungated, so the rates are clean). It announces its agenda in nine seconds — **"I tested GPT-6 Astra with four use cases and here's how it went. The four use cases were 3d rendering, animated 3d website, film storyboard and viral content writing."** (hook, 0.0–9.5 s) — labels each case aloud, and earns a verdict: **"Very good model for sure, but the API cost has shot up so high that any business trying to build a real product on Astra is walking into a train wreck"** (payoff, 80.8–92.6 s). The deliverable here is a decision, delivered whole, and it is the closest thing in the corpus to what an M2 Radar reel should be.

Visual reading. `resource_handoff` reels are visually specific: the artefact is on screen and countable. `fa-v1`'s read of [Dct6Op3n6Zd](https://www.instagram.com/reel/Dct6Op3n6Zd/) (<https://www.instagram.com/reel/Dct6Op3n6Zd/>) (75.9 saves per 1 000, the second-highest in the corpus) records bold repo logo cards appearing above the host as he opens "4 GitHub Repos went viral this week and barely anyone's actually using them yet", with a before/after pipeline diagram and an agent-count dashboard as its two proof visuals. `framework_mental_model` reels have nothing equivalent to show, which is very likely why they are not kept: there is no object. Note that [Dct6Op3n6Zd](https://www.instagram.com/reel/Dct6Op3n6Zd/) holds a split screen for all nine samples — copy the ordering, not the layout (chapter 23).

The part of the script this lives in

`solution_s` is the **only** script part whose strong-versus-weak difference clears $p<0.01$: **12.25 s in the strong `robust_z` quartile against 8.4 s in the weak one (Mann-Whitney $p=0.0065$)**, and 21.1 % of speech against 15.3 %. It is present in only 70 of 264 parsed reels (27 %), at a median 10.4 s / 36 words, and the median time to reach it is 19.7 s. The pooled correlation `solution_s` → `view_lift` is +0.343 ($n=70$) but it does **not** survive creator normalisation ($cn +0.085$) and neither does the `robust_z` version (+0.342 → $cn +0.106$) — so the quartile contrast is the evidence, and the correlation is a statement about which creators write long solution statements.

Strategic implication

(1) Every M2 reel ships an artefact, and publishes it. A checklist, a process map, a prompt, a one-page teardown — named in the reel and available without asking, because RULES.md bans the gate and chapter 18 shows the ban costs no reach. The save then comes from there being something to collect.

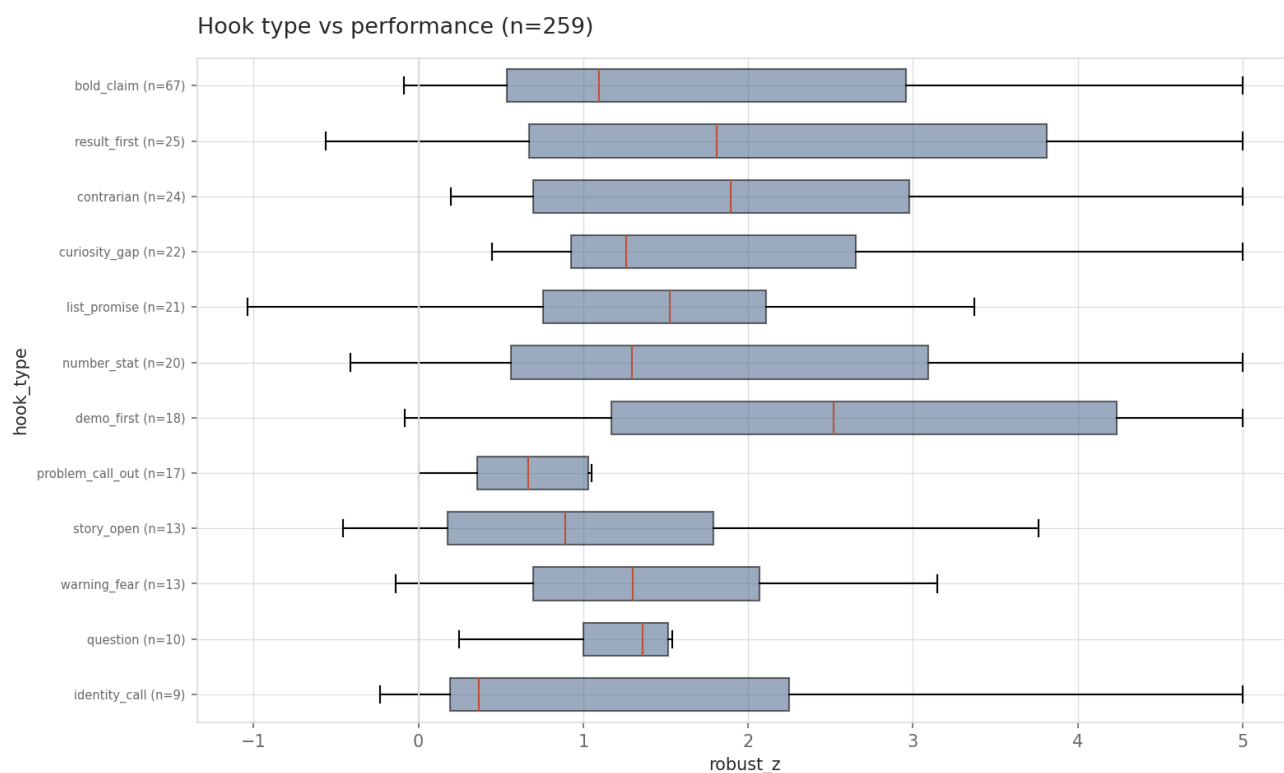
(2) Never let the solution be a framework. "Here is how to decide" is the lowest-saved cell in the table; if M2 has a mental model to teach, it must arrive attached to a downloadable object. **(3) Give the solution sentence room** — around 12 seconds, taken out of the mechanism block, not added to the runtime. It is the one part-length instruction this dataset actually supports.

Confidence: PROBABLE for the `resource_handoff` versus `framework_mental_model` save contrast — both cells are large enough to report (38/28 and 25/16 creators) but a categorical median admits no creator-normalised check, and 33 of the 40 pool reels are gated, which inflates saves $\times 2.8$ unevenly across cells. **PROBABLE** for `agent_build` and `comparison` reach, which rest on 23 and 12 reels. The `solution_s` contrast is the strongest single numeric result of the wave ($p=0.0065$) and travels inside RELIABLE insight [I-22], but its own cell is 18 reels against 19 inside a 67-per-side split, so on its own it is **PROBABLE**. No causal claim.

17. Hook Analysis

Denominators. 266 analysis-ready reels carry a ta-v1 `hook_type` from a closed list of 13 values; the boxplot draws the twelve cells with $n \geq 5$, **n=257**. Hook *length* statistics divide by **260** — the codes with a beat-derived hook (`hook_s` is never written from the clock-based fallback). First-word and n-gram counts run over the **264** hook beats. Visual first-frame statistics divide by 266. **Only `bold_claim` clears the SUFFICIENT_SAMPLE threshold**, so every individual cell below is PROBABLE and only the grouped contrasts should be quoted [I-30].

Hook type against performance



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Hook type vs performance – twelve cells, n=257

HOOK_TYPE	N	CREATORS	MEDIAN VIEW_LIFT	MEDIAN ROBUST_Z	SHARE/1K	SAVE/1K
demo_first	18	14	9.55	2.52	17.1	21.1
result_first	24	18	4.20	1.74	20.0	33.1
list_promise	21	18	2.72	1.52	17.4	46.3
question	10	10	2.37	1.36	5.4	19.8
curiosity_gap	22	19	2.16	1.26	15.5	30.1
warning_fear	13	12	2.15	1.30	10.4	25.5
bold_claim	67	43	1.88	1.09	16.9	30.6
contrarian	23	19	1.71	2.04	7.6	13.9
story_open	13	11	1.43	0.89	0.5	1.7
number_stat	20	15	1.04	1.30	16.0	33.9
problem_call_out	17	15	0.49	0.67	9.8	22.0
identity_call	9	8	0.41	0.37	14.4	18.4

Grouped, the contrast is the largest hook effect measured in the wave. **Show-first** (demo_first + result_first , n=42, 29 creators) runs at median robust_z **2.17** and view_lift **5.81** against **1.21** and **1.80** for everything else (Mann-Whitney p=0.021 and 0.008). **Setup-first** (identity_call + story_open + problem_call_out , n=39, 31 creators) runs at **0.67** and **0.80** against **1.36** and **2.25** (p=0.002 on both) [I-05, I-06]. bold_claim — the niche's default, used in 67 reels by 43 creators — sits exactly at the corpus median on both metrics. It is the baseline, not an advantage.

The same-creator evidence is the most persuasive part of this. edhillai published [DNA7-d3o5sQ](https://www.instagram.com/reel/DNA7-d3o5sQ/)(https://www.instagram.com/reel/DNA7-d3o5sQ/), which opens on the artefact speaking its own first words — "Hello, how can I assist you? This voice assistant can replace your receptionist. I'll show you how to build it in three easy steps." (hook, 0.0–6.4 s) — at robust_z **+4.69**, and [DXb8P08Dat](https://www.instagram.com/reel/DXb8P08Dat/)i(https://www.instagram.com/reel/DXb8P08Dat/), which opens on the series — "Day one of two AI tips to make you an N8N expert." (hook, 0.0–9.4 s) — at robust_z **–1.04**. Same creator, same subject area, same production; the difference is whether second one belongs to the thing or to the programme.

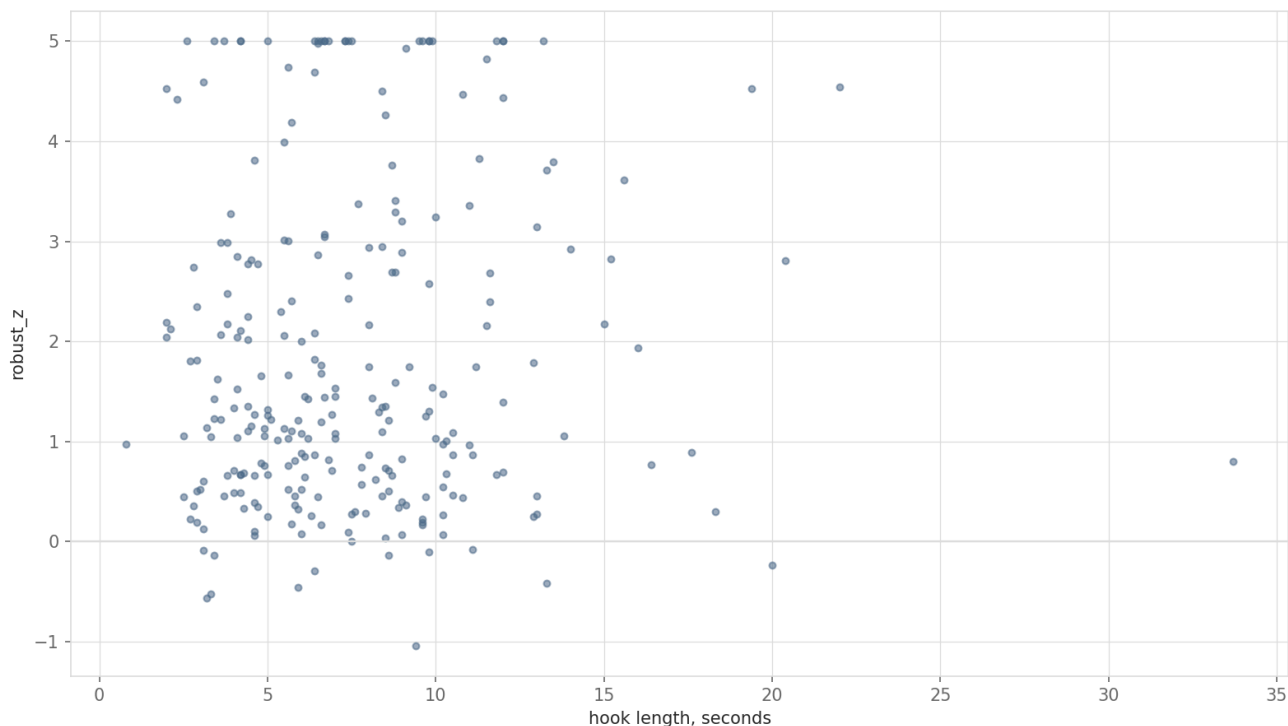
Question hooks are rare and expensive

Only **24 of 264** opening beats (9 %) contain a question mark. Their median view_lift is higher (3.00 against 1.91) and their save rate is less than half (**0.0130 against 0.0284**) — and the script-level question count is one of the strongest RELIABLE negatives in the entire run: questions_n → save_rate rho **–0.229** (p=0.00031, creator-normalised –0.162, survives), lex_questions_per_10s **–0.235** (cn **–0.186**) and lex_rhetorical_questions_n **–0.186** (cn **–0.151**) [I-07]. The corpus median questions_n is **0**. The exception proves the rule by being specific: [Db9MT-axxcO](https://www.instagram.com/reel/Db9MT-axxcO/)(https://www.instagram.com/reel/Db9MT-axxcO/) opens "What happens when you ask Claw to apply to 500 jobs for you? I tried this and I landed six interviews in 24 hours." (0.0–7.0 s) — the

question is answered inside the same breath, and the reel carries the corpus's third-best combined share+save (47.9 and 68.9 per 1 000) on a 1.6× lift.

Hook length is a distribution, not a target

Hook length vs performance (n=262)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Hook length vs performance, n=260

Median hook **6.7 s** (p25 4.6, p75 9.6), mean 7.5 s, median **21.5 words** and **13.2 % of spoken time**. Only **6 %** of reels close the hook inside 3 seconds, 31 % inside 5, and 19 % run past 10. Hook duration correlates with nothing: `robust_z` rho +0.063 (p=0.31), `view_lift` +0.075 (p=0.23), and the strong quartile's median hook is 7.6 s against 7.0 s (p=0.162). What *does* separate the quartiles is **hook words: 26.5 against 22.0 (p=0.028)** — how much is promised, not how long the promising takes [I-03]. Any audit that scores hooks against a 0–3 second rule is measuring something this niche does not do, and that is the one criticism of Max's card set that should be withdrawn (see chapter 32).

The visual hook

First-frame type across the 266 analysed reels: SPLIT_SCREEN 73, A_ROLL_MEDIUM 57, A_ROLL_CLOSE_UP 45, B_ROLL_CONTEXT 18, SCREEN_RECORDING 16, OVERLAY 14, A_ROLL_WIDE 13, A_ROLL_TALKING_HEAD 9, UI_DEMO 8, B_ROLL_PROCESS 8, SCREENSHOT 7, MOTION_GRAPHIC 7, HOOK_VISUAL 6, DATA_VISUAL 5, MEME 4, TEXT_ONLY 4, TRANSITION 1.

COARSE FIRST STATE	N	MEDIAN VIEW_LIFT	MEDIAN ROBUST_Z	SHARE / 1 K
screen (recording / screenshot / UI demo / data visual)	35	5.25	2.13	21.3
text or graphic	31	2.12	1.43	—
A-roll	11 4	1.99	1.15	—
B-roll	23	1.60	1.25	—
split screen	63	1.34	1.05	13.5

Opening on a screen rather than a face is worth roughly **half again the forwarding rate** (share 0.0213 vs 0.0140), and the creator-normalised correlation (**+0.285, p=0.00025, n=161**) is *stronger* than the pooled one (+0.208, p=0.00063) — the strongest form of evidence this dataset can produce, because it means the effect holds inside individual creators' own output and is not a fact about which creators like screens [I-09]. Individual sub-cells are small but point the same way: UI_DEMO first (n=8, 7 creators) median `view_lift` 29.2, DATA_VISUAL first (n=5) 18.7. One caution: the `is_visual_hook` flag is set on frame 0 of all 295 codes, so it is a labelling artefact and carries no signal.

Verbal × visual combinations with n ≥ 5. `result_first` on `A_ROLL_MEDIUM` (n=8, 6 creators, `robust_z` 2.98, `view_lift` 8.07) is the strongest; `contrarian` on `A_ROLL_CLOSE_UP` (6/5, 1.86/4.98) and `curiosity_gap` on `A_ROLL_CLOSE_UP` (5/5, 1.76/13.42) follow. **bold_claim on SPLIT_SCREEN is the single most common combination in the niche (n=22, 17 creators) and one of the weakest** (`robust_z` 0.84, `view_lift` 1.21) — the default verbal hook over the default layout, and it performs like the default.

Strategic implication

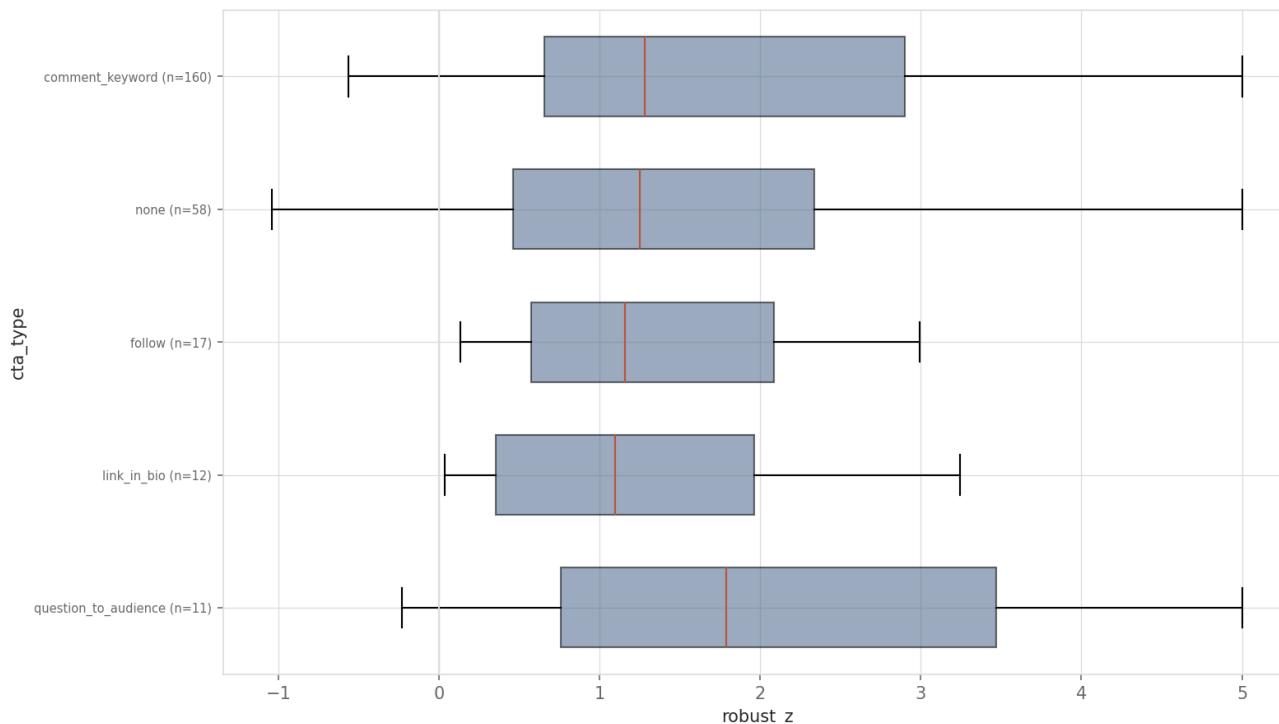
(1) Frame one is the artefact mid-action, with the promise already burned in as a top banner; the presenter arrives at the first explanation beat, not before. **(2) The hook is one complete promise in 5–10 seconds and 20–30 words** — never truncated to satisfy a three-second rule, and judged on whether the promise is concrete. **(3) Write it as a statement.** A question is permitted only when the same sentence answers it. **(4) Ban series preambles and "if you are a business owner" addresses:** the identity-call shape is the worst-performing hook in the corpus, and the same-creator pair above shows the cost is not a sampling accident. **(5) Do not open on split screen** (chapter 23).

Confidence: PROBABLE for the grouped show-first and setup-first contrasts (n=42 and 39, 29 and 31 creators, p=0.002–0.021, but no single cell clears the size threshold). **RELIABLE** for the questions-depress-saves finding (n=245, 95 creators, p=0.0003, survives creator normalisation) and for the screen-first share-rate finding (n=266/161, p<0.001, normalised effect stronger than pooled). **RELIABLE** for the hook-length null. **INSUFFICIENT** for every verbal × visual combination (n=5–8).

18. CTA Analysis

Denominators. 256 of the 266 analysis-ready reels with a ta-v1 read carry a `cta_type` from a closed list of nine values (the boxplot drops cells under 5, leaving the five named below). CTA *position and length* statistics divide by the **216** reels whose beat parse contains a `cta` or `closing` beat — 82 % of the 264 parsed reels; **58 reels have no CTA at all**. Gate-versus-no-gate comparisons run on n=214 for the rate tests [I-12]. Baselines: median `view_lift` 2.03, `robust_z` 1.26, comment rate, share rate 0.0152 and save rate 0.0278.

CTA type vs performance (n=258)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

CTA type vs performance – five cells with n≥5, n=256

CTA_TYPE	N	CREATORS	MEDIAN VIEW_LIFT	MEDIAN ROBUST_Z	COMMENT / 1K	SHARE / 1K	SAVE / 1K	CONFIDENCE
comment_keyword	158	71	2.08	1.2475	21.7	17.0	34.0	SUFFICIENT_SAMPLE
none	58	30	1.72	1.2481	0.9	9.6	12.4	SUFFICIENT_SAMPLE
follow	17	13	2.25	1.16	—	15.5	35.4	PROBABLE
link_in_bio	12	6	1.02	1.10	—	13.7	8.8	PROBABLE
question_to_audience	11	9	4.08	1.79	—	18.3	13.4	PROBABLE
dm / next_video / free_resource / save_share	3/3/2 /2	≤3	—	—	—	—	—	INSUFFICIENT

The central finding: the gate buys engagement, not distribution

59 % of the analysed corpus (158 of 266) runs a comment-keyword gate, and it is the most consequential single fact in this report — not because of what it does to reach, but because of what it does to M2's own selection machinery.

	GATED	UNGATED	RATIO	P
comment_rate	0.0217	0.0009	×23	4e-23
save_rate	0.0340	0.0120	×2.8	4e-10
share_rate	0.0170	0.0096	×1.8	0.00015
view_lift	2.08	1.72	—	0.58
robust_z	1.2475	1.2481	—	0.63

Two medians agreeing to four decimal places (1.2475 against 1.2481) is as flat as a comparison gets. The gate multiplies comment rate twenty-three-fold, roughly triples saves, nearly doubles shares — and moves reach by nothing measurable. Every one of those engagement effects survives creator normalisation, so they are properties of the mechanic rather than of the accounts that use it.

The consequence for the radar is a selection defect, not an editorial one. `cards.py` ranks candidate references on `resh_1k` and `save_1k` — exactly the two rates the gate inflates. With 59 % of the analysed corpus gated, the radar is currently part-ranking DM funnels rather than ideas. The fix is mechanical: **record `cta_type` on every reference and compare a candidate's share and save rates only against other reels carrying the same `cta_type`**. The reference pool in `06-creators-and-references.md` makes the scale of the distortion visible — **33 of its 40 rows are gated**, and the seven ungated rows (`DNA7-d3o5sQ`, `Dc_TX1CttIc`, `DbVKVz0y8xo`, `DTC0U9DkRTG`, `DX9kWtYzssE`, `DYSKnz3TxPZ`, `DZ-wujrTJ0u`) are the only ones whose hi-intent rates can be read at face value.

The second consequence is liberating. `RULES.md` §1.2 already bans the gate from M2 scripts on positioning grounds; this measurement says **the ban is free**. Nothing in reach is forfeited. What is forfeited is the inflated share and save rate — which means M2's own published numbers will be lower than the references' and should not be compared against them.

Position and length are settled

The CTA starts at a median **90 % of spoken length** (p25 0.84, p75 0.94). **202 of 216 sit in the final quarter**; exactly **three** of 216 place it before halfway. Median length **4.95 s / 21 words**, and the strong `robust_z` quartile runs slightly longer (5.7 s against 4.4 s, p=0.075 — not significant). There is no positional experiment left to run: the niche has converged, and converged sensibly.

Stacking asks is the one thing the data visibly punishes.

[Da3uSkrNjli](https://www.instagram.com/reel/Da3uSkrNjli/)(<https://www.instagram.com/reel/Da3uSkrNjli/>) spends its last twenty seconds on four separate requests — **"So if you're interested in selling automations like this local dentist near you, then comment the word automation in the comment section below. But if you really want to..."** (`cta_beat`, 68.4–88.5 s) — and lands at `view_lift` **−0.05**, below its own author's median. At the other end, [DZPYR_0y4VV](https://www.instagram.com/reel/DZPYR_0y4VV/)(https://www.instagram.com/reel/DZPYR_0y4VV/) is 27 seconds carrying **one** claim and **one** ask, with the CTA occupying over a third of the runtime, at 139.3× its author's median. One ask, clearly placed, is the whole of the instruction.

The lexical signature

"Comment" is the single most frequent imperative in the corpus — 41 occurrences, three times the next verb (follow 14, let 13, go 12, look 9, think 9, click 7). Sentences in this niche end on a destination or an availability claim rather than a conclusion: "right now" 13, "for free" 11, "for you" 11, "on github" 10, "the link" 5. Where the gate keyword itself is recoverable from the CTA beat text, the most common is a generic placeholder ("comment *the word below*" patterns, 14 occurrences), then AI 7, agent 4, section 3, and single letters such as W. That last detail matters: a one-letter gate is not a topic signal, it is a funnel trigger, and it tells you the comment was never about the content.

Two lexical findings that are about the closing line rather than the gate survive creator normalisation and belong here: **CTA verbs** → save_rate rho +0.241 (cn +0.155) and **direct address per 100 words** → save_rate +0.174 (cn +0.127), both RELIABLE. Concrete asks written in the second person are saved; the emotion around them is not what does it.

Visual reading. is_cta_visual is set on only 3 % of the 2 645 labelled frames — the CTA is almost never given its own visual. fa-v1 records cta_visual: none on the strongest template in the pool, DNA7-d3o5sQ(<https://www.instagram.com/reel/DNA7-d3o5sQ/>), whose ungated close is simply the artefact confirming the booking aloud: "Perfect. Ed Hill, your appointment has been successfully booked for tomorrow at 4 p.m." (closing, 62.8–67.9 s). The most effective closing line in the corpus is the machine finishing its job.

Strategic implication

(1) **Keep the gate banned** — confirmed free by measurement, not only by policy. (2) **Hold cta_type constant in reference selection**, or record it on the card so a gated reel's rates are never compared with an ungated one's. (3) **Close in the last 10 % of the script, about 5 seconds and 20 words, with exactly one ask.** (4) **Put the ask on the face, not on the screen** (chapter 22's A>SCREEN>A pattern), and make the final visual the artefact having finished. (5) **Do not A/B the CTA's position** — the niche converged at 90 % of spoken length and there is nothing left to learn there.

Confidence: RELIABLE for the gate's engagement effects and its reach null (n=214–256, 71 and 30 creators, p from 4e-23 to 0.00015, all surviving creator normalisation; the reach null at p=0.58 and 0.63). RELIABLE for CTA position and length as a description (n=216). PROBABLE for question_to_audience being the highest-lift cell (n=11, 9 creators). INSUFFICIENT for dm, next_video, free_resource and save_share (n=2–3 each), reported with their suppressed_reason rather than deleted.

19. Script Part Lengths

Denominators. Part lengths run on **264** codes — those whose script block was built from ta-v1 beats (`features_json.script.part_source == 'beats'`). The four analysis-ready codes without a beat-derived block are excluded, and the clock-based timing fallback never writes these columns. Each row reports the median **only over the codes where that part exists**, so the n differs by row by design and the shares do not sum to 1.0. `total_s` is **spoken** length, always shorter than `reels.dur` for a reel with a silent opening or an outro card; the two are never swapped. Strong-versus-weak splits are quartiles of `robust_z` *inside* the analysed tier, so the weak side still beats its own authors' medians [I-01].

Typical lengths and shares

PART	N	% OF 264	MEDIAN S	MEAN S	MEDIAN SHARE OF SPEECH	MEDIAN WORDS
hook	260	98 %	6.7	7.50	13.2 %	21.5
setup	34	13 %	7.8	9.41	14.0 %	29
problem	64	24 %	8.95	12.04	16.4 %	32
explanation	208	79 %	18.6	23.54	38.3 %	64
solution	70	27 %	10.4	12.95	20.1 %	36
proof	133	50 %	17.0	23.19	30.5 %	57
payoff	150	57 %	8.2	10.24	17.0 %	26.5
cta	216	82 %	4.95	6.27	9.7 %	21

Whole-reel reference values: median spoken length **53.6 s, 182 words, 3.52 words per second**; median video duration 54.2 s in the analysed set against 47.6 s across all 3 211 ingested reels. `unbucketed_s` (objection + rehook + other) is non-zero in only **35 of 264** reels, so the eight columns account for essentially all spoken time.

The table's shape is the finding. **Two parts are near-universal** (hook 98 %, CTA 82 %) and **explanation is the load-bearing middle** — present in 79 % of reels and taking 38 % of the speech, more than twice any other part's share. **Four parts are minority moves**: solution 27 %, problem 24 %, setup 13 %, and proof only 50 %. Half of this niche never proves anything and three quarters never state a problem. For M2 that is the structural gap: the mandatory editorial filter requires friction and an AI boundary, which means M2's script will spend seconds the niche's does not, and those seconds have to come out of the mechanism block rather than being added to the runtime.

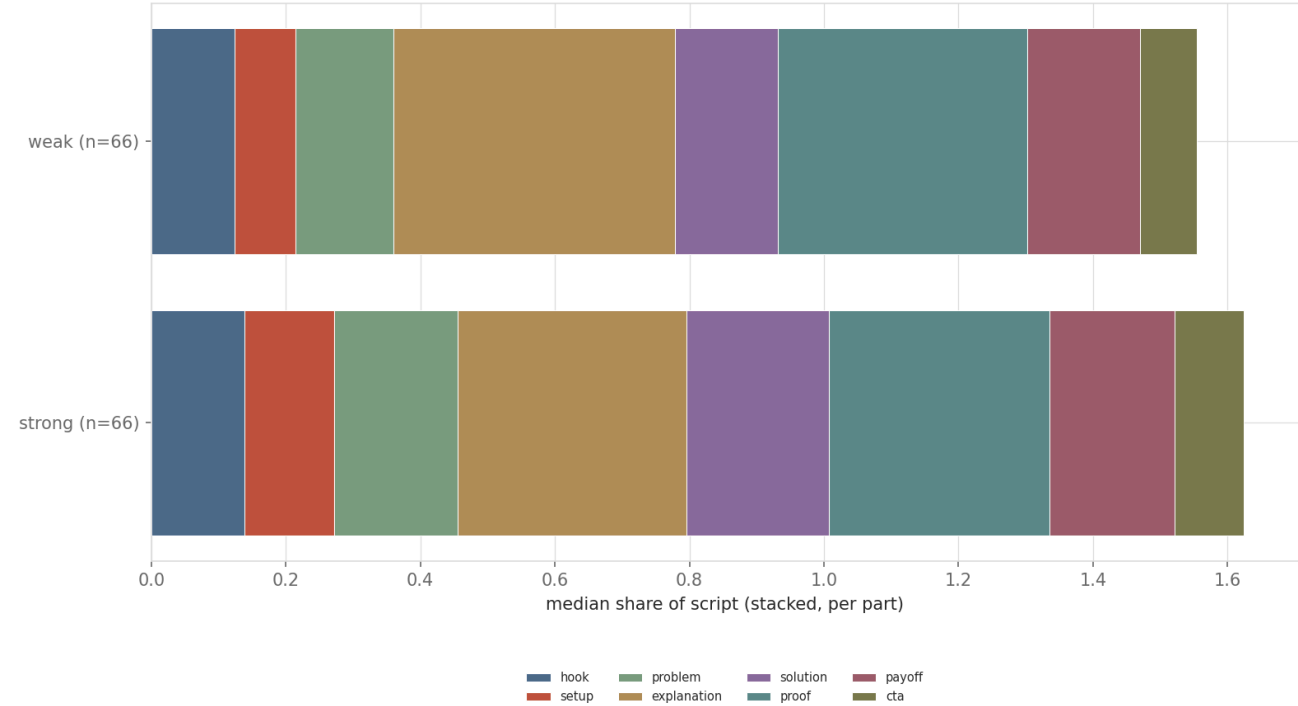
Distributions, not just medians

QUANTITY	P 25	MEDIAN	P 75	NOTES
hook_s	4.6	6.7	9.6	6 % under 3 s, 31 % under 5 s, 19 % over 10 s
total_s	39.1	53.6	69.7	min 0.8 s (a five-word ASR row), max 259.0 s
total_words	128.8	182	237	max 1 034
wps	3.07	3.52	3.83	min 0.65, max 6.25
time to first proof beat	11.1	22.0	32.0	n=133
time to solution beat	7.7	19.7	29.0	n=70
CTA start, share of speech	0.84	0.90	0.94	94 % in the final quarter; 3 of 216 before halfway
problem start, share of speech	0.15	0.21	0.39	56 % inside the first quarter

Three of these are production instructions in disguise. The first proof beat arrives at a median **22.0 seconds** — that is where this audience expects evidence, and a reel that proves nothing by then has spent its credibility window. The problem, where it exists, lands at 21 % of spoken length. The CTA lands at 90 %. Those three timings are the only placement facts this dataset supports, and all three are descriptions of convergence rather than measured advantages.

Strong versus weak: reallocation, not extension

Script part shares, strong vs weak (n=132)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Script part shares, strong vs weak robust_z quartile (n=66 per side)

PART	N S	MEDIAN S S	SHARE S	N W	MEDIAN S W	SHARE W	Δ S	MANN-WHITNEY P
hook	66	7.6	13.8 %	64	7.0	12.4 %	+0.6	0.162
setup	7	11.0	13.4 %	11	4.7	9.1 %	+6.3	0.057
problem	18	8.6	18.4 %	16	9.15	14.5 %	-0.55	0.809
explanation	54	17.3	34.1 %	53	19.5	41.9 %	-2.2	0.236
solution	18	12.25	21.1 %	19	8.4	15.3 %	+3.85	0.0065
proof	32	19.25	32.9 %	34	23.9	37.1 %	-4.65	0.551
payoff	37	10.3	18.6 %	38	7.95	16.8 %	+2.35	0.123
cta	56	5.7	10.3 %	51	4.4	8.4 %	+1.3	0.075

Read the segments individually, never as a stacked total: each share is a median taken independently per role over a different n, so the bars need not sum to 1.0. **Total spoken length is identical** between the quartiles — 56.9 s against 58.0 s, $p=0.98$ — and so are word count (198 vs 179, $p=0.33$) and pace (3.53 vs 3.45 wps, $p=0.44$). What changes is the allocation: the strong quartile takes 2.2 seconds and 7.8 points of share out of the mechanism and puts them into the solution statement (+3.85 s), the payoff (+2.35 s) and the closing line (+1.3 s). **solution_s is the only part clearing $p<0.01$ in the entire 115-feature test.** Proof length moves the *other* way ($p=0.55$) and the problem statement does not move at all ($p=0.81$) — what matters about proof in this corpus is whether it is on screen, not how long it is spoken [I-16, I-18].

The plainest reading: **a longer "how" is usually an unclear "what"**. The weak quartile gives 41.9 % of its speech to the mechanism; the strong one gives 34.1 % and spends the difference saying what the thing is and what it changes.

By creator reliability, topic and archetype

Joining the 268 analysed reels to `creator_stats.reliability`: 28 reels from 15 CONSISTENT creators, 240 from 87 HIGH_VARIANCE creators. **Script shape is identical** — hook 6.7 s both sides, total 53.8 s against 53.5 s, 3.4 against 3.5 wps — while median `robust_z` is 0.60 against 1.37 and `view_lift` 0.65 against 2.38. Two real differences in shape: consistent creators spend **less on proof (11.2 s vs 18.6 s)** and **more on the closing line (6.3 s vs 4.8 s)**.

By narrative archetype, `demo_walkthrough` is the largest ($n=62$, 42 creators), the highest-lift (`view_lift` **3.70**, `robust_z` **1.71**) and the most evenly balanced — 7.8 s hook / 19.9 s explanation (38 %) / 19.3 s proof (34 %) / 8.0 s payoff / 5.0 s CTA. Listics run the shortest hooks (4.6 s) and by far the largest proof share (72 %, because every list item is tagged as an example). Myth-bust carries the longest hooks (8.5 s) and the second-highest `robust_z` (1.93) on the smallest explanation share (28 %). By topic, news reels carry the longest proof (31.3 s) and longest hooks (8.1 s); lead-generation reels the shortest hooks (5.2 s); agency-business reels by far the longest problem statements (20.9 s, but on $n=4$ — INSUFFICIENT, and the only topic where problem is the biggest part). Every topic cell is PROBABLE or INSUFFICIENT.

Transcript reading. The reallocation is audible. [DYC-x8DogOI](https://www.instagram.com/reel/DYC-x8DogOI/) (399.4×, the corpus's highest lift) gives 11.8 s to a hook that already contains the result ("I just built a website that looks like it costs \$10,000 with literally one line of code and honestly the whole thing took Claude about two minutes"), **17 s** to the mechanism, **10 s** to a payoff that restates what it means ("But here's the crazy part. We wrapped all four into one single repo"), and 3.6 s to one ask. Mechanism is a third of the reel. Compare [DaDn5L-xKWj](https://www.instagram.com/reel/DaDn5L-xKWj/)

xKWi/), five identically shaped ten-second tricks with no solution statement at all — five open loops, no reallocation, and a reel that flattens.

Visual reading. DYC-x8DogOI's frame read is `SCREENSHOT > A_ROLL_CLOSE_UP > SCREENSHOT` : the finished websites carry the mechanism, the face carries the payoff, and eight of eight observed sample transitions are hard cuts. The seconds taken out of the spoken mechanism are visible on screen instead.

Strategic implication

Budget the M2 script as: hook 6–8 s · friction 8–10 s in the first quarter · mechanism capped at **one third** · solution statement **~12 s** · screen proof arriving by **~22 s** · payoff ~10 s naming what changes on Monday · closing line ~5 s and 20 words at 90 % of the script. The friction block is M2's addition to the niche's shape and must be paid for out of the mechanism, not bolted onto the runtime. Do not set a words-per-second target; set a words-per-idea target (chapter 26).

Confidence: RELIABLE for the part-length distributions as a description (n=264, direct measurement of 1 527 beats). **PROBABLE** for the reallocation finding — `solution_s` clears $p < 0.01$ but on 18 against 19 reels, and the explanation and payoff differences do not reach $p < 0.05$. **INSUFFICIENT** for every topic-level part table (largest cell n=27, most under 10) and for `setup_s` (7 against 11 reels). No causal claim: a longer solution statement is associated with a higher creator-relative score inside a pre-selected sample.

20. Script Frequency Patterns

Denominators. Orderings and beat-role counts run on **264** codes and **1 527 beats** (median 6 beats per reel, min 0, max 12). Phrase, opening and n-gram counts come from `features_json.lexical` over the same codes; the hook n-grams are the first words of the **264** hook beats. Open-loop and pattern-interrupt counts are ta-v1's own tallies over **266** reels. Sentence-level counts must be read against `sentence_source` : **255** transcripts are punctuated by Whisper and split on punctuation, **11** have no punctuation at all and are split on Whisper's own VAD phrase boundaries; the two groups are never pooled silently.

Orderings: two fixed endpoints and chaos in between

Collapsed to the eight script parts and de-duplicated, there are **146 distinct orderings across 264 reels**. The hook is the first beat in **259 of 264**; a CTA or closing beat is last in **215**.

ORDERING	N	% OF 264
HOOK > EXPLAIN > PAYOFF > CTA	27	10.2 %
HOOK > EXPLAIN > CTA	13	4.9 %
HOOK > EXPLAIN > PROOF > CTA	10	3.8 %
HOOK > PROOF > CTA	8	3.0 %
HOOK > PROOF > PAYOFF > CTA	6	2.3 %
HOOK > EXPLAIN > PROOF > PAYOFF > CTA	6	2.3 %
HOOK > SOLUTION > PAYOFF > CTA	5	1.9 %
HOOK > PROOF (no CTA)	5	1.9 %
HOOK > EXPLAIN > PROOF > EXPLAIN > CTA	5	1.9 %
HOOK > SETUP > EXPLAIN > PROOF > PAYOFF	4	1.5 %

No ordering covers more than 10 % of the corpus. The stable facts are the two endpoints and the dominance of explanation in the middle, which is why the honest description of this niche's default is four moves, not eight: **HOOK > EXPLANATION > PAYOFF > CTA** [I-04]. The eight-part model in the method doc is a vocabulary for measurement, not a template anyone follows. Raw beat-role frequencies confirm where the mass sits: hook 259 · mechanism 250 · cta 205 · example 157 · payoff 145 · proof 104 · explanation 96 · solution 84 · context 40 · setup 31 · tension 31 · problem 29 · closing 23 · objection 19 · rehook 15 · pain 12 · transformation 10 · other 5 · audience 3 · hook_extension 2.

For M2 the implication is permissive rather than restrictive: there is no ordering to copy, so problem, proof and objection are **deliberate additions that must earn their seconds** — which for M2 they do, because POSITIONING.md §8 requires friction and an AI boundary that this corpus omits.

The three hook grammars

POSITION MOST FREQUENT OPENINGS, OVER THE 264 HOOK BEATS

first word	"I" 22 · "this" 18 · "if" 16 · "you" 15 · "so" 11 · "the" 10 · "here's" 8 · "someone" 6 · "I'm" 6 · "Claude" 5 · "hey" 5
------------	--

first bigram	"you can" 10 · "if you" 9 · "I just" 9 · "this is" 7 · "if you're" 4 · "here's how" 4 · "Claude can" 3 · "most people" 3
--------------	--

first trigram	"you can now" 7 · "if you want" 4 · "this is what" 3 · "I just built" 3 · "Claude can now" 2 · "nobody is talking" 2 · "most people think" 2
---------------	--

Three templates carry almost all of it. **"You can now X"** announces a new capability. **"I just X"** is a first-person demonstration — the builder register, and the register chapter 28 says M2 should adopt. **"If you X, then Y"** is conditional address, and it is also the identity-call family that underperforms worst in chapter 17 (`identity_call` median `view_lift` 0.41). That the third template is both common and weak is the most actionable frequency fact in this chapter: **a shape's popularity in this niche is not evidence that it works.**

Sentence openings, endings and imperatives

FAMILY COUNTS OVER THE ANALYSED CODES

sentence openings	"this is" 26 · "you can" 23 · "if you" 21 · "it's called" 20 · "here's how" 14 · "so you" 14 · "so this" 10 · "let me" 10 · "i just" 10 · "now you" 7 · "the first" 7 · "but here's" 7 · "you just" 7 · "most people" 7
-------------------	---

sentence endings	"right now" 13 · "for free" 11 · "for you" 11 · "on github" 10 · "to you" 7 · "do it" 6 · "use it" 6 · "it works" 5 · "right here" 5 · "the link" 5
------------------	---

imperative verbs	comment 41 · follow 14 · let 13 · go 12 · look 9 · think 9 · imagine 7 · do 7 · click 7 · tell 6 · make 6 · open 6 · start 5 · write 5 · build 5 · see 5
------------------	---

keyphrase bigrams	"claude code" 11 · "it will" 11 · "you will" 10 · "open source" 8 · "to build" 7 · "click on" 7 · "your business" 7 · "cloud code" 7 · "chat gpt" 6 · "ai agent" 6 · "gpt 6" 6
-------------------	---

Two readings. Sentences in this niche **end on a destination or an availability claim, not on a conclusion** — the same finding chapter 16 reaches from the solution vocabulary and chapter 26 from the platform-name correlation, arrived at from three independent directions. And "comment" is three times more frequent than the next imperative: the lexical fingerprint of the keyword gate (chapter 18), and the clearest measure of how much of this niche's apparent call-to-action vocabulary is one funnel mechanic. The "cloud code" row is an ASR defect, not a product (chapter 13).

Repetition: the counter-intuitive result

Deliberate phrase repetition within a reel is **rare**: "if you want to" 3 · "you want to learn" 2 · "you don't have to" 2 · "exactly how to use" 2, then a long tail of one-offs. Median `lex_repeated_phrases_n` is **0** in both the strong and the weak quartile (p=0.286).

What the strong quartile repeats is not phrases but **vocabulary**, and it does so measurably more: type-token ratio **0.605 against 0.628 (p=0.039)** and MATTR50 **0.795 against 0.811 (p=0.044)** — both *lower* in the strong quartile. Two of the seven features that separate the quartiles at $p < 0.05$ therefore run in the direction most writing advice forbids: **winners keep saying the same tool name and the same step word rather than reaching for variety**. Read alongside the entity findings (distinct entities 2 against 1, $p = 0.011$), the pattern is coherent: name the thing, then name it again.

Retention devices

DEVICE	FREQUENCY
reels with ≥ 1 open loop	240 of 266 ; median exactly 1 (198 reels), 36 carry two, 26 carry none
pattern interrupts	140 reels have none, 110 have one
explicit re-hook beat	15 of 264 (6 %)
explicit objection beat	19 of 264 (7 %)

The niche's retention device is a single unresolved promise held from the hook to the payoff, not repeated re-hooking [I-26]. Where a re-hook does appear it is very short and placed at the moment the mechanism stops being new: [DNA7-d3o5sQ](https://www.instagram.com/reel/DNA7-d3o5sQ/) inserts a **1.8-second** rehook — **"But how about the results?"** (32.0–33.8 s) — between the architecture and the live call, and carries 36.5 saves per 1 000 on an ungated close. [DK9pFUjPQcx](https://www.instagram.com/reel/DK9pFUjPQcx/) uses the same move as an escalation — **"And the best part? It's not just these four. I've collected over 10 such free api platforms"** (rehook, 31.0–37.0 s) — immediately before its gate.

The counter-example is instructive. [DaDn5L-xKWi](https://www.instagram.com/reel/DaDn5L-xKWi/) opens **five** loops ("Five AI image tricks top realtors are using...") and closes them in five identically shaped ten-second blocks; it flattens. [Db3SERzzOqG](https://www.instagram.com/reel/Db3SERzzOqG/), the corpus's 178-second outlier, also opens four loops in its hook — but closes them **in the exact order they were opened**, and runs at 88.5× its author's median. Multiple loops are survivable only with explicit ordering discipline; one loop is the default for a reason.

Rhetorical devices. `semantics.rhetorical_devices` is free text and produced **975 distinct strings across 266 reels**, so no device reaches a countable frequency. The recurring families by repeated wording: numbered steps or lists (17 combined), borrowed authority (6), price contrast or anchoring (6), future pacing (4), objection pre-emption (3), rule of three (3), keyword gate (3), benefit stacking (3), delayed reveal (2), mid-roll rehook (2), myth vs reality (2), share-with-a-peer close (2). The countable equivalents live in the lexical features: median `questions_n` **0**, `second_person_n` **8**, `first_person_n` **4–5**, `imperatives_n` **1**, `superlatives_n` **1**, `intensifiers_n` **1**, `negations_n` **2**, `modals_n` **3**, `numbers_n` **3** (strong) against 1 (weak).

Strategic implication

(1) Default the M2 script to four moves — hook, mechanism, payoff, CTA — and add friction, proof and the AI boundary as deliberate, budgeted insertions. **(2) Write hooks in the "I just X" or "you can now X" grammar and never in the "if you are a..." grammar.** **(3) Repeat the tool name, the step number and the promise rather than varying them** — the strong quartile's lower lexical diversity is a finding, not a

flaw. **(4) Open exactly one loop**, and add one short re-hook only past about 45 seconds, placed where the mechanism stops being new. **(5) End sentences on a destination.**

Confidence: RELIABLE for the ordering counts, beat-role frequencies and the two endpoints (n=264, 1 527 beats, direct count). **PROBABLE** for the lexical-diversity reversal (p=0.039 and 0.044, inside a 67-per-side split, and the direction is counter-intuitive enough to deserve replication). **PROBABLE** for the open-loop description (n=266, ta-v1's own tallies, not an independent measurement). **INSUFFICIENT** for anything about rhetorical devices as categories — 975 distinct free-text strings is not a vocabulary.

21. Frame Analysis

Denominators and the limitation that governs them. The legacy corpus carries **9 fixed frames per reel** — sampled at 0.4 / 1.2 / 2.4 / 4.0 s and then evenly spaced. **295** codes have an fa-v1 read, producing **2 645 labelled frames**; 266 of those 295 also have a transcript and appear in the performance joins; 243 of those have a save rate. Every roll share below is a fraction of **labelled samples, not of screen time** (`features_json.visual.share_basis` says so on every row), so a screen shown for three seconds between two samples is invisible to this entire chapter. A "scene" is a run of consecutive samples with the same `frame_type` — `boundary_reason = 'sampled'`, `boundary_confidence = 'low'` — and can read as 4.5 s long when the real shot was 0.6 s.

What a reel in this niche looks like

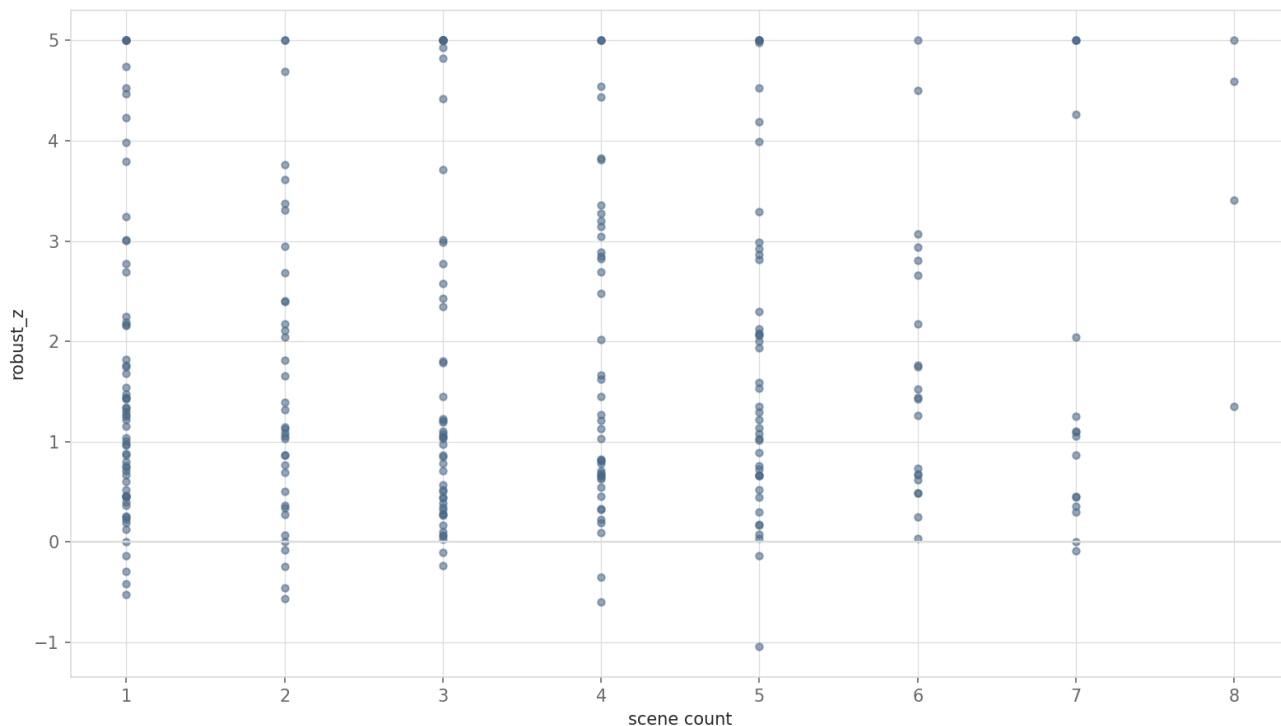
FRAME_TYPE	FRAMES	% OF 2 645
SPLIT_SCREEN	591	22.3
A_ROLL_MEDIUM	363	13.7
A_ROLL_CLOSE_UP	353	13.3
SCREENSHOT	294	11.1
SCREEN_RECORDING	258	9.8
UI_DEMO	127	4.8
OVERLAY	123	4.7
B_ROLL_CONTEXT	113	4.3
A_ROLL_WIDE	78	2.9
B_ROLL_PROCESS	72	2.7
DATA_VISUAL	64	2.4
MOTION_GRAPHIC	54	2.0
A_ROLL_TALKING_HEAD	47	1.8
TEXT_ONLY	46	1.7
everything else	55	2.1

Rolled up: **A-roll 33 %**, **SPLIT 26 %**, **SCREEN 26 %**, **B-roll 7 %**, **other 5 %**, **text 2 %**. Per frame, **text_overlay 86 %**, **face_present 65 %**, **ui_present 58 %**, **is_proof_visual 41 %**, **is_cta_visual 3 %**. Framing: medium 1 043 · screen 728 · close-up 440 · wide 321. Caption placement: bottom 1 099 · top 571 · middle 568 · none 403. Visual density: medium 1 148 · high 857 · low 640. Median per-reel shares in the analysed set: `a_roll_share` 0.333, `screen_share` 0.222, `split_share` 0.000, `b_roll_share` 0.000, `text_overlay_density` **1.000**.

Three facts in that table are worth isolating. **Captions are table stakes, not a lever**: the median `text_overlay_density` is 1.0 in both the strong and the weak `robust_z` quartile ($p=0.418$) and the pooled save-rate correlation ($+0.180$, $p=0.005$) does not survive creator normalisation (cn $+0.080$). **Forty-one per cent of frames are flagged as proof visuals** while only 50 % of reels carry a proof *beat* — proof in this niche is visual more often than it is spoken. And **the CTA almost never has a visual** (3 % of frames), which is consistent with chapter 18's finding that the closing line is delivered on the face.

Visual complexity is not a lever

Scene count vs performance (n=282)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Scene count vs performance, n=282

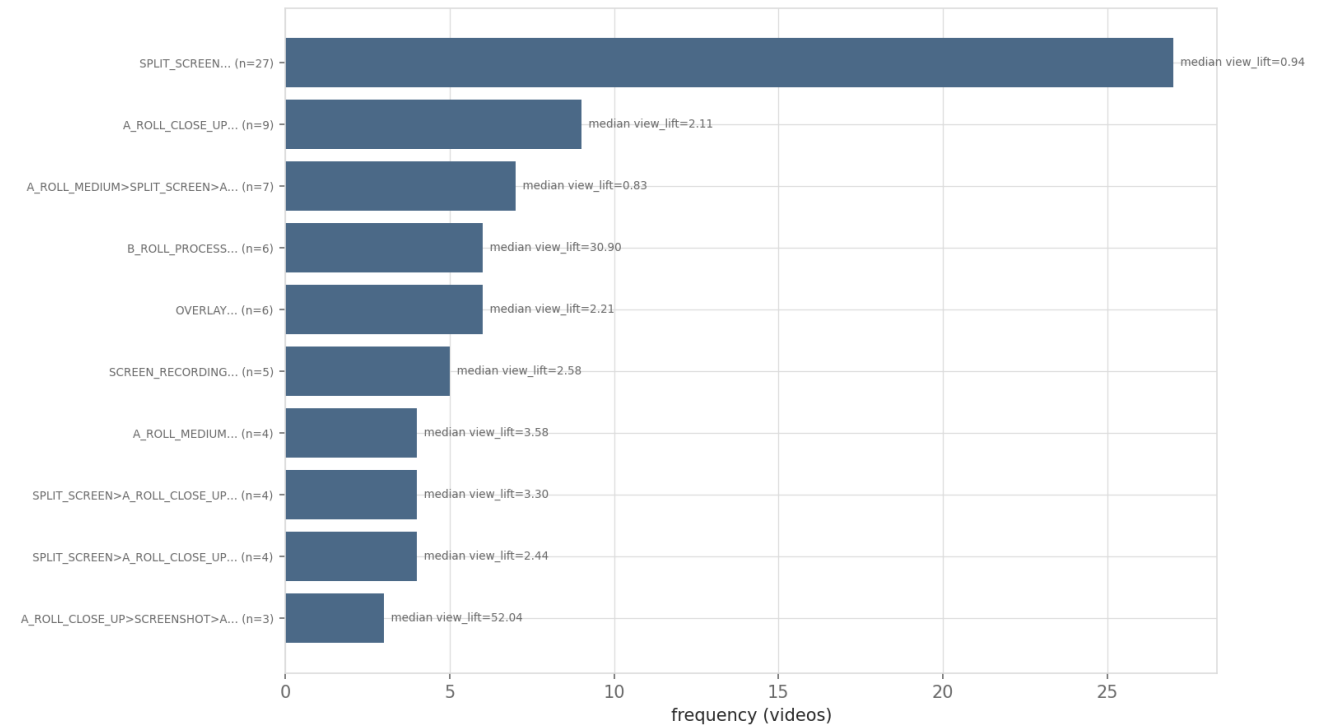
Median `scenes_n` is 3 (p25 2, p75 5). Across the 282 codes with a scene count the Spearman correlation against `robust_z` is **+0.085 (p=0.17)** and against `view_lift` **+0.052 (p=0.40)**; the strong quartile's median is 4 scenes against 3 ($p=0.110$). Grouping instead by the number of **distinct coarse visual states** per reel gives the same answer across the whole observable range:

COARSE STATES PER REEL	REELS	MEDIAN ROBUST_Z
1	68	1.25
2	40	1.21
3	52	1.14
4	41	1.63
5	36	1.39
6	18	1.35
7	9	—
8	2	—

From one state to six the median `robust_z` moves between 1.14 and 1.63 with no trend. **Visual complexity is not a lever**, and the measurement floor reinforces rather than undermines the null: `scenes_n` is a lower bound on real cuts, so if anything the real range of complexity is wider than plotted and still shows nothing.

The literal vocabulary fragments

Visual sequence frequency, top 10 (n=75)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Visual sequence frequency, top 10 – 75 of 295 reels

295 reels produce 206 distinct `visual_sequence` strings, so the median sequence occurs exactly once, and the ten most frequent cover only **75 of 295 reels (25 %)**. That is why every visual finding in this report is stated on a **coarse four-way collapse** (A = any A-roll · SCR = screen recording / screenshot / UI

demo / data visual · SPL = split screen · B = B-roll · TXT = text, meme, motion graphic, overlay), which reduces 295 reels to 121 strings of which 15 cover 53 % of the corpus. Two reels edited very differently between samples can still produce the same string.

Within that caveat, the chart's two endpoints are the chapter's real content. **SPLIT_SCREEN** held throughout is the single most common arrangement (n=27, 18 creators) and among the weakest (median `view_lift` 0.94 against a corpus median of 2.03). **B_ROLL_PROCESS** held throughout is the strongest (n=6, 4 creators, median `view_lift` 30.90, `robust_z` 4.13, median play 554 907), and the strongest multi-state form is `A_ROLL_CLOSE_UP > SCREENSHOT > A_ROLL_CLOSE_UP` (n=3, 3 creators, `view_lift` 52.04, `robust_z` 5.0). Both extreme cells are INSUFFICIENT on their own and are dominated by large outliers; they are reported because they point the same way as the much better-powered screen findings in chapters 17 and 22, not as evidence in themselves.

What the frames see that the transcript cannot

Transcript reading, with its visual counterpart.

[DZgBxjnBzMe](https://www.instagram.com/reel/DZgBxjnBzMe/)(<https://www.instagram.com/reel/DZgBxjnBzMe/>) speaks **14 words in 27 seconds** — hook "Welcome back." (5.3–7.3 s), then "Unlock my phone. Open Geohot Star in my phone. Call my sister." (7.3–23.0 s) — and reaches 207× its author's median. Its ta-v1 confidence is INSUFFICIENT, so it contributes nothing to any script finding. The frame read is where the reel actually lives: a held **B_ROLL_PROCESS**, a camera filming a laptop running a Jarvis Python script, a "J.A.R.V.I.S V6.5" HUD appearing, and a phone changing state beside the laptop, under one white title pinned for the whole runtime — **seven of eight observed sample transitions are `same_shot`**. The evidence is continuous, so there is no cut where a trick could hide.

The mirror case is [DcxV37-CJOC](https://www.instagram.com/reel/DcxV37-CJOC/)(<https://www.instagram.com/reel/DcxV37-CJOC/>) (559 515 plays, 72.2 saves per 1 000) whose transcript is four words and whose frame read is complete: a host holding a printed card with a pink dot pattern, one held **A_ROLL_MEDIUM**, **no proof visual at all**. Two reels invisible to the transcript pipeline, both legible to the frame pipeline, and they reach the same conclusion from opposite directions: the frame read is not a decoration on the script read, it is an independent measurement.

Strategic implication

(1) Fix one caption system and never revisit it — bold white on a solid box, bottom-anchored for speech, plus a persistent top banner carrying the promise. It is universal in the niche (86 % of frames) and has no measured performance effect, which means its absence is a penalty and its presence is not a gain. **(2) Do not set a scene count**. Max's fixed 60-second / 7-scene template is wrong to be fixed at all: the observable range is 1 to 8 coarse states with no performance difference, so the one thing that should vary with the topic is being held constant. **(3) Let 41 % of frames be proof**. That is the niche baseline for evidence on screen, and M2's own footage — its own build, its own sheet, its own inbox — should at minimum match it. **(4) Never inherit a reference's layout without checking it** (chapters 22 and 23).

Confidence: RELIABLE as a description of the visual baseline (2 645 frames, direct labelling) and for the caption-density null. **RELIABLE** for the scene-count and visual-complexity nulls (n=282 and 266).

INSUFFICIENT for every literal `visual_sequence` cell except the top few — 9 of 15 reported cells fall below the size floor, and the two extreme cells are n=6 and n=3. All shares are fractions of nine samples, not of screen time; no number here may be quoted as a screen-time percentage.

22. A-roll / B-roll Analysis

Denominators. Roll shares are computed over the **295** codes with an fa-v1 read; performance joins use the **266** that also carry a transcript, and save-rate joins **243**. A roll share is the fraction of **nine labelled samples** carrying that roll, never a fraction of screen time. The quartile split in `visual_patterns.json` uses 74 per side (q1 `robust_z` 0.543, q3 2.943); the 115-feature Mann-Whitney test in `associations.json` uses 67 per side (q1 0.596, q3 2.855). The two denominators are reported separately and never mixed.

The baseline and the quartile test

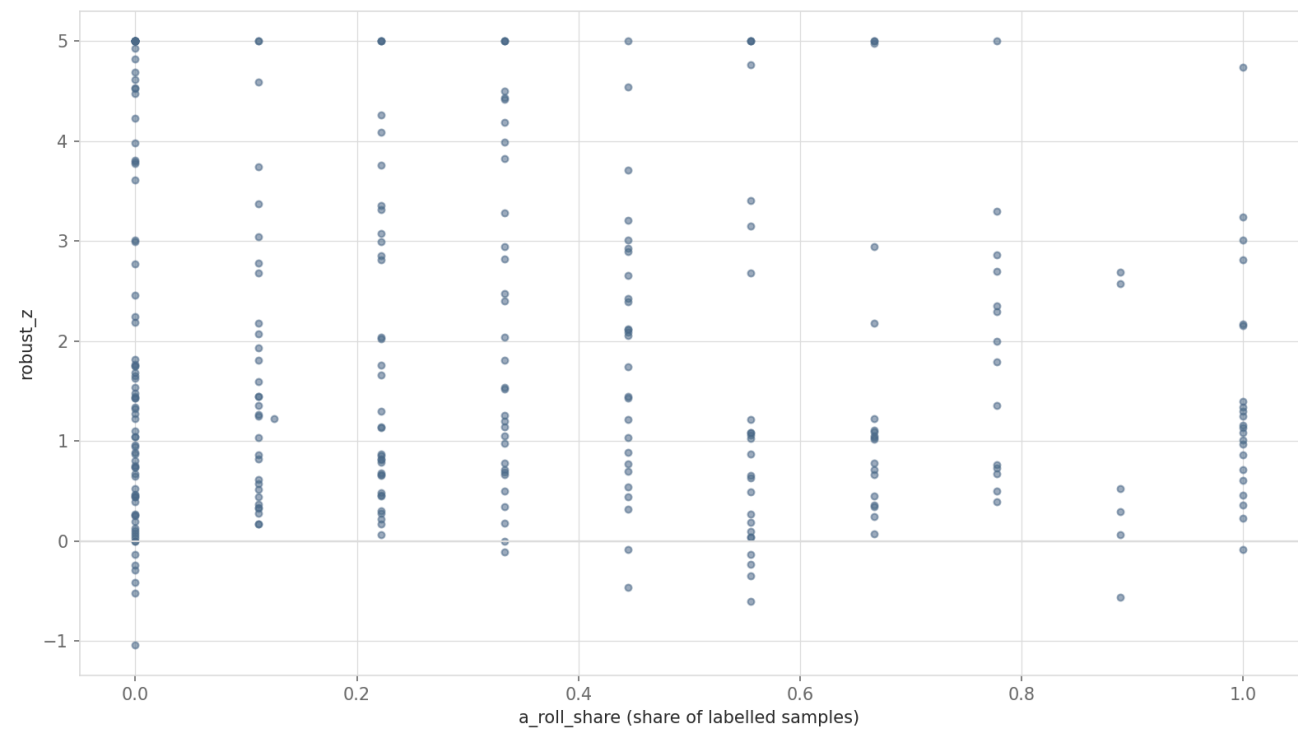
A-roll is the niche's largest roll — **33 % of all 2 645 labelled frames**, with a face present in 65 % of them — and the median per-reel `a_roll_share` in the analysed set is **0.333**. B-roll is almost absent: **7 % of frames** (B_ROLL_CONTEXT 113, B_ROLL_PROCESS 72) and a median per-reel `b_roll_share` of **0.000**.

SHARE	STRONG MEDIAN (N=74)	WEAK MEDIAN (N=74)
a_roll_share	0.2222	0.2222
b_roll_share	0.0000	0.0000
split_share	0.0000	0.0000
screen_share	0.2222	0.1111

Three of the four rolls are identical on both sides of the quartile split. Only screen share differs, and even it does not clear $p < 0.05$ in the 67-per-side Mann-Whitney ($p = 0.234$) — a test that simply lacks power inside a pre-selected corpus. The stronger evidence for the screen is the correlation, not this table.

More face is weakly worse, not better

A-roll share vs performance (n=295)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

A-roll share vs performance, n=295

PAIR	N	CREATOR S	RHO	P	CREATOR-NORMALISED	SURVIVE S
a_roll_share → share_rate	266	102	-0.156	0.011	-0.094	yes
a_roll_share → save_rate	243	95	-0.155	0.015	-0.052	no
a_roll_share → robust_z	266	102	-0.046	0.46	—	—
screen_share → share_rate	266	102	+0.203	0.00085	+0.129	yes
screen_share → save_rate	243	95	+0.139	0.030	+0.120	yes
b_roll_share → any metric	266	102	no signal	—	—	—

The scatter is a cloud with a faint downward tilt, and the table says what the cloud cannot: **more face on camera is associated with fewer forwards**, and the association survives creator normalisation, so it is not merely a statement about which creators talk to camera. Against reach there is nothing at all (−0.046, p=0.46). The mirror finding is the one to act on: **screen_share** is **the only roll with a positive RELIABLE association anywhere in the 460-pair run** [I-08]. Reels whose visual sequence contains any screen state carry median share rate **0.0168 against 0.0118** (p=0.00013) and save rate **0.0308 against**

0.0220 ($p=0.001$), and the *first-frame* version of the same finding is stronger inside creators than between them (+0.285 against +0.208 pooled, $n=161$) [I-09].

The pattern that uses both: talk, show, come back

A > SCREEN > A — presenter, cut to a full-frame screen, cut back to the presenter — is the strongest named visual arrangement in the corpus.

	VALUE
reels containing the run	39 of 266 , across 32 creators
reels that are exactly A > SCR > A and nothing else	11 , one each from 11 different creators
containing-the-run performance	median share rate 0.0213 vs 0.0140 ($p=0.00076$); median robust_z 2.09 vs 1.22 ($p=0.10$)
the exact 11-reel form	median robust_z 3.01 , view_lift 9.86 , median play 218 733
the literal A_ROLL_CLOSE_UP > SCREENSHOT > A_ROLL_CLOSE_UP	$n=3$, 3 creators, robust_z 5.0, view_lift 52.04

The eleven exact-form reels come from eleven different creators — `tenfoldmarc`, `buildingwithstring.com`, `kevinfremon`, `shrug.manny`, `manthanjethwani`, `codewithnishant` among them — which is the best protection available here against a single-account artefact.

Transcript reading. The division of labour is explicit in the beats: **the face carries the promise and the payoff, the screen carries the mechanism.** [DWCPx0PkQzA](https://www.instagram.com/reel/DWCPx0PkQzA/)(<https://www.instagram.com/reel/DWCPx0PkQzA/>) (1 074 517 plays, 142.8×, 57.3 saves per 1 000) opens to camera with a bold claim, hands the two mechanism beats to the screen ("copying this line into your terminal... step three is to download the UI UX pro max skill", 14.4–47.0 s), then returns to the face for the payoff: **"Normandy would take you four steps to create beautiful websites like this But I've created a github repo where all you need to do is copy one line of code"** (47.4–58.5 s). The "so what" is delivered by a person. [DZ-wujrTJ0u](https://www.instagram.com/reel/DZ-wujrTJ0u/)(<https://www.instagram.com/reel/DZ-wujrTJ0u/>) (51.6×, 64.9 saves per 1 000) runs the same shape over a 172-second runtime, demoing before the instructions and again after, so the reward bookends a long middle.

Visual reading. fa-v1's read of DWCPx0PkQzA is A_ROLL_CLOSE_UP > SCREENSHOT > A_ROLL_CLOSE_UP with a black rounded-box title ("CLAUDE CODE MAKE \$10,000 WEBSITES FOR FREE") held over the opening beats and four proof visuals: the Claude Code quickstart page, the npm `framer-motion` page, a `ui-ux-pro-max-skill` repository and an MCP install guide. Five of eight observed sample transitions are hard cuts. Every claim in the spoken mechanism has a page behind it.

Why it plausibly works — stated as interpretation, not cause. A screen is a claim a viewer can check in the moment; a face is where the interpretation can be delivered. Reels that end on the screen have nowhere to put the "so what", and the "so what" is what gets forwarded.

B-roll: nearly absent, and its one strong cell is process footage

`b_roll_share` shows no signal against any metric and its median is zero on both sides of the quartile split. The interesting cell is not the share but the content: **pure B_ROLL_PROCESS** (n=6, 4 creators — `alassafi.ai`, `bennett.spooner`, `jarvis_ai_spark`, `olivermerrick__`) carries median `view_lift` **30.90** and `robust_z` **4.13**, the highest of any visual sequence, while single-state reels overall show no advantage at all (median `robust_z` 1.25 against 1.26 for multi-state, $p=0.47$). The effect therefore belongs to the *content* — the device, dashboard or map visibly doing the thing — and not to holding one shot. [DbVKVz0y8xo](https://www.instagram.com/reel/DbVKVz0y8xo/)(https://www.instagram.com/reel/DbVKVz0y8xo/) is the ungated example: a camera filming a curved monitor showing a "THE CLIP MACHINE" workflow diagram with a hand on a drawing tablet below, and a spoken objection block that spends thirty seconds killing the obvious answer before offering its own.

B-roll *context* — stock clips of people at computers — is the other 4.3 % of frames and carries nothing. The distinction matters for M2: process footage is evidence, context footage is decoration, and only one of them is worth a shooting day.

Strategic implication

The M2 storyboard is **face (promise, ~8 s) → screen (the run, ~25 s) → face (what it changes on Monday, plus the closing line, ~10 s)**, with the CTA on the face and never on the screen. Two limits: do not cut back to the face more than twice — the four-and-more-state group shows no additional gain (chapter 21) — and do not use a screen that is not M2's own. Never a stock UI, never a slide of a UI. Where the build genuinely runs end to end, the `B_ROLL_PROCESS` variant is licensed: record the whole run in one take and narrate over it, which also satisfies the one-take production rule at no extra cost. For a Teardown whose subject is a process rather than a tool, the screen is the artefact of that process — the spreadsheet, the inbox, the paper form — not a face.

Confidence: RELIABLE for `screen_share` → `share_rate` and for any-screen-present → `save_rate` (n=243–266, 95–102 creators, $p<0.01$, creator-normalised survives). **PROBABLE** for `a_roll_share` → `share_rate` ($p=0.011$, survives normalisation but fails the $p<0.01$ bar). **PROBABLE** for the `A > SCREEN > A` pattern: pooled $p<0.001$ on share rate only, and the exact-sequence cell is n=11. **INSUFFICIENT** for pure `B_ROLL_PROCESS` (n=6, 4 creators, dominated by two very large outliers). Shares are fractions of nine samples; none of this is causal.

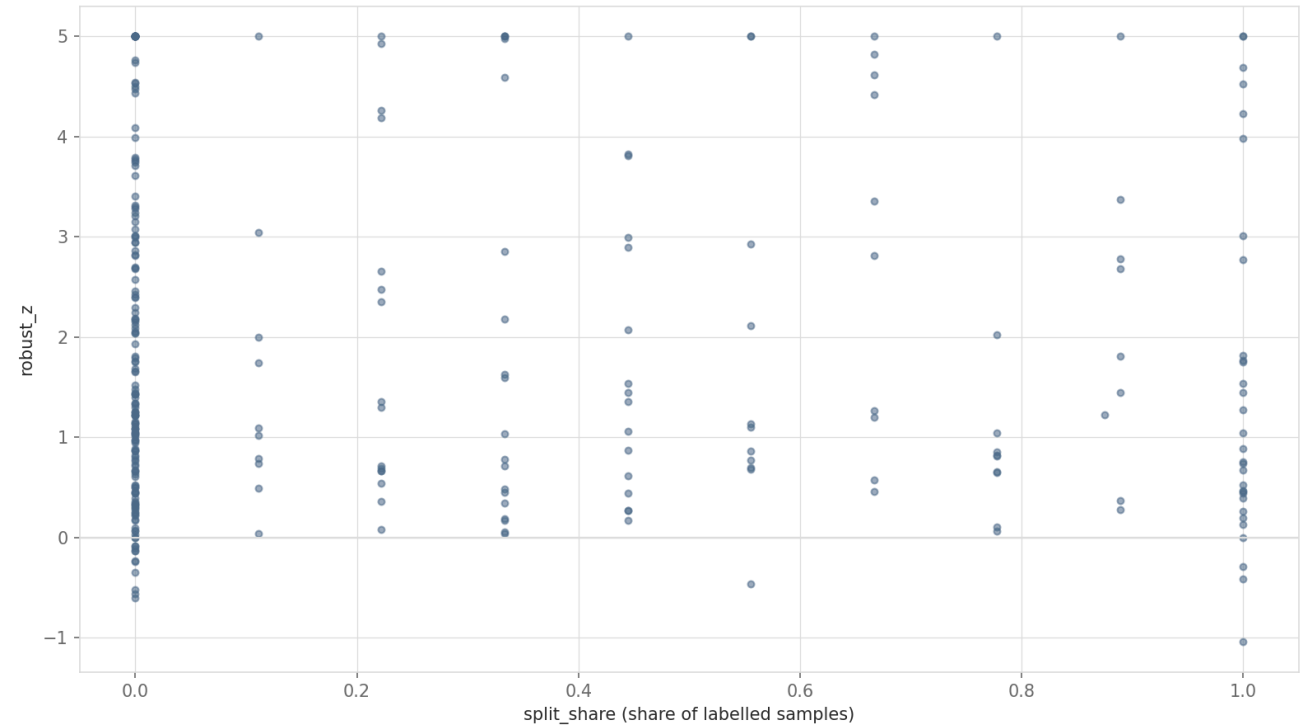
23. Split-Screen Analysis

Denominators. `split_share` is computed over the **295** fa-v1 codes as a fraction of nine labelled samples; performance joins use the **266** with a transcript and **243** with a save rate. First-frame counts use 266. The quartile contrast in `visual_patterns.json` uses 74 per side; the Mann-Whitney feature test uses 67 per side. Baselines: analysed-set median `view_lift` **2.03**, `robust_z` **1.26**, share rate **0.0152**, save rate **0.0278**.

Split screen is the niche's most-used and least-rewarded arrangement

	VALUE
share of all labelled frames	591 of 2 645 = 22.3 % — the single largest frame type
reels opening on it	73 of 266 — the most common opening state
reels containing any split state	99
reels holding it across all nine samples	27 , from 18 creators
creators using a split state at all	51
median per-reel <code>split_share</code> (analysed set)	0.000

Split-screen share vs performance (n=295)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Split-screen share vs performance, n=295

COMPARISON	SPLIT	NO SPLIT	P
any split state, median <code>robust_z</code>	1.03	1.43	0.044
any split state, median <code>view_lift</code>	1.38	2.49	0.093
pure <code>SPLIT_SCREEN</code> sequence (n=27, 18 creators), median <code>view_lift</code>	0.94	corpus 2.03	—
pure <code>SPLIT_SCREEN</code> , median <code>robust_z</code>	0.76	corpus 1.26	—
<code>A_ROLL_MEDIUM > SPLIT_SCREEN > A_ROLL_MEDIUM</code> (n=7, 6 creators), <code>view_lift</code>	0.83	corpus 2.03	—
first frame is <code>SPLIT_SCREEN</code> (n=73, 40 creators), <code>view_lift</code> / <code>robust_z</code>	1.63 / 1.10	corpus 2.03 / 1.26	—
<code>split_share</code> , strong vs weak quartile	0.0	0.0	0.628

The scatter and the table agree on an unusual combination: **the layout the niche uses most is the only roll share with a negative association to reach that reaches $p < 0.05$** , and its pure form sits at less than half the corpus median on `view_lift`. The one positive pooled association — `split_share` → `save_rate` **+0.193 ($p = 0.0025$)** — **collapses under creator normalisation (cn +0.036)**, which means it is a statement about *which creators use split screen*, not about the layout. The median `split_share` is 0.0 in both quartiles ($p = 0.628$), so the quartile test adds nothing either way.

Two honest limits on this finding. First, $p = 0.044$ on a single Mann-Whitney inside a pre-selected corpus is a weak result, and split use is heavily creator-specific — 18 accounts supply all 27 pure-form reels. Second, 9 samples cannot distinguish a held split from a reel that cuts to full-frame screen between samples, so the measured `split_share` is noisy. What survives both caveats is the *absence of any positive evidence* for a layout that covers 22.3 % of all frames in the niche. With 99 reels and 51 creators, an advantage of any size should have been visible here.

Why it plausibly does not pay

Stated as interpretation, not as cause: a split halves the pixels available to the evidence at the exact moment the evidence is the argument, and it signals reaction content rather than first-hand work. Chapter 22's measurement is the complement — full-frame screen is the only roll with a positive RELIABLE association, and opening on it is worth roughly half again the forwarding rate.

Transcript reading. The split-screen reels in this corpus are usually **talking over footage the creator did not make**. [DcnHtUGS_tE](https://www.instagram.com/reel/DcnHtUGS_tE/) is the clearest case: "You can now turn a single idea into an entire AI film. Script, characters, scenes, all of it. This is called Buzzy." (hook, 0.0–7.3 s). It reached **65×** its author's median on views and carries **share rate 0.0016 and save rate 0.0005** — the lowest hi-intent rates anywhere in the top thirty by reach. Plenty of attention, almost no intent. The pattern recurs: [DVwI55QDinY](https://www.instagram.com/reel/DVwI55QDinY/) holds a split for all nine samples and its ta-v1 read is INSUFFICIENT — there is no spoken script to analyse at all.

The corpus's strongest pure-split reel is the exception that still does not rescue the layout. [Dct6Op3n6Zd](https://www.instagram.com/reel/Dct6Op3n6Zd/) carries **75.9 saves per 1 000**, the second highest in the corpus, on a held split. Its hook — "4 GitHub Repos went viral this week and barely

anyone's actually using them yet." (0.0–4.4 s) — is followed by four explanation beats ordered by increasing scale, so retention survives to the last item, which sits immediately before the gate. **Copy the ordering, not the layout:** the reel works because of a countdown structure and a gated artefact, and its save rate carries the gate's $\times 2.8$ inflation (chapter 18).

Visual reading. fa-v1's read of Dct6Op3n6Zd shows bold Graft and GitHub logo cards appearing above the host, with a before/after pipeline diagram and an agent-count dashboard as its only two proof visuals across nine samples — two pieces of evidence in 38 seconds, each at half size. DcnHtUGS_tE is the same geometry applied to generated footage: a cinematic AI clip above a reacting host, with the host occupying the lower half throughout. In both, the thing the viewer is being asked to believe never gets the full frame.

What this explains about M2's existing cards

Nine of main's 23 legacy source reels open on SPLIT_SCREEN — the corpus's weakest opening state. That is not an editorial preference; cards.py ranks candidates on metrics only and has no visual input at all, so the selection was blind to layout. The three shot-plan columns the card writer fills automatically (shot_banner, shot_frame, shot_screen) therefore had nothing to inherit from the reference. The fix is structural: **a card must not inherit a reference's layout without checking it**, and the reference's first_frame_type and coarse visual_sequence must travel onto the card so the check is possible.

Strategic implication

Split screen has no place in the M2 template. If a reaction layout is genuinely needed — commenting on someone else's demo, for instance — **cut between full-frame face and full-frame screen instead**, which is chapter 22's A > SCREEN > A pattern and the one arrangement with positive evidence behind it. Where a reference M2 wants to borrow is a split-screen reel, borrow its script ordering and its topic and replace its geometry. And do not spend production time testing split variants: the corpus has already run that experiment across 99 reels and 51 creators, and the result is at best nothing.

Confidence: PROBABLE. The headline contrast reaches $p=0.044$ pooled on robust_z — above the $p<0.01$ bar the method requires for RELIABLE — the creator-normalised save-rate check does not survive, and split use is concentrated in 18 accounts. The supporting cells (pure split $n=27$; A > SPL > A $n=7$) are small. Shares are fractions of nine samples, not screen time, so a reel that cut to full-frame screen between samples may be miscoded. No causal claim is licensed: this is an association between a layout and a creator-relative score inside a corpus pre-selected for strength.

24. Transition Analysis

Denominators, and a warning that is larger than the chapter. Two different transition counts exist in this dataset and they are not interchangeable. The **database** records **663** `hard_cut_or_more` and **15** `unknown` — 678 in total, which is exactly `960 scenes - 282 codes`, i.e. one `transition_in` per scene after the first.[^1] The **raw fa-v1 reads** of the same corpus record **1 306** `hard_cut_or_more`, **844** `same_shot` and **16** `unknown` across 295 codes, because they judge every adjacent *sample pair* rather than every scene boundary.[^2] Cut-frequency statistics are a third quantity again, computed by ffmpeg over **282** codes. Every number below says which of the three it came from.

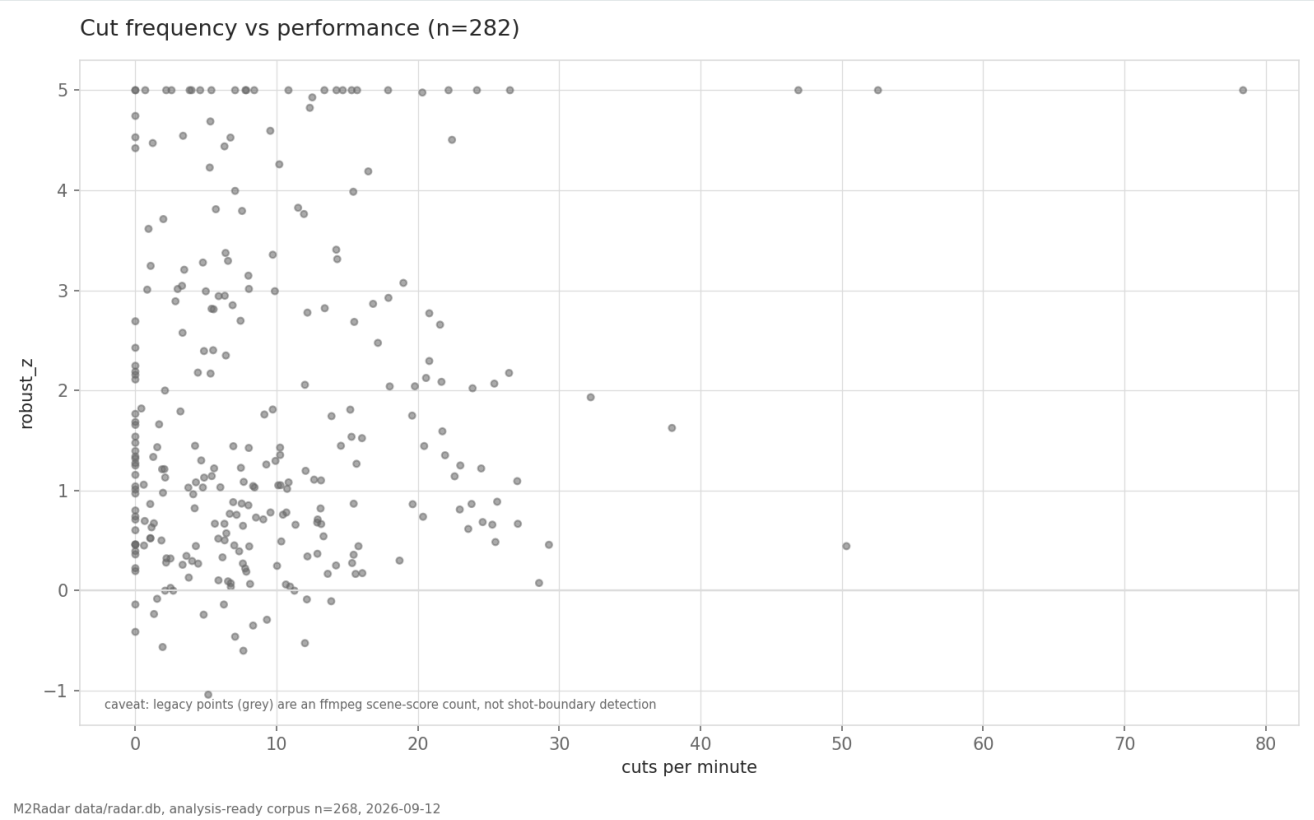
Only one transition type is observed anywhere in the corpus

Per SPEC §5 a transition is recorded **only where fa-v1 actually saw two adjacent labelled samples differ**; `same_shot` and everything unobserved becomes `unknown`. The consequence is absolute and must be stated before anything else: **there is no evidence in this dataset for any transition type other than a hard cut**. No dissolves, whips, match cuts, zoom transitions or speed ramps are recorded anywhere across 2 645 labelled frames. **Their absence is a measurement artefact of the nine-sample grid, not a finding about the niche** — a 0.4-second whip pan between two samples is invisible by construction, and a dissolve between two visually similar shots is coded `same_shot`.

SOURCE	HARD_CUT_OR_MORE	SAME_SHOT	UNKNOWN	UNIT
database (<code>scenes.transition_in</code> , 282 codes)	663	not stored	15	one per scene boundary
raw fa-v1 reads (295 codes)	1 306	844	16	one per adjacent sample pair

The gap between 663 and 1 306 is the cost of the scene collapse: consecutive samples sharing a `frame_type` are merged into one scene, so a reel that alternates A-roll and screen twice within a run of identical labels loses its internal changes. **Both numbers are floors on the real cut count**, and the larger one is still a floor — nine samples over a 54-second median runtime cannot see eight cuts between samples. Per reel, the raw reads observe a median of **5** hard cuts (mean 4.43, max 8), with **28 of 295** codes showing none at all.

Cut frequency: no relationship, and a metric that cannot carry a claim



Cut frequency vs performance, n=282

	VALUE
median <code>cuts_per_min</code>	7.26 (p25 2.49, p75 13.68, max 78.4, n=282)
Spearman vs <code>robust_z</code>	+0.056 (p=0.36)
Spearman vs <code>view_lift</code>	+0.003 (p=0.96)
strong vs weak quartile	7.06 vs 6.75 (p=0.420)
rows reporting exactly zero cuts	38 of 282
<code>cut_metric_quality</code> on every row	<code>ffmpeg_scene_0.35_count_only</code>

There is no relationship in any metric, and the distribution is so wide (2.49 to 13.68 at the quartiles, maximum 78.4) that a target would be meaningless even if one existed. The 38 zero rows are indistinguishable from a detector that never fired.

The metric itself is the real finding. Every point on the chart comes from `ffmpeg -filter:v "select='gt(scene,0.35) ',showinfo"` counted as `pts_time`: lines — a **scene-score threshold count, not shot-boundary detection**. It misses hard cuts between visually similar shots (same speaker, same set, same framing — precisely the talking-head format that dominates this niche) and over-counts whip pans, flashes, heavy zooms and screen-recording scrolls, with a fixed 0.35 threshold and no per-clip calibration. It is usable as a relative busy-versus-calm signal **inside this dataset only**. **No M2 output may**

quote a cut rate from it, and no claim comparing M2's editing density with the niche's is permitted from this wave.

What the reels themselves show

Visual reading. The two visual extremes of the corpus are also its transition extremes, and both work. [DZgBxjnBzMe](https://www.instagram.com/reel/DZgBxjnBzMe/) — 207× its author's median, 1 570 719 plays — shows **seven same_shot transitions and one hard cut** across its nine samples: one continuous take of a laptop running a Jarvis script beside a phone changing state. At the other end, [DYC-x8DogOI](https://www.instagram.com/reel/DYC-x8DogOI/) — the corpus's highest lift at 399.4× — shows **eight hard cuts out of eight observed pairs**, cutting between website mockups, two Claude Code terminal sessions and the presenter. Zero observed cuts and maximum observed cuts, both at the top of the distribution. Whatever separates these reels, it is not their edit.

Transcript reading. What the two share is that the cut is never asked to carry meaning. [DZgBxjnBzMe](https://www.instagram.com/reel/DZgBxjnBzMe/) speaks 14 words and lets the artefact act; [DYC-x8DogOI](https://www.instagram.com/reel/DYC-x8DogOI/)'s mechanism beat names a step per cut ("Step two, you add the UI UX Pro Max skill so Claude builds clean modern layouts automatically", 11.8–28.9 s) so each cut coincides with a new named object rather than with a rhythmic beat. That is the only transition-level craft observation this dataset supports, and it is an observation, not a measurement.

Strategic implication

(1) Keep the one-take rule. Not because the data argues for it — the data argues for nothing here — but because nothing argues against it, and at five publications a week with one filming day, editing is a consumable resource rather than a requirement. A change of speaker stays the only cut M2 allows. **(2) Never publish a claim about M2's cut rate, or about the niche's.** The metric does not support it, and quoting it would be the exact kind of confident wrong number this method document exists to prevent. **(3) Do not spend editing budget aligning cuts to script beats** (chapter 25). **(4) If the transition question is ever to be answered properly, it needs the new scene-v1 detector over a random sample of the 2 914 unprocessed codes** — more features on the same 295 reels will not help, because the limitation is the nine-sample grid, not the feature set.

Confidence: RELIABLE for the statement that only hard cuts are observed (direct count, and the reason for the absence of other types is structural and documented). **RELIABLE** for the cut-frequency null (n=282, all four metrics, plus the quartile test). **INSUFFICIENT** for any quantitative statement about *how many* cuts a reel in this niche contains: the DB count (663) and the raw fa-v1 count (1 306) differ by a factor of two for documented reasons, both are floors, and `cut_metric_quality` is count-only on all 282 rows. No causal claim.

[^1]: Database counts from `reports/data/visual_patterns.json` → `transitions_observed`, built from `scenes.transition_in`. Row counts from `reports/data/manifest.json` → `row_counts : scenes` 960, `deepdives` 282; $960 - 282 = 678 = 663 + 15$.

[^2]: Computed for this chapter over the per-video frame analyses — the source fa-v1 wrote before ingest collapsed adjacent identical samples into scenes:

```
import json, glob, collections
c = collections.Counter()
files = glob.glob('data/analysis/frames/*.json')
for p in files:
    d = json.load(open(p))
    c.update(t['kind'] for t in (d.get('transitions_observed') or []))
print(c, len(files))
```

Returns `Counter({'hard_cut_or_more': 1306, 'same_shot': 844, 'unknown': 16})` over 295 files. The per-reel median (5), mean (4.43), maximum (8) and zero-count (28 codes) come from the same pass.

25. Transcript-to-Frame Alignment

Denominators. The alignment pass checks **1 527 beat boundaries** across **264** codes — every beat in the corpus, each with a scene id attached. A boundary counts as **cut-aligned when a scene in the same code starts within 0.3 s of the beat's start_s**. Scenes come from the legacy nine-sample grid (a run of consecutive samples with the same `frame_type`, `boundary_confidence = 'low'`), so the resolution of this test is bounded by that grid and not by the editors' actual work. The 264 codes here are the beat-parsed set, not the full 268.

The measurement

	VALUE		
beat boundaries checked	1 527		
beats carrying a scene id	1 527 (100 %)		
codes with alignment computed	264		
boundaries with a scene start within 0.3 s	55		
semantic boundary cut rate	0.036 — 3.6 %		
codes with no aligned boundary at all	215 of 264 (81 %)		
PERFORMANCE OF REELS WITH ≥1 ALIGNED BOUNDARY (N=49) VS NONE (N=215)	ALIGNED	NONE	P
median <code>robust_z</code>	1.45	1.25	0.603
median share rate	0.0168	0.0149	0.28
median save rate	—	—	0.64

Two statements, and the order matters. **Descriptively: editors in this niche do not cut on meaning.** Fewer than four beat boundaries in a hundred coincide with a visible shot change, and four reels in five contain no coincidence at all. **Evaluatively: nothing follows from that**, because reels that do align show no advantage on any metric, and the null itself sits on a measurement floor [I-21].

Why the honest reading is a null, not a finding

With nine samples over a 54-second median runtime, the average gap between observable scene boundaries is roughly six seconds, while the alignment window is 0.3 seconds — twenty times narrower than the measurement's own resolution. A perfectly beat-cut reel edited at, say, one cut every four seconds would still register almost no alignments, because eight of its nine sampled boundaries fall wherever the sampler happened to land rather than where the editor cut. The 3.6 % is therefore **at least partly an artefact of the grid**, and the correct label is INSUFFICIENT rather than "editors do not do this".

The same reasoning cuts the other way for the performance test. A null computed on a floor is not evidence of absence: it is evidence that the question was asked at the wrong resolution. Answering it properly needs the `scene-v1` detector (real scene intervals, $\text{cuts} = \text{scenes} - 1$) rather than the count-only legacy metric, over a corpus that has not been pre-selected for strength.

What the aligned reels look like, for what it is worth

The 49 codes with at least one alignment are not a coherent group. They include [DWCPx0PkQzA](https://www.instagram.com/reel/DWCPx0PkQzA/) (1 aligned of 5 checked), the corpus's best-evidenced `A > SCREEN > A` reel, whose spoken payoff — **"Normandy would take you four steps to create beautiful websites like this But I've created a github repo where all you need to do is copy one line of code"** (47.4–58.5 s) — does begin as the frame returns from the screenshot to the presenter's close-up. They also include [DUduffIE5w5](https://www.instagram.com/reel/DUduffIE5w5/), the one reel in the reference pool that is disqualified on content grounds under RULES.md §2a. Alignment is not a quality signal; at this resolution it is close to a coin toss.

Conversely, several of the strongest reels in the corpus register **zero** aligned boundaries: [DNA7-d3o5sQ](https://www.instagram.com/reel/DNA7-d3o5sQ/) (0 of 7), [DYC-x8DogOI](https://www.instagram.com/reel/DYC-x8DogOI/) (0 of 4), [DLzPQzGNEmY](https://www.instagram.com/reel/DLzPQzGNEmY/) (0 of 6), [DZgBxjnBzMe](https://www.instagram.com/reel/DZgBxjnBzMe/) (0 of 2). DNA7-d3o5sQ is the most instructive: its **1.8-second re-hook — "But how about the results?" (32.0–33.8 s)** — is exactly the beat a meaning-cut editor would cut on, and fa-v1 records a hard cut between samples 3 and 4 of that reel. The alignment test does not see it because neither the beat boundary nor the cut lands where the sampler looked. That single case is the clearest available illustration of what the 3.6 % is measuring.

Visual reading. The related free-text field `visual_to_script_sync` — fa-v1's own prose judgement of whether the pictures match the words — was **never tested against performance**, because it holds free prose rather than a category. Its per-reel content is often precise (DNA7-d3o5sQ: "Captions across the samples ... read as fragments describing the assistant routing a spoken request to an AI agent that books an appointment in Google Calendar, matching the post caption exactly"), and it is usable when reading a single reference by hand. It is not usable as a variable, and no statistic in this report rests on it.

Strategic implication

(1) Do not spend M2's editing budget aligning cuts to script beats. There is no measured return, and under the one-take rule the question does not arise at all — a reel with one speaker change has one cut to place, and it should be placed where the speaker changes. **(2) Do not read the 3.6 % as a craft criticism of the niche** and never publish it as one; it is a statement about a nine-sample grid. **(3) If M2 ever wants this question answered** — for instance to decide whether a screen insert should land on the solution beat or before it — it must be answered on the new detector, and the decision until then should follow chapter 22's storyboard (screen enters with the mechanism, face returns for the payoff) because that pattern *does* have measured support.

Confidence: INSUFFICIENT, and deliberately so. Both the 3.6 % alignment rate and the null performance result (robust_z 1.45 vs 1.25, $p=0.603$) are provisional: the 0.3-second window is twenty times finer than the ~6-second resolution of the nine-sample scene grid, 215 of 264 codes contribute a zero by construction, and the analysed corpus is pre-selected for strength. Nothing in this chapter may be quoted as a fact about how this niche is edited, and nothing here licenses a causal claim.

26. Performance Analysis

Denominators. Four metrics: `view_lift` ($\text{plays} \div \text{the creator's accumulated median} - 1$), `robust_z` (a MAD-based z on $\ln(1+\text{plays})$ against the creator's own distribution, clipped to ± 5), `share_rate` and `save_rate`. **Every rate is NULL below 100 plays** (`stats.MIN_PLAY_FOR_RATE`). **HikerAPI drops the save field on ~9–10 % of rows (486 of 5 147) and 226 codes have never had a save value**, so `save_rate` always runs on a smaller n — 245 inside the analysed tier, 2 899 across the corpus where both rates clear the guard. The association run covers **460 Spearman pairs (115 features × 4 metrics)** over n=268 rows and 102 creators, plus **1 135 categorical contrasts** with bootstrap CIs (1 000 replicates, seed 20260911, **resampling creators, not videos**). `robust_z_floored` is set on **0** rows, so no magnitude here is an upper bound for that reason.

The headline: 13 of 460 pairs are RELIABLE, and none of them is against reach

FEATURE	METRIC	N	CREATOR S	RHO	P	CREATOR- NORMALISED
lex_platforms_n	save_rate	24 5	95	+0.33 1	1.1e- 07	+0.173
lex_cta_verb_n	save_rate	24 5	95	+0.24 1	1.4e- 04	+0.155
lex_questions_per_10s	save_rate	24 5	95	-0.23 5	2.0e- 04	-0.186
questions_n	save_rate	24 5	95	-0.22 9	3.1e- 04	-0.162
lex_second_person_n	save_rate	24 5	95	+0.20 6	1.2e- 03	+0.177
blk_hook_wps	save_rate	24 0	94	+0.20 5	1.4e- 03	+0.156
screen_share	share_rate	26 6	102	+0.20 3	8.5e- 04	+0.129
blk_hook_segments	save_rate	24 5	95	+0.19 8	1.9e- 03	+0.197
lex_rhetorical_questions_n	save_rate	24 5	95	-0.18 6	3.5e- 03	-0.151
lex_first_person_n	save_rate	24 5	95	-0.18 0	4.8e- 03	-0.124
lex_direct_address_per_100w	save_rate	24 5	95	+0.17 4	6.3e- 03	+0.127
lex_entities_distinct	view_lift	26 8	102	+0.16 1	8.3e- 03	+0.198
first visual state is a screen	share_rate	16 1	—	+0.20 8	6.3e- 04	+0.285
any screen state present	save_rate	24 5	—	+0.21 0	9.8e- 04	+0.213

Nine of the thirteen are save-rate correlations. One is share rate. One is view lift. None is **robust_z**. Read with the strong-versus-weak result — of 115 features, none separates the top from the bottom **robust_z** quartile at $p < 0.01$ — the conclusion is unavoidable: **what the video controls is the save and the share, not the view** [I-22]. Reach is driven by things this dataset does not contain.

Phrase and word analysis: the save is bought with second person and named destinations

The save-rate block above is not eleven findings; it is one behaviour measured eleven ways. A save is a viewer filing away something they intend to *do*. The language that produces it is **concrete, addressed and destination-bearing**: platform names (+0.331), CTA verbs (+0.241), second person (+0.206), direct address per 100 words (+0.174). The language that suppresses it is **interrogative and self-referential**: questions per 10 s (−0.235), question count (−0.229), rhetorical questions (−0.186), first person (−0.180). All eight survive creator normalisation [I-15, I-07].

The reach side is thinner. `lex_entities_distinct` → `view_lift` is the **only reach correlation in 460 pairs that is RELIABLE**, and it is *stronger* inside creators (+0.198) than pooled (+0.161). Two PROBABLE pairs behave the same: `lex_tools_n` → `view_lift` +0.157 (cn **+0.223**) and `specificity` → `view_lift` +0.132 (cn **+0.205**), → `robust_z` +0.121 (cn +0.197). The quartile test agrees — 2 distinct entities against 1 (p=0.011), specificity 4.26 against 3.45 (p=0.080) [I-13].

Two counter-directions. **Pressure language runs negative on everything measured**: urgency → `view_lift` −0.164 (p=0.0072) and → `share_rate` −0.150 (cn **−0.180**), fear → `share_rate` −0.157 (cn −0.086), weak quartile 2 urgency words against 1 (p=0.056) [I-14]. And **lexical diversity runs backwards**: `lex_ttr` 0.605 against 0.628 (p=0.039), `lex_mattr50` 0.795 against 0.811 (p=0.044), both lower in the strong quartile (chapter 20).

What vanished under creator normalisation must travel with the table above, because until it is tested it looks like the same kind of number.

PAIR	POOLED RHO	CREATOR-NORMALISED
<code>info_density</code> → <code>save_rate</code>	+0.303	+0.132
<code>lex_technical_n</code> → <code>save_rate</code>	+0.268	−0.060
<code>lex_entities_distinct</code> → <code>save_rate</code>	+0.256	+0.030
<code>specificity</code> → <code>save_rate</code>	+0.237	+0.047
<code>wps</code> → <code>save_rate</code>	+0.213	+0.071
<code>split_share</code> → <code>save_rate</code>	+0.193	+0.036
<code>text_overlay_density</code> → <code>save_rate</code>	+0.180	+0.080
<code>solution_s</code> → <code>save_rate</code>	+0.343	−0.217 (sign flip)
<code>setup_s</code> → <code>view_lift</code> (n=34)	+0.395	−0.318 (sign flip)

These are statements about **which creators post what**, not about videos, and quoting any of them as a production instruction would be an error.

Transcript reading. [DVtgbY8Av6A](https://www.instagram.com/reel/DVtgbY8Av6A/)(<https://www.instagram.com/reel/DVtgbY8Av6A/>) is these findings in 84 words: fifteen named objects in thirteen seconds ("My cost reducer skill. My internet skill. My scalability skill...", 2.3–15.3 s), zero questions, an imperative close, **57.3 saves per 1 000**. Its opposite is [DcvBtiNtbYO](https://www.instagram.com/reel/DcvBtiNtbYO/)(<https://www.instagram.com/reel/DcvBtiNtbYO/>), a sung reel with an identity-call hook, no destination and no ask, carrying the corpus's second-highest **share** rate (59.2 per 1 000) and a save rate of **7.3**. The two metrics are not the same instruction.

Proof analysis

PROOF_TYPE	N	CREATORS	MEDIAN VIEW_LIFT	MEDIAN SAVE/1K	CONFIDENCE
screen_demo	84	50	2.64	27.6	SUFFICIENT_SAMPLE
none	68	46	2.09	29.4	SUFFICIENT_SAMPLE
numbers	36	28	1.81	33.5	SUFFICIENT_SAMPLE
personal_story	28	20	1.82	29.9	PROBABLE
authority_claim	25	20	1.25	14.8	PROBABLE
third_party_data	16	12	1.72	24.9	PROBABLE
before_after	5	5	0.78	21.2	PROBABLE
client_story	4	3	3.82	15.1	INSUFFICIENT

Proof is optional in this niche and the corpus does not punish its absence. Sixty-eight of 266 reels carry no proof beat and perform indistinguishably from the 84 with a screen demonstration (2.09× against 2.64×; both SUFFICIENT_SAMPLE at 46 and 50 creators). Proof *length* does not separate the quartiles either (19.25 s against 23.9 s, p=0.551). Where proof exists it is **visual rather than spoken**: 41 % of 2 645 labelled frames are flagged `is_proof_visual` against 50 % of reels carrying a proof *beat*, and the median time to the first proof beat is **22.0 s** (p25 11.1, p75 32.0) [I-18].

Two absences are the finding. `before_after` is the **second-rarest** proof type (n=5) and below the corpus median on lift; and **nothing in the 266 analysed reels shows a failure**. For M2 that is an opportunity, not a cost: POSITIONING.md §6 already requires evidence M2 owns, and a reel showing its own run *including the part that did not work* does something no measured competitor does — delivered as screen footage at 20–25 seconds, where this audience expects evidence to arrive.

Temporal and momentum analysis

	VALUE
median <code>log_play_slope_last10</code>	-0.052 (mean -0.138, p10 -0.589, p90 +0.155)
creators with a full 10-post window	131
of those, trending up	34 (26 %)
median <code>delta_play_pct</code> against the previous post	-5.2 %
median <code>ratio_to_rolling_median</code>	0.885
posts above 5× their creator's rolling median	269 of 3 078 (8.7 %), 101 creators
posts above 10×	150 (4.7 %)

DIRE TOP 12 BY LOG_PLAY_SLOPE_LAST10
CTIO
N

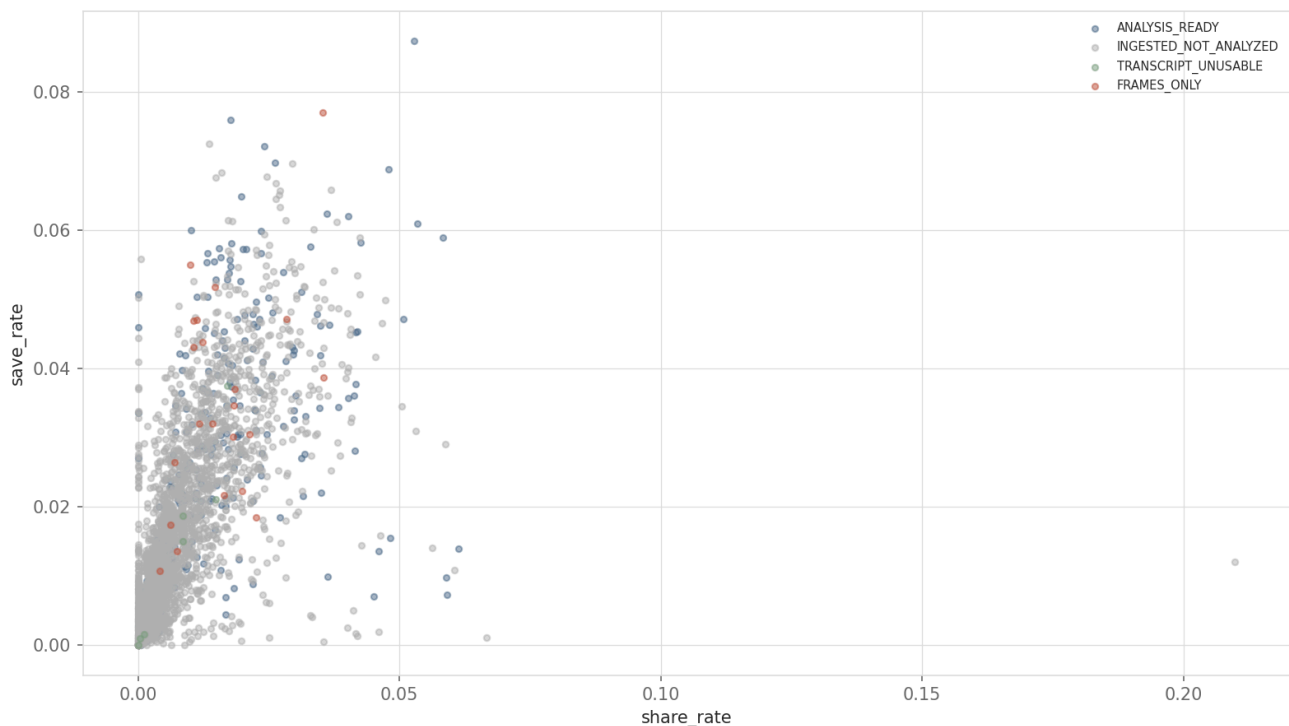
rising swapsays_wtf +0.373 · manthanjethwani +0.337 · gannon.meyer +0.198 · harshsharma_ai +0.197 ·
soojintech +0.174 · valeridoesai +0.167 · protips.ai +0.163 · gennaroautomates +0.162 ·
rence_ur_hands +0.159 · wokecoder +0.145 · kayvon.ai +0.144 · chase.h.ai +0.140

fallin kraya.ai -0.706 · jackroberts___ -0.428 · cashflow.automations -0.413 · andreapalacio -0.403 ·
g kallaway -0.401 · michaelpkocher -0.349 · lukebuildsai -0.311 · petergriffin.ai -0.310 · publicity.ai
-0.303 · davi.d_roberts -0.282 · sanji.chien -0.275 · techflowanjani -0.264

Three quarters of tracked creators are losing reach over their last ten posts; the typical post lands at 0.885 of its author's rolling median; 8.7 % of posts carry the reach and 87 % of creators cannot repeat them [I-24]. Only `harshsharma_ai` is both rising and CONSISTENT. Four limits bind: the slope is a **trend line over post index, not a derivative**, and one hit bends it; there are only **three snapshots** (2026-09-01, -07, -10); the `followers` table is empty, so creator growth is unanswerable; and `engine/stats.py` does **not** control for age, so `view_lift` for very recent reels is biased downwards. Emerging *formats* cannot be separated from emerging *topics* here. What the data does support is the measured justification for RULES.md's 14-day freshness threshold: **a competitor's recent reach is not a stable benchmark.**

Cross-metric relationships

Share rate vs save rate (n=2899)



M2Radar data/radar.db, analysis-ready corpus n=268, 2026-09-12

Share rate vs save rate, 2 899 reels clearing the 100-play guard

	VALUE
Spearman(share_rate, save_rate)	+0.781 (n=2 899)
reels whose save rate exceeds their share rate	86 %
medians across the corpus	save 0.0131 vs share 0.0056 (p95 0.0458 vs 0.0281)
medians inside the analysed tier	save 0.0278 (n=245) vs share 0.0152 (n=268)
HIGH outliers (n=375) vs the rest	share rate ×2.62 , save rate ×2.09
Spearman with <code>robust_z</code>	share +0.326 , save +0.277

Share and save are not independent selection axes. At rho +0.781 they move together, so ranking on one and then the other adds far less information than it appears to, and `cards.py` 's split (Radar on `resh_1k` , Builds and Teardown on `save_1k`) is closer to one ranking than to two. Share remains the marginally better discriminator (×2.62 against ×2.09; rho +0.326 against +0.277), which keeps "why would anyone forward this?" as the right first production question — but **the published magnitudes are wrong**: RULES.md §7's 3.6× and 2.8× become **2.62× and 2.09×** on the 2 899 reels clearing the guard, the original having come from a 100-reel top slice of an earlier 2 352-reel export [I-28].

One relationship dominates and is invisible on this chart: **the comment gate** (chapter 18) multiplies save rate ×2.8 and share rate ×1.8 while leaving `view_lift` and `robust_z` untouched (p=0.58 and 0.63). Any comparison of hi-intent rates that does not hold `cta_type` constant is partly measuring funnels.

Strategic implication

Judge M2's own reels on **shares and saves per 1 000, never on views**, and never promise that a script change will move reach. Write in the second person, name three distinct destinations rather than one three times, delete questions from the script (the corpus median `questions_n` is 0), and drop the urgency register after the hook. Deliver proof visually at ~22 seconds and make it M2's own run including its failure — the one evidence move nobody in this corpus makes. Re-derive the reference pool weekly. And correct RULES.md §7 to 2.62× and 2.09× before anyone quotes it again.

Confidence: RELIABLE for the 13-pair table (every row meets $n \geq 30$, ≥ 8 creators, $p < 0.01$ and a surviving creator-normalised check), for the cross-metric relationships (n=2 899) and for the temporal aggregates as a description of three snapshots. **PROBABLE** for the specificity and tool-name reach findings ($p < 0.05$, survive normalisation, fail the $p < 0.01$ bar) and for the proof table (three SUFFICIENT_SAMPLE cells, but categorical medians admit no normalisation check and 59 % of the corpus is gated). **INSUFFICIENT** for `before_after` (n=5) and `client_story` (n=4). Everything above is associational; method §8 forbids reading any of it as a cause.

27. Reference Selection Logic

What this chapter is. Selection is where the analysis becomes an instruction, and it is the place a measurement error turns into a production error. This chapter states the rules that govern which competitor reel may be used as a reference, how they are implemented in the engine, and what the completed selection for this wave looks like. The per-hypothesis tables live in `reports/analysis/09-reference-selection.md` and in the card book; **41 references across 10 selected hypotheses, drawn from 27 distinct reels, ingested into `hypothesis_refs` with 0 rejected.**

The four selection rules

- 1. Consistent creators and high-variance creators are not selected the same way.** Only **17 of 132** creators are CONSISTENT ($CV < 1$ and $n \geq 8$); **115 are HIGH_VARIANCE** [I-23]. A CONSISTENT creator may contribute several structural references, because their output is predictable and a modest lift from them is a finding. A HIGH_VARIANCE creator may contribute only reels that clear a meaningfulness floor ($view_lift \geq 0.5$ or $robust_z \geq 1.5$), and their multipliers may never be read as evidence that an idea travels. The arithmetic reason is chapter 11's measurement trap: reels by CONSISTENT creators score *lower* on creator-relative metrics (median $robust_z$ 0.60, $view_lift$ 0.65) than reels by HIGH_VARIANCE creators (1.37, 2.38) **with identical script shape**, because both metrics divide by the creator's own spread. Wherever a large multiplier rests on a tiny author median — `DUJYKENjZc5` at 388× on a 1 374-view median, `DVZGBgTk6zz` at 304× on the same, `DZLDk9ySi-7` at 51× on 3 421 — the reference is used for cadence or ordering only, and the reason field says so.
- 2. Gated and ungated rates are not the same quantity.** The comment-keyword gate moves comment rate ×23, save ×2.8 and share ×1.8 while moving reach not at all [I-12], and **33 of the 40 pool rows carry one**. `cta_type` must therefore be held constant in any comparison of hi-intent rates, or recorded on the card so the reader can hold it constant. The seven ungated pool rows were weighted up and six are used; where a gated reel is used, the reason states which of its numbers can be read at face value.
- 3. Cross-creator best video, not best account.** The pool is built on **mean percentile rank across `robust_z`, `share_rate` and `save_rate` inside the analysis-ready corpus, capped at two reels per creator** — 40 reels from 30 creators — so that the reference set is a survey of the niche rather than a portrait of three accounts. The cap is the only thing standing between this method and a pool dominated by whichever account happened to have a good fortnight.
- 4. Selection is hypothesis-specific and function-based.** A reel is never adopted whole. It is attached to one hypothesis for **one named function** from the `hypothesis_refs` vocabulary — `topic`, `hook`, `pain`, `explanation`, `proof`, `cta`, `a_roll`, `b_roll`, `split`, `screen_proof`, `rhythm`, `transition` — with the verbatim beat that makes it useful, the real beat and scene ids, the performance behind it, and an explicit statement of **what is borrowed structurally against what changes** for M2's positioning. A reel may serve several hypotheses only with a *different* function; no `(code, function)` pair may appear twice.

How the rules are implemented

LAYER	MECHANISM
-------	-----------

candidate pool	<code>score.py</code> ranks reels against their own author inside an age band (7/30/90 days), excluding the focal row from its own baseline, minimum baseline 5 videos, <code>Z_CLIP</code> ± 5 , <code>LOG_MAD_FLOOR</code> 0.10
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freshness and format	<code>cards.py</code> : <code>FRESH_DAYS = 14</code> ; three independent selections — M2 Radar (2 slots, ranked on <code>resh_1k</code>), M2 Builds (2 slots, <code>save_1k</code>), M2 Teardown (1 slot, <code>save_1k</code>)
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suitability	RULES §2a check after storyboarding, before the card is written : not about AI in business processes; other people's characters, cloned voices or IP; promotion of a model-restriction bypass
-------------	--

storage	<code>hypothesis_refs</code> (<code>hypothesis_id</code> , <code>code</code> , <code>function</code> , <code>reason</code> , <code>useful_beat_ids_json</code> , <code>useful_scene_ids_json</code> , <code>performance_json</code> , <code>transformation</code>) , primary key (<code>hypothesis_id</code> , <code>code</code> , <code>function</code>) — the schema itself enforces rule 4
---------	---

ingest	<code>engine/ingest_insights.py</code> loads <code>data/analysis/hypothesis_refs.json</code> ; a row failing validation is rejected, never silently repaired
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Two defects in the current implementation are worth naming because they are fixable and consequential. `cards.py` ranks on `resh_1k` and `save_1k` with no `cta_type` guard, so with 59 % of the analysed corpus gated it part-ranks DM funnels rather than ideas (chapter 18). And `deepdives.suitable` is NULL for every code except four 2026-09-01 rejections (4 of 282 deep-dive rows; NULL on all 23 drafted cards' sources), so the RULES §2a compliance record does not exist anywhere in the database even though the rule has been on paper since 1 September.

What the completed selection looks like

	VALUE
--	-------

hypotheses with references	10
----------------------------	----

references written	41 (4 per hypothesis, 5 for H-11)
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distinct reels used	27 — 9 carry two functions each, 18 carry one
---------------------	---

reels from CONSISTENT creators	1 (<code>DcoE5ZsK4I7</code> , <code>damini.knows</code>)
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ungated reels used	6 of the 7 available
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gated reels used	21 of 27
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reels read and rejected	13, of which 3 on content grounds under RULES §2a
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Transcript reading — the rules doing visible work. The two highest-rate reels available were both refused. [DK9pFUjPQcx](https://www.instagram.com/reel/DK9pFUjPQcx/)(<https://www.instagram.com/reel/DK9pFUjPQcx/>) carries the best combined hi-intent rates in the pool (42.5 shares and 58.2 saves per 1 000) and is excluded because its solution beat says "First, open route up with over 60 free api keys, you can even try the Uncensored ones." (8.9–31.0 s) — promotion of a model-restriction bypass, RULES §2a clause 3. [DUduffIE5w5](https://www.instagram.com/reel/DUduffIE5w5/)(<https://www.instagram.com/reel/DUduffIE5w5/>) is structurally the strongest reel in its

batch and is excluded on two §2a clauses at once. **The rule has to fire on the numbers, not in spite of them**, and these two rejections are the evidence that it does.

Visual reading. Layout is now part of selection, and it changes outcomes.

[Dct6Op3n6Zd](https://www.instagram.com/reel/Dct6Op3n6Zd/)(<https://www.instagram.com/reel/Dct6Op3n6Zd/>) carries the corpus's second-highest save rate (75.9 per 1 000) and was rejected partly because it is held in `SPLIT_SCREEN` for all 37 samples — the only roll share with a negative association to `robust_z` clearing $p < 0.05$ [I-10] — and its countdown rhythm is already better represented without the layout problem. This is the direct fix for the defect chapter 23 records: **9 of main's 23 legacy source reels opened on split screen**, because the old selector was blind to visual structure. Every reference now carries its `first_frame_type` and coarse `visual_sequence`, and each hypothesis carries a planned visual grammar (predominantly `SCR>A>SCR>A` or `A>SCR>A`, and no split screen anywhere).

What the selection is weak on

Four weaknesses are recorded rather than hidden. **One CONSISTENT creator in 27** — a consequence of creator-relative metrics favouring high-variance accounts, which means most quoted lifts describe a creator's spread rather than a reel; read the rates, not the lifts. **Twenty-one of 27 reels are gated**, so their save and share rates carry a known inflation. **No reference shows a failure, because none exists** — `before_after` is the second-rarest proof type ($n=5$) and not one reel in 266 shows its author's build breaking, so for the four hypotheses that turn on exactly that the reference set supplies grammar and never precedent. And **every scene id comes from a nine-sample interpolation** with `boundary_confidence = 'low'`: use scene ids to locate a moment, never to plan a cut.

Strategic implication

Three changes to make before the next card run. **(1) Recompute every headline multiplier against the accumulated creator median and carry the baseline** `n`, `robust_z`, `percentile_in_creator`, `outlier_status` and `reliability` **onto the card** — two of 23 legacy cards had materially wrong headline numbers without it. **(2) Record** `cta_type` **on every reference and compare hi-intent rates only within the same** `cta_type`. **(3) Populate** `deepdives.suitable` **before an angle is written**. Beyond the fixes, the standing rule is the one the function vocabulary encodes: **borrow a beat, never a topic, and write down the transformation** — which is RULES §1.2 made executable.

Confidence: RELIABLE for the mechanics (schema, ranking code and ingest counts are direct observations of the engine). **PROBABLE** for the evidence behind individual rules — rule 1 rests on I-23, rule 2 on I-12 (both RELIABLE), while the visual constraint behind the layout check rests on I-10 at $p=0.044$. **INSUFFICIENT** as a claim that this selection produces better reels: `our_posts` is empty, nothing has been shot or published, and the loop that would test it does not yet have a single data point.

28. Best Creators

Denominators, and what "best" can mean here. 132 creators have a `creator_stats` row at snapshot 3; 102 contribute reels to the analysis-ready set. "Best" has to be defined before it is measured, because the three obvious definitions rank almost disjoint sets of accounts. **Biggest** (median play) is a statement about audience, not craft. **Steadiest** ($CV < 1$, $n \geq 8 \rightarrow$ `CONSISTENT`) is a statement about repeatability. **Most forwarded and kept** (median share and save rate) is the only one that measures what a reel does to the person who watched it — and it is the definition M2 should use, because reach in this corpus is not a video property (chapter 26).

The seven consistent creators with a real audience

USERNAME	N VIDEOS	MEDIAN PLAY	CV	CONSISTENCY	POSTS/WEEK	HIGH-PERFORMER SHARE	SHARE/1K	SAVE/1K
angus.sewell	12	89 439	0.98	0.505	4.39	0.250	9.6	23.1
harshsharma_ai	12	44 377	0.96	0.509	2.59	0.333	13.9	—
chris.raroque	12	36 820	0.80	0.555	5.60	0.167	6.8	23.3
decodingai.vs	29	30 932	0.73	0.580	4.09	0.138	12.6	24.5
ajsahni.ai	12	28 067	0.95	0.512	1.64	0.250	4.4	—
justyn.ai	30	19 173	0.91	0.524	0.72	0.133	5.8	17.9
divyannshisharma	27	18 948	0.78	0.561	1.47	0.185	22.0	36.1

Six of these seven are steady and unremarkable on intent. **divyannshisharma** is the exception that **matters**: consistent *and* above the analysed-set medians on both hi-intent rates (0.0152 share, 0.0278 save). The eighth name belongs on this list even though it falls below the play threshold — **damini.knows** (14 videos, median play 12 678, the roster's second-lowest CV at **0.64**, 19.9 shares and 31.4 saves per 1 000). Those two are the only creators in the entire roster that are simultaneously predictable and better than the median at being forwarded and kept. **They are the two accounts worth studying properly.**

The rest of the `CONSISTENT` list is a warning. `automationatlas.co` posts **32.1 times a week** to a **459-play** median and `hcmettelus` is steady at 410 — consistency without distribution is not a model, and cadence explains nothing across the roster (the range runs from 32.1 to 0.13 posts a week with no relation to performance).

The high-variance top, and how to read a multiplier from it

USERNAME	N	MEDIAN PLAY	CV	OUTLIER SHARE	TOP - 1 CODE	TOP - 1 LIFT	TOP - 1 PLAY
adamstewartmarke ting	3 1	6 415	4.7 7	0.161	CrvV4zk- Mth4	218.8×	1 409 789
mikemeansbusines s_ai	2 7	1 314	4.6 5	0.037	DVRu6Gl- DNHO	281.2×	370 807
dainikautomates	2 7	3 619	4.5 3	0.148	Db- 3ruzKgzw	256.8×	932 923
shrug.manny	2 7	14 448	4.4 6	0.222	DYQkKkx- R3Et	1 210×	17 496 130
gregisenberg	2 6	18 949	4.0 1	0.077	Dal9I7KRA1 L	159.1×	3 033 819
jarvis_ai_spark	2 2	7 544	3.9 4	0.091	DZgBxjn- BzMe	207.2×	1 570 719
valeridoesai	3 0	4 946	3.8 3	0.200	DYC- x8DogOI	399.4×	1 980 128
davi.d_roberts	2 4	3 862	3.5 9	0.208	DKzbAvyo- CMJ	1 190×	4 597 656
loganwelbaum	2 4	3 170	3.5 4	0.083	Cq3Lkw1JaB v	2 381×	7 550 238
sharmadhruveshh	2 8	1 375	3.4 5	0.250	DUJYKENj- Zc5	388.2×	534 953
kraya.ai	3 2	28 122	3.2 1	0.031	DcvrU4_xFjU	37.8×	1 090 634

loganwelbaum 's 2 381× and davi.d_roberts 's 1 190× are statements about a **3 170-view** and a **3 862-view** median, not about a reel. Per RULES §6 the honest label for these is "резонировало у своих" at best; at worst it is noise. **A multiplier is interpretable only alongside the creator's CV and baseline n**, and this table exists to make that inseparable.

Momentum: who is rising

Rising, top 12: swapsays_wtf +0.373 · manthanjethwani **+0.337** · gannon.meyer +0.198 · harshsharma_ai +0.197 · soojintech +0.174 · valeridoesai +0.167 · protips.ai +0.163 · gennaroautomates +0.162 · rence_ur_hands +0.159 · wokecoder +0.145 · kayvon.ai +0.144 · chase.h.ai +0.140. Only harshsharma_ai appears on both the rising list and the CONSISTENT list. The slope is a trend line over post index, not a derivative; one hit inside a ten-post window bends it, and there are only three snapshots.

manthanjethwani **(+0.337)** is the single most interesting account in the corpus for M2. He is rising, he contributes multiple reels to the top of the pool, and his work is the clearest example of demo-first,

screen-evidenced craft on a rising trajectory. He also owns the corpus's highest share rate: [Dc_tsSeAjBy](https://www.instagram.com/reel/Dc_tsSeAjBy/)(https://www.instagram.com/reel/Dc_tsSeAjBy/) at **61.3 shares per 1 000** on 2 350 035 plays and 66× his accumulated median.

Transcript reading. What that account does is name the exact object and then show it. Dc_tsSeAjBy's proof beat is **"Look at the other use cases. This is going to Excel, the whole Excel is controlling. And you can automate whatever you do in Excel."** (29.0–48.0 s), and its close is a genuine question rather than a gate — **"And this is just the beginning, what do you think? How many people's jobs will be perfect because of this?"** (85.0–90.0 s), `cta_type = question_to_audience`, so its share rate carries no gate inflation. His other high reel, [Dbi-yxNgrhG](https://www.instagram.com/reel/Dbi-yxNgrhG/)(https://www.instagram.com/reel/Dbi-yxNgrhG/) (240 255 plays, 58.4 shares and 58.9 saves per 1 000), does the same with a universal account and a bounded count: **"If you don't have these four settings, you can access any private documents or bank statements on your Gmail."** (0.6–7.6 s). A system everyone already uses, a countable promise, instructions too fast to follow live — so the reel is forwarded and saved rather than watched.

Visual reading. Both reels run `A_ROLL_MEDIUM > SCREEN_RECORDING > ...` — face for the claim, full-frame screen for the evidence, which is chapter 22's pattern. fa-v1 records Dbi-yxNgrhG's proof visuals as a real Gmail inbox with the Gemini sidebar, the settings dialog and the phone settings page: three screens, all of them the viewer's own software, none of them a product demo.

Strategic implication

Study two accounts properly and copy neither's topic. `damini.knows` and `divyannshisharma` are the only creators who are both predictable and above median on forwarding and saving — they are the model for *how to be a reliable account*. `manthanjethwani` is the model for *what a forwardable reel looks like*: a system the audience already owns, a bounded count, screen evidence, an ungated close. On selection, carry `reliability`, CV and baseline `n` onto every reference card, prefer creators with more than one entry in the pool, and treat any multiplier over ~100× as a fact about a small median. On measurement, expect M2's own numbers to look like the CONSISTENT rows — small lifts, decent rates — and do not read that as failure.

Confidence: RELIABLE for the roster statistics and reliability labels ($n=132$, complete, direct measurement). **PROBABLE** for "who is worth studying", which rests on two accounts and on share/save comparisons that do not hold `cta_type` constant. **INSUFFICIENT** for the momentum ranking as a predictor: three snapshots, a trend line rather than a derivative, and an empty `followers` table mean nothing here can be said about growth.

29. Best Individual Reels

Denominators. Every reel below is inside the analysis-ready corpus (**n=268**, 102 creators). Share rate is available on all 268; **save rate on 245** (HikerAPI drops the field on ~9–10 % of rows). All four metrics carry the 100-play denominator guard. `robust_z` is clipped at ± 5 , so **25 reels tie at 5.0** and the magnitude there is a floor, not a value (`06-creators-and-references.md` §5 says 23; the count in `reports/data/analysis_ready.csv` is 25). Above all: this is the top of `score.py` 's own ranking, so "best" means best inside an already-selected set [I-01].

Best per metric

METRIC	REEL	CREATOR	VALUE	CONTEXT
highest <code>view_lift</code>	DYC-x8DogOI (https://www.instagram.com/reel/DYC-x8DogOI/)	valeridoesai	399.4×	1 980 128 plays on a 4 946 median
second	DUJYKENj-Zc5 (https://www.instagram.com/reel/DUJYKENjZc5/)	sharmadhruveshh	388.2×	534 953 plays on a 1 374 median
highest <code>robust_z</code>	25 reels tied at the ± 5 clip	—	5.0	the clip makes the magnitude a floor
highest share rate	Dc_tsSe-AjBy (https://www.instagram.com/reel/Dc_tsSeAjBy/)	manthanjethwani	61.3 / 1 000	2 350 035 plays, 66× median, ungated (<code>question_to_audience</code>)
second	DcvBti-NtbYO (https://www.instagram.com/reel/DcvBtiNtbYO/)	leadgenman	59.2 / 1 000	79 235 plays, 2.6× median, ungated, save rate only 7.3
highest save rate	DbO4zv-JR1k8 (https://www.instagram.com/reel/DbO4zvJR1k8/)	shrug.mannny	87.4 / 1 000	209 121 plays, 13.5× median, gated
second	Dct6Op3n6Zd (https://www.instagram.com/reel/Dct6Op3n6Zd/)	valeridoesai	75.9 / 1 000	90 775 plays, 17.4× median, gated, held split screen
highest absolute play	Dc9GZ0Gzf6o (https://www.instagram.com/reel/Dc9GZ0Gzf6o/)	anshmehra.in	6 043 051	18.3× median, no save value, topic is entertainment
best combined share + save	DbO4zv-JR1k8 (https://www.instagram.com/reel/DbO4zvJR1k8/)	shrug.mannny	52.9 + 87.4 = 140.3	gated; the ungated best is <code>Dc_tsSeAjBy</code>
best combined at ordinary reach	Db9MT-axxcO (https://www.instagram.com/reel/Db9MT-axxcO/)	shrug.mannny	47.9 + 68.9 = 116.7	only 1.6× its own median

Three rows in this table correct `reports/analysis/06-creators-and-references.md` §5, which was written from the reference pool rather than from the full analysed set: the highest share rate is Dc_tsSeAjBy at 61.3, not Dbi-yxNgrhG at 58.4; the highest absolute play is Dc9GZ0Gzf6o at 6 043 051, not DRXZJeHiAES at 3 349 555; and the best combined share+save is DbO4zvJR1k8 at 140.3, not Db9MT-axxcO at 116.7. All four differences are recorded in this report's source-verification note.

The instructive row is the last one. Db9MT-axxcO carries near-best hi-intent rates on a reel that barely beat its own author's median. RULES §6 already has a label for this case — "резонировало у своих", resonated with its own audience — and it is exactly the reference a views-based filter drops and a share/save filter promotes. Both filters are wrong on their own; the card must carry all three numbers.

The five reels worth studying, and why

1. DNA7-d3o5sQ(<https://www.instagram.com/reel/DNA7-d3o5sQ/>) — edhillai, 451 491 plays, 55.4×, 23.3 shares / 36.5 saves per 1 000, `cta_type: none`. The best overall template in the corpus, and it is **ungated**, so every rate can be read at face value. Screen-first: the artefact speaks the first four words — "Hello, how can I assist you? This voice assistant can replace your receptionist. I'll show you how to build it in three easy steps." (hook, 0.0–6.4 s). The mechanism names each connector at the moment it becomes necessary ("it uses Google Calendar to check your schedule and Google Sheets to save any details"). A **1.8-second rehook** — "But how about the results?" (32.0–33.8 s) — sits exactly where the mechanism stops being new, and is followed by a complete recorded call ending in a confirmed booking. The closing line is the machine finishing its job. fa-v1 records a held n8n workflow canvas with a small facecam, `first_frame_type: UI_DEMO`, and Google Calendar showing the booked appointment as the second proof visual.

2. DYC-x8DogOI(<https://www.instagram.com/reel/DYC-x8DogOI/>) — valeridoesai, 399.4×, 59.9 saves per 1 000. The corpus's highest lift, and 43 seconds long. The hook states the result, the exact time and the step count in one breath — "I just built a website that looks like it costs \$10,000 with literally one line of code and honestly the whole thing took Claude about two minutes." (0.0–11.8 s) — and every step afterwards states its own benefit. `SCREENSHOT > A_ROLL_CLOSE_UP > SCREENSHOT`: eight of eight observed transitions are hard cuts, and the screen carries the payoff while the face carries only the interpretation. Caution: our own analysis flags this reel and DWCPx0PkQzA as **the same template running on two accounts**, so the pair is one structure, not two pieces of evidence.

3. Dc_TX1CttIc(https://www.instagram.com/reel/Dc_TX1CttIc/) — v.i.s.h.ai, 562 094 plays, 24.1×, 31.7 shares per 1 000, **ungated**. The closest thing in the corpus to an M2 Radar reel. Agenda announced in nine seconds, four cases labelled aloud, and a verdict that is economic rather than promotional: "Very good model for sure, but the API cost has shot up so high that any business trying to build a real product on Astra is walking into a train wreck" (payoff, 80.8–92.6 s). Conditions are stated before the conclusion and the concession comes before the claim — "Every one of these was one shot. And yes, more context makes it better..." This is what a decision delivered whole looks like.

4. DbO4zvJR1k8(<https://www.instagram.com/reel/DbO4zvJR1k8/>) — shrug.manny, 87.4 saves per 1 000, **the corpus maximum**. Seventy-one seconds, one unbroken `A_ROLL_CLOSE_UP`, **no screen at all**, and it never flattens because every item has an identical shape: the consequence, then the exact clause that prevents it. "Number one and most important, if your terms and service doesn't cap your liability, there's no limit to how much somebody can sue you for." (5.6–18.2 s), repeated four times. Proof that a control reel can hold on a face if the cadence is strict — and a reminder that its save rate carries the gate's ×2.8 inflation.

5. **Db9MT-axxcO**(<https://www.instagram.com/reel/Db9MT-axxcO/>) — shrug.manny, 1.6× lift, 47.9 shares / 68.9 saves per 1 000. Result at 3 seconds, three steps in thirty, ask at forty-three; simultaneously a story and a procedure. Its question hook works because the same breath answers it: "What happens when you ask Claw to apply to 500 jobs for you? I tried this and I landed six interviews in 24 hours." (0.0–7.0 s). A > SCR > A > SCR > A — the face for interpretation, the screen for the action — with a phone calendar of booked interviews as the first proof visual.

Two reels that top a metric and must not be copied

DcvBtiNtbYO(<https://www.instagram.com/reel/DcvBtiNtbYO/>) is the corpus's second-highest share rate (59.2 per 1 000) and a **sung** reel — "I'm maxing out my fable 5 on extra high" (hook, 0.3–4.7 s) — with no destination, no ask and a save rate of **7.3**. It demonstrates that share and save are different instructions (chapter 26) and nothing else. **DUduffIE5w5**(<https://www.instagram.com/reel/DUduffIE5w5/>) is structurally strong and **disqualified on content** under RULES §2a (cloned cartoon voices, promotion of an uncensored model). It is kept in the pool solely so the suitability check has something to fire on.

Strategic implication

Use **DNA7-d3o5sQ** as the **M2 Builds template** (screen-first, one rehook, ungated close, artefact speaking first), **Dc_TX1CttIc** as the **M2 Radar template** (agenda, labelled cases, economic verdict), and **Db9MT-axxcO** as the **Teardown clock** (outcome by 5 s, steps in one run, ask in the last tenth). Read **Db04zvJR1k8** for cadence and discount its save rate by the gate. And put all three numbers — lift with its baseline **n**, share per 1 000, save per 1 000, plus **cta_type** and **reliability** — on every card, so that a 399× on a 4 946 median and a 1.6× on a 14 448 median are never mistaken for the same event.

Confidence: RELIABLE for the per-metric maxima (direct measurement over **reports/data/analysis_ready.csv**, n=268/245). **PROBABLE** for the reading of any individual reel: each is a single case, four of the seven named are gated, and a reel's structure is a hypothesis about why it worked, never a measurement of it. **INSUFFICIENT** for anything about the 25 reels tied at **robust_z** 5.0 — the clip destroys the ordering among them.

30. Cross-Video Conclusions

What this chapter does. Chapters 11–29 each answered one question about one layer. This chapter states what holds when the layers are read together; every figure below has already appeared with its denominator. Those remain **n=3 211** for metric-only statements, **n=268** for anything joining script and visuals, **266/264** for semantic and beat statements, **295** for frames, and **9 fixed samples** per reel throughout.

1. Format is not the lever, and the negative result is the strongest finding of the wave

Of **115 numeric features tested**, none separates the top from the bottom `robust_z` quartile at $p < 0.01$ and only seven reach $p < 0.05$, two of them in the counter-intuitive direction. Of **460 Spearman pairs**, 13 are RELIABLE and **none is against** `robust_z`. Duration, spoken length, words per second, hook duration, cut rate, scene count, A-roll share, split share and caption density are flat on both tests [I-22].

The negative was obtained under conditions that should have made a difference *easier* to find: the analysed corpus is the top of `score.py`'s ranking (median `robust_z` +1.26 against 0.00 for all 3 211), so the "weak" quartile still beats its own authors' medians at about +0.6. A comparison between the strongest and the merely strong found nothing structural. **Once a reel is competently made, what remains is subject, specificity and evidence.**

2. What a video controls is the save and the share, not the view

Eleven of the 13 RELIABLE correlations (ten distinct features; one is counted twice) are save-rate correlations, one share rate, one view lift, none `robust_z`. Share and save themselves correlate at **+0.781** across 2 899 reels — close to one axis rather than two — with share the marginally better discriminator (**×2.62** against **×2.09** for the 375 HIGH outliers). Reach is driven by factors this dataset does not contain, and any promise that a script change will move it is unsupported.

3. The one production instruction with RELIABLE evidence is: put a screen on camera

`screen_share` → `share_rate` **+0.203** ($p=0.00085$, creator-normalised +0.129) and any-screen-present → `save_rate` **+0.210** (cn +0.213) are the only visual findings that clear the RELIABLE bar. Opening on the screen rather than the face is worth roughly **half again the forwarding rate** (0.0213 against 0.0140), and — uniquely in this wave — the creator-normalised correlation (**+0.285**) is *stronger* than the pooled one (+0.208), which means the effect holds inside individual creators' own output [I-08, I-09]. Its mirror is equally consistent: more face is weakly worse (`a_roll_share` → `share_rate` -0.156, cn -0.094, survives), and the niche's most-used arrangement, split screen (22.3 % of all frames), is its least-rewarded (`robust_z` 1.03 against 1.43, $p=0.044$).

4. Three findings converge on the same sentence from three directions

DIRECTION	MEASUREMENT
lexical	<code>lex_platforms_n</code> → save_rate +0.331 (cn +0.173), <code>lex_entities_distinct</code> → view_lift +0.161 (cn +0.198)
categorical	<code>resource_handoff</code> save rate 0.0429 against <code>framework_mental_model</code> 0.0135
linguistic	sentences end on a destination — "right now" 13, "for free" 11, "on github" 10 — not on a conclusion

All three say: **name the destination and hand over the object**. The audience saves what it intends to use and does not save what it is asked to believe — the desire category "certainty that a thing actually works" carries save rate **0.0140 against 0.0282** (n=19, p=0.017, negative). This is the strongest cross-layer agreement in the report, and it holds across a lexical correlation, a categorical median and a phrase count computed independently of one another.

5. The corpus's shape is four moves, and its two structural omissions are M2's opening

The modal ordering is **HOOK > EXPLANATION > PAYOFF > CTA** (27 of 264, 10 %); the hook opens 259 of 264 reels and a CTA closes 215. Explanation is present in 79 % of reels and takes 38 % of the speech. Against that, **only 24 % of reels state a problem at all** and **half carry no proof beat**; `before_after` is the second-rarest proof type (n=5) and **not one reel in 266 shows its author's own build failing**. Those two omissions are exactly what POSITIONING.md §8 requires of M2 — friction, and an AI boundary with real evidence — which means M2's differentiation is structural rather than stylistic, and has to be paid for in seconds taken out of the mechanism block.

Transcript reading across videos. The same three sentences recur at the top of the corpus regardless of topic: a result with a quantity in it, a step naming a specific product, and a payoff restating what changed. [DYC-x8DogOI](https://www.instagram.com/reel/DYC-x8DogOI/)(https://www.instagram.com/reel/DYC-x8DogOI/): "I just built a website that looks like it costs \$10,000 with literally one line of code and honestly the whole thing took Claude about two minutes." [DNA7-d3o5sQ](https://www.instagram.com/reel/DNA7-d3o5sQ/)(https://www.instagram.com/reel/DNA7-d3o5sQ/): "This agent understands your goal but needs data. Therefore, it uses Google Calendar to check your schedule and Google Sheets to save any details." [DWCPxOPkQzA](https://www.instagram.com/reel/DWCPxOPkQzA/)(https://www.instagram.com/reel/DWCPxOPkQzA/): "But I've created a github repo where all you need to do is copy one line of code." Result, named destination, object handed over. The reels at the bottom of the corpus have none of the three: [DXb8P08Dati](https://www.instagram.com/reel/DXb8P08Dati/)(https://www.instagram.com/reel/DXb8P08Dati/) opens on the series ("Day one of two AI tips...", `robust_z` -1.04) and [DaDn5L-xKWi](https://www.instagram.com/reel/DaDn5L-xKWi/)(https://www.instagram.com/reel/DaDn5L-xKWi/) opens five loops and closes them in five identically shaped blocks.

Visual reading across videos. The visual vocabulary fragments so badly — 295 reels, **206 distinct sequences**, median sequence occurring once — that only the coarse four-way collapse supports any claim. On that collapse the picture is stable across every chapter: full-frame screen up, face down, split screen down, and one arrangement (`A > SCREEN > A` , 39 reels across 32 creators) carrying share rate 0.0213 against 0.0140. Everything else about the picture — scene count 1 to 8, caption density universal at 1.0, cut rate 2.5 to 13.7 per minute — is flat.

6. Two selection defects contaminate the analysis of everything above

The comment gate. 59 % of the analysed corpus runs one. It multiplies save $\times 2.8$ and share $\times 1.8$ and moves reach by nothing (`robust_z` 1.2475 against 1.2481). Since `cards.py` ranks candidates on `resh_1k` and `save_1k`, the radar has been part-ranking DM funnels, and **every share/save comparison in this report that does not hold `cta_type` constant carries that inflation unevenly.**

Creator-relative metrics favour lottery accounts. Reels from CONSISTENT creators score *lower* (median `robust_z` 0.60, `view_lift` 0.65) than reels from HIGH_VARIANCE creators (1.37, 2.38) with identical script shape, purely because both metrics divide by the creator's own spread. That is why 33 of 40 pool rows are gated and only 1 of 27 selected references comes from a CONSISTENT creator — a property of the machinery, not of the niche.

7. Most of the category grid cannot answer a question at all

Of 159 reported categorical cells across eight vocabularies, **18 are SUFFICIENT_SAMPLE, 96 PROBABLE, 45 INSUFFICIENT.** All 20 `subtopic` cells are INSUFFICIENT; 9 of 27 `topic` cells are; no topic cell and only one hook type reaches $n \geq 30$ with 8 creators. **The thinness is itself the finding:** the next wave must buy breadth — more codes per cell — not more features per code [I-30].

Strategic implication

Stop looking for a format that wins. Spend M2's effort on three things the data does support: **subject choice** (a named repeated act, not a condition), **specificity** (name the exact tool, number and step, and repeat the name rather than varying it), and **evidence M2 owns** (own screen, own run, including the failure — the one move no measured competitor makes). Judge M2's reels on shares and saves per 1 000, hold `cta_type` constant when comparing anything, and expect small multipliers once the account becomes consistent.

Confidence. The negative results (§1) and the metric structure (§2) are **RELIABLE** — large n , direct measurement, and obtained under conditions favourable to finding a difference. The screen findings (§3) are **RELIABLE**. The convergence in §4 is **RELIABLE** on its lexical leg and **PROBABLE** on its categorical leg. §5's structural description is **RELIABLE** as a count and **PROBABLE** as a reading. §6 is **RELIABLE**. §7 is **RELIABLE**. Everything here is associational and computed inside a pre-selected 8.3 % of the database; **nothing in this chapter licenses a causal claim.**

31. What Works

What "works" can mean here. Nothing in this chapter is a cause. Every line is an association measured inside the analysis-ready corpus (**n=268**, 102 creators, median `robust_z` +1.26 against 0.00 across all 3 211), which is the top of `score.py` 's own ranking. "Works" therefore means: **associated with a higher creator-relative score, or a higher share or save rate, among reels that were already strong** [I-01]. Where a contrast is present it is probably understated; where one is absent, that is not proof the feature is irrelevant in the niche at large.

The evidence, ranked by strength

#	WHAT WORKS	EVIDENCE	CONFIDENCE
1	Put a real screen on camera, anywhere in the reel	screen_share → share_rate +0.203 (p=0.00085, cn +0.129); any screen state present → save_rate +0.210 (cn +0.213); any-screen reels share 0.0168 vs 0.0118 (p=0.00013), save 0.0308 vs 0.0220 (p=0.001)	RELIABLE
2	Open on the screen, not the face	35 of 266 first frames are a screen: share 0.0213 vs 0.0140, view_lift 5.25 vs 1.88, robust_z 2.13 vs 1.21; pooled +0.208, creator-normalised +0.285 (p=0.00025) — stronger within creators than between them	RELIABLE
3	Name the exact tool, number and step	lex_entities_distinct → view_lift +0.161 (cn +0.198) — the only reach correlation surviving normalisation; strong quartile 2 distinct entities vs 1 (p=0.011); lex_tools_n → view_lift +0.157 (cn +0.223); specificity → view_lift +0.132 (cn +0.205)	RELIABLE (entities) / PROBABLE (rest)
4	Write in the second person with concrete asks	second person → save_rate +0.206 (cn +0.177); CTA verbs +0.241 (cn +0.155); direct address per 100 w +0.174 (cn +0.127); first person -0.180 (cn -0.124)	RELIABLE
5	Write the hook as a statement, not a question	questions_n → save_rate -0.229 (p=0.00031, cn -0.162); questions per 10 s -0.235 (cn -0.186); rhetorical questions -0.186 (cn -0.151); corpus median questions_n = 0	RELIABLE
6	Open on the artefact doing the work (show-first)	demo_first + result_first, n=42, 29 creators: median robust_z 2.17 / view_lift 5.81 vs 1.21 / 1.80 (p=0.021 and 0.008)	PROBABLE
7	Come back to the face for the payoff and the close	A > SCREEN > A, 39 reels / 32 creators: share 0.0213 vs 0.0140 (p=0.00076); the exact three-state form (n=11, 11 creators) median robust_z 3.01	PROBABLE
8	Give the solution statement room (~12 s)	strong quartile solution_s 12.25 s vs 8.4 s (p=0.0065) — the only script part clearing p<0.01 among 115 features; strong reallocates from explanation (34.1 % vs 41.9 %) rather than extending	PROBABLE
9	Hand over an artefact	resource_handoff save rate 0.0429 , highest of any solution type (n=38, 28 creators), vs framework_mental_model 0.0135 (n=25); ready-made-artefact desire save 0.0346 vs 0.0270 (p=0.052)	PROBABLE
10	Name the pain as an act, not a condition	manual_repetition view_lift 6.94 / save 0.0390 (n=15) and time_waste 3.54 / 0.0365 (n=20) vs chaos_no_process 0.87 (n=17) and slow_response_to_leads 0.47 (n=5)	PROBABLE
11	Give the hook 5–10 s and 20–30 words	median hook 6.7 s / 21.5 words / 13.2 % of speech (n=260); hook words separate the quartiles (26.5 vs 22.0, p=0.028) while hook seconds do not (p=0.162)	PROBABLE
12	Open exactly one loop	240 of 266 reels carry ≥1; median exactly 1 (198 reels); only 15 of 264 carry an explicit re-hook	PROBABLE

#	WHAT WORKS	EVIDENCE	CONFIDENCE
1 3	Adopt the builder register	<code>builder_positioning</code> n=38, 27 creators, median <code>view_lift</code> 3.42 vs <code>educator</code> 1.73 (n=138); <code>demo_walkthrough</code> is the largest archetype (n=62, 42 creators) and the highest-lift (3.70)	PROBABLE
1 4	Repeat vocabulary rather than varying it	strong quartile <code>lex_ttr</code> 0.605 vs 0.628 (p=0.039), <code>lex_mattr50</code> 0.795 vs 0.811 (p=0.044) — winners repeat the tool name and the step word	PROBABLE
1 5	Close in the last 10 %, ~5 s and 20 words, one ask	CTA starts at a median 90 % of spoken length; 202 of 216 in the final quarter; median 4.95 s / 21 words; strong quartile 5.7 s vs 4.4 s	PROBABLE
1 6	Show a single unbroken process shot when the build really runs	pure <code>B_ROLL_PROCESS</code> n=6, 4 creators: median <code>view_lift</code> 30.90 , <code>robust_z</code> 4.13 — the highest of any visual sequence	INSUFFICIENT

Rows 1–5 are the only ones that meet the method's RELIABLE bar ($n \geq 30$, ≥ 8 creators, $p < 0.01$ and a surviving creator-normalised check). Rows 6–15 are real but weaker: they rest on categorical medians, which admit no normalisation test, or on cells of 15 to 42 reels. Row 16 is printed because it points the same way as rows 1 and 2 and because suppressing it would hide a cell, not because six reels can carry an instruction.

What this sounds like when a reel does most of it at once

[DNA7-d3o5sQ](https://www.instagram.com/reel/DNA7-d3o5sQ/)(<https://www.instagram.com/reel/DNA7-d3o5sQ/>) — 451 491 plays, 55.4× its author's median, 36.5 saves per 1 000, **ungated** — does rows 1, 2, 3, 6, 7, 9, 12 and 15 in 68 seconds. The artefact speaks the first four words ("**Hello, how can I assist you? This voice assistant can replace your receptionist. I'll show you how to build it in three easy steps.**", hook 0.0–6.4 s): show-first, screen first frame, statement not question, and a bounded promise. The mechanism names each destination at the moment it becomes necessary — "**it uses Google Calendar to check your schedule and Google Sheets to save any details**" (6.4–20.0 s) — which is row 3 executed literally. A single 1.8-second re-hook ("But how about the results?", 32.0–33.8 s) closes the one open loop, and the reel ends on the machine finishing its job rather than on an ask.

Visual reading. `fa-v1` records `first_frame_type`: `UI_DEMO` — the n8n "AI Voice Assistant" canvas with a pulsing "Listening" widget and the presenter's facecam small at the bottom — and two proof visuals: the live workflow run and the Google Calendar day view showing the appointment the assistant created. The screen is the argument; the face is a footnote on it. That is rows 1, 2 and 7 in one storyboard.

The single clearest same-creator demonstration

`edhillai` published [DNA7-d3o5sQ](https://www.instagram.com/reel/DXb8P08Dati/) at `robust_z` **+4.69** and [DXb8P08Dati](https://www.instagram.com/reel/DXb8P08Dati/)(<https://www.instagram.com/reel/DXb8P08Dati/>) — "**Day one of two AI tips to make you an N8N expert.**" — at `robust_z` **-1.04**. Same creator, same subject area, same production quality, both with `UI_DEMO` first frames. What separates them is which sentence comes first: the artefact, or the programme. Because it is one creator, the comparison holds the audience, the account and the algorithmic history constant in a way no pooled statistic can.

Strategic implication

Rows 1–5 are **binding production rules** for M2 and should be written into the card template as checks: own screen present, screen in frame one, three distinct named destinations, second person, zero questions. Rows 6–15 are **defaults to follow unless a specific reel argues otherwise**, and each should carry its confidence label onto the card so a writer knows which ones are negotiable. Row 16 is a **licence, not an instruction**: when an M2 build genuinely runs end to end, record it in one take and narrate over it — it also happens to be the cheapest thing to shoot.

Confidence: **RELIABLE** for rows 1–5 as associations inside this corpus. **PROBABLE** for rows 6–15. **INSUFFICIENT** for row 16. All of it is associational, all of it is computed inside a pre-selected 8.3 % of the database, and none of it licenses a causal claim or a promise about reach.

32. What Does Not Work

Three different kinds of "does not work" appear below, and conflating them would be an error. (a) **Measured negatives** — the association runs the wrong way. (b) **Measured nulls** — the feature was tested at adequate power and nothing was found. (c) **Unmeasurable** — the instrument cannot answer, so no claim is permitted in either direction. Denominators as always: 3 211 for metric-only, 268 for script-and-visual, 266/264 for semantics and beats, 295 for frames, 9 fixed samples per reel. And the standing caveat cuts hardest here: a **null** measured inside a corpus pre-selected for strength is weaker evidence of absence than it looks [I-01].

(a) Measured negatives

WHAT	EVIDENCE	CONFIDENCE
Split screen — the niche's most-used arrangement	22.3 % of all 2 645 frames, 73 of 266 opening frames, 99 reels, 51 creators; any split state median robust_z 1.03 vs 1.43 (p=0.044); pure split (n=27, 18 creators) view_lift 0.94 vs corpus 2.03; the one positive pooled link (split_share → save_rate +0.193) collapses under normalisation (cn +0.036)	PROBABLE
Setup-first openings — series preambles, "if you are a...", story openings	identity_call + story_open + problem_call_out, n=39, 31 creators: median robust_z 0.67 / view_lift 0.80 vs 1.36 / 2.25 (p=0.002 on both); identity_call alone 0.41 view_lift (n=9)	PROBABLE
Questions in the script	questions_n → save_rate -0.229 (p=0.00031, cn -0.162); questions per 10 s -0.235 (cn -0.186); rhetorical questions -0.186 (cn -0.151); question hooks save 0.0130 vs 0.0284	RELIABLE
First person as the subject of the payoff	lex_first_person_n → save_rate -0.180 (cn -0.124)	RELIABLE
Urgency and fear vocabulary	urgency → view_lift -0.164 (p=0.0072); urgency → share_rate -0.150 (cn -0.180 , survives); fear → share_rate -0.157 (cn -0.086, survives); weak quartile uses 2 urgency words vs 1	PROBABLE
Income framing	the 71 of 266 reels (27 %) framing the outcome as money run at median view_lift 1.19 vs 2.12 ; save rate unchanged (0.0286 vs 0.0264)	PROBABLE
More face on camera	a_roll_share → share_rate -0.156 (p=0.011, cn -0.094, survives)	PROBABLE
Frameworks and "how to think about it"	framework_mental_model save rate 0.0135 (n=25) against resource_handoff 0.0429; the "certainty" desire carries save 0.0140 vs 0.0282 (n=19, p=0.017)	PROBABLE
News without a work decision	announcement_news n=30, 21 creators: view_lift 1.29, robust_z 0.96; model-news topic save rate 0.0097 , roughly a third of the corpus median	PROBABLE
Stacking asks	Da3uSkrNjli (https://www.instagram.com/reel/Da3uSkrNjli/) runs four asks in its last twenty seconds and lands at view_lift -0.05	INSUFFICIENT (n=1)

(b) Measured nulls — features that simply do not separate

Of 115 numeric features, none separates the top from the bottom robust_z quartile at $p < 0.01$. The list of things that show nothing is longer than the list of things that do: duration ($|\rho| \leq 0.09$ over 3 211 reels; strong 56.9 s vs weak 58.0 s, $p=0.981$) · spoken length (198 vs 179 words, $p=0.333$) · words per second (3.53 vs 3.45, $p=0.444$) · hook seconds (7.6 vs 7.0, $p=0.162$) · cuts and cuts per minute (7.06 vs 6.75, $p=0.420$) · scene count (4 vs 3, $p=0.110$) · average scene length · A-roll share (0.2222 both,

$p=0.573$) · B-roll share · split share (0.0 both, $p=0.628$) · caption density (1.0 both, $p=0.418$) · average sentence length · every lexical-diversity measure against reach · `proof_s` , `problem_s` and `explanation_s` against reach.

The comment gate is the most consequential null. It moves comment rate $\times 23$, save $\times 2.8$ and share $\times 1.8$ — and reach not at all: view_lift 2.08 vs 1.72 ($p=0.58$), robust_z **1.2475 vs 1.2481** ($p=0.63$). Nothing in reach is bought by it, which is why banning it costs M2 nothing.

A null that vanished on retest deserves its own line: nine pooled associations disappear under creator normalisation, including `info_density` $\rightarrow$ save_rate (+0.303 $\rightarrow$ cn +0.132), `specificity` $\rightarrow$ save_rate (+0.237 $\rightarrow$ +0.047) and `solution_s` $\rightarrow$ save_rate (+0.343 $\rightarrow$ **-0.217**, sign flip). These describe which creators post what, not what videos do.

(c) Rules people expect, which this data does not support

The 0–3 second hook rule. Median hook is **6.7 s**; only **6 %** of reels close inside 3 seconds; hook duration correlates with nothing ($p \geq 0.23$ on all four metrics). An audit scoring hooks against a three-second rule is measuring something this niche does not do — which is why the 8 September criticism of Max's card set on hook timing **should be withdrawn**: the rule was wrong, not the scripts [I-03].

A fixed scene template. Max's fixed 60 s / 7-scene template is not wrong about duration (duration predicts nothing) but is wrong to be fixed: coarse visual states range from 1 to 8 with median `robust_z` between 1.14 and 1.63 and no trend. Fixing the scene count holds constant the one thing that should vary with the topic.

"Cut faster." No signal in any metric — and the metric itself, `ffmpeg_scene_0.35_count_only` , is a scene-score threshold count, not shot detection. 38 of 282 rows report zero. **No M2 output may quote a cut rate from it.**

Cutting on script beats. Only **55 of 1 527** beat boundaries (3.6 %) have a scene starting within 0.3 s, 215 of 264 reels have none, and aligned reels show no advantage (robust_z 1.45 vs 1.25, $p=0.603$). With a 0.3-second window against a ~6-second sampling resolution this is a measurement floor, so the honest label is **INSUFFICIENT**, not "editors do not do this".

What the failures sound like

[DaDn5L-xKWi](https://www.instagram.com/reel/DaDn5L-xKWi/)(https://www.instagram.com/reel/DaDn5L-xKWi/) opens **five** loops — "Five AI image tricks top realtors are using to sell listings faster" (0.0–6.4 s) — and closes them in five identically shaped ten-second blocks; the reel flattens because nothing escalates.

[DcbnTAtxb8C](https://www.instagram.com/reel/DcbnTAtxb8C/)(https://www.instagram.com/reel/DcbnTAtxb8C/) describes three departments in three grammatically identical sentences ("Marketing researches what's going on... Sales takes a lead from our CRM... Customer handles onboarding...", 13.4–37.8 s) and never names a person responsible for anything. [DXb8P08Dati](https://www.instagram.com/reel/DXb8P08Dati/)(https://www.instagram.com/reel/DXb8P08Dati/) spends its first sentence on the programme rather than the subject and sits at `robust_z` -1.04, the bottom of this corpus, from a creator whose demo-first reel sits at +4.69.

Visual reading. [DcnHtUGS_tE](https://www.instagram.com/reel/DcnHtUGS_tE/)(https://www.instagram.com/reel/DcnHtUGS_tE/) is the visual failure mode in one frame: a held split screen with an AI-generated cinematic clip above a reacting host. It reached **65×** its author's median and carries share rate **0.0016** and save rate **0.0005** — the lowest hi-intent rates in the top thirty by reach. Attention without intent, and the geometry is why: the thing the viewer is asked to believe never gets the full frame.

Strategic implication

Ban four things outright from the M2 template — **split screen, setup-first openings, questions in the script, and the comment gate** — and drop the urgency and income registers after the hook. Stop enforcing three rules that the data does not support: the 0–3 second hook, the fixed scene count, and any editing target. And write the nulls into the card review as *permissions*: a reviewer may not reject a script for being 43 seconds or 92 seconds, for having three cuts or eleven, or for speaking at 3.1 or 3.9 words per second, because none of those separates anything in a corpus of 3 211 reels.

Confidence. RELIABLE for the question and first-person negatives, for the gate's reach null and for the 115-feature quartile null. PROBABLE for split screen ($p=0.044$), setup-first ($p=0.002$ but no creator-normalised test exists for a categorical median), urgency, income framing and A-roll share. INSUFFICIENT for the alignment result and for any single-reel example. Absence of evidence here is not evidence of absence: the corpus is 8.3 % of the database and pre-selected for strength.

33. Recommendations for Our Positioning

Denominators, one last time. Everything below rests on 268 analysed reels (8.3 % of 3 211 ingested), 266 semantic reads, 264 beat parses, 295 frame reads at 9 fixed samples each, and 245 rows with a save rate. The analysed corpus is the top of `score.py` 's ranking, so it describes what the *already-successful* part of this niche sells [I-01]. Nothing here is causal.

1. What viewers in this niche are actually looking for

Across 266 semantic reads the demand is not for understanding. It is for **a thing you can have** — a repo, a file, a workflow, a prompt, a map — that removes a named piece of manual work, named tool by named tool, with the screen showing it running. Nine of the thirteen RELIABLE correlations in the whole association run are save-rate correlations, and a save is a bet that the thing will be used.

THE DEMAND	MEASUREMENT	WHAT IT MEANS FOR M2
an artefact to take away	ready-made-artefact desire n=27: save 0.0346 vs 0.0270 (p=0.052); <code>resource_handoff</code> save 0.0429 (n=38, 28 creators), the highest of any solution type	publish a checklist, process map or prompt with every reel
a named repeated act, not a condition	<code>manual_repetition</code> view_lift 6.94 / save 0.0390 (n=15); <code>time_waste</code> 3.54 / 0.0365 (n=20); <code>chaos_no_process</code> 0.87 (n=17); <code>slow_response_to_leads</code> 0.47 (n=5)	"you retype the same numbers out of a PDF every Monday", never "your process is chaotic"
named destinations	<code>lex_platforms_n</code> → save_rate +0.331 (cn +0.173), RELIABLE; <code>lex_entities_distinct</code> → view_lift +0.161 (cn +0.198)	name Claude, then the Google Sheet, then the inbox — three destinations, not one repeated
time back, in minutes	time-saving desire n=33: view_lift 3.52 vs 1.88, save 0.0330 vs 0.0275 (p=0.025)	quantify in minutes in the hook
screen evidence at ~22 s	median time to first proof beat 22.0 s ; 41 % of 2 645 frames are proof visuals; <code>screen_demo</code> is the largest proof type (n=84)	the run on screen, at the twenty-second mark
not certainty on its own	"does it actually work" desire n=19: save 0.0140 vs 0.0282 (p=0.017, negative)	a decision must arrive attached to an object
not news	model-news topic save 0.0097 , a third of the corpus median; <code>announcement_news</code> view_lift 1.29 (n=30)	already excluded by POSITIONING §8; the data agrees
not status as an entry ticket	<code>identity_call</code> hook: median view_lift 0.41 (n=9), the worst in the corpus	status may be a payoff clause, never the opening
not urgency	urgency → view_lift -0.164; fear → share_rate -0.157; both negative on every metric	name a real risk once, then drop the register

The uncomfortable half of that table: **M2's declared territory is the least-served and lowest-performing labelled cell in the measured niche.** AI в конкретном бизнес-процессе is n=8 across 8 creators at view_lift **0.65**, robust_z 0.85 and share rate 0.0040 — a quarter of the corpus median. The topic M2 is

positioned to refuse, agent building, is $n=27$ at $6.63\times$ [I-25]. Eight reels cannot prove the subject fails, and the selection bias runs the other way; what they do establish is that nobody is producing it at volume. **The audience narrowing itself is free:** developer-audience reels run at median view_lift 1.87 against 1.89 for business-audience reels — no measurable difference [I-27].

2. The three pillars, with what the data changes in each

PIL LAR	AUDIENCE QUESTION	WHAT THE DATA ADDS
M2 Rad ar	"Is this solution relevant to me?"	Rank on shares per 1 000, not views. The model is Dc_TX1CttIc (https://www.instagram.com/reel/Dc_TX1CttIc/): agenda in nine seconds, each case labelled aloud, and an economic verdict earned after the evidence — "the API cost has shot up so high that any business trying to build a real product on Astra is walking into a train wreck" (80.8–92.6 s). Ungated, 31.7 shares/1 000. Never a model-news reel without a work decision (save 0.0097).
M2 Builds	"What does a first working version look like?"	Screen-first, one re-hook, ungated close — the template is DNA7-d3o5sQ (https://www.instagram.com/reel/DNA7-d3o5sQ/), where the artefact speaks the first four words and a 1.8-second re-hook at 32 s precedes the proof. Adopt the builder register ($n=38$, view_lift 3.42) over the educator register ($n=138$, 1.73) [I-27].
M2 Tea rdo wns	"Is this process suitable for AI, and how would it work?"	The hardest pillar and the most defensible. Its beat clock is Db9MT-axxcO (https://www.instagram.com/reel/Db9MT-axxcO/): outcome by 5 s, three steps in thirty, ask at forty-three, 47.9 shares and 68.9 saves per 1 000 on only $1.6\times$ its author's median. The pain block must be acts with minutes attached, placed in the first quarter as the niche's own minority does (median problem start 21 % of speech, 56 % inside the first quarter).

The one thing no competitor does. `before_after` is the second-rarest proof type ($n=5$) and **not one reel in 266 shows its author's own build failing**. POSITIONING §6 already requires evidence M2 owns and forbids client claims. That constraint, which reads as a limitation, is the only genuinely uncopyable asset the analysis found: **show the run, including the part that broke, on screen, at around 22 seconds.**

3. The mandatory four-part filter, now measurable

Do not publish a reel unless it makes clear: **(1) which repeated work process is being discussed, (2) where the real friction is, (3) what AI should do and what remains human-owned, (4) what the viewer should do next.** Three of the four are now checkable against the data rather than against taste.

Part 1 is the pain-as-act rule ($6.94\times$ against $0.87\times$). Part 2 costs 8–10 seconds inside the first quarter, and M2 pays for them out of the mechanism block — the strong quartile already spends **34.1 %** of speech on explanation against 41.9 % for the weak one, reallocating into solution (**12.25 s vs 8.4 s, $p=0.0065$**) and payoff. Part 3 has exactly one precedent in 266 reels, and it is worth quoting because it is the entire M2 thesis said by a competitor in passing:

[DbVKVz0y8xo](https://www.instagram.com/reel/DbVKVz0y8xo/)(https://www.instagram.com/reel/DbVKVz0y8xo/) — **"Because the AI knows how to cut a video but it doesn't know the exact moment that needs to be a clip. It has no context and no content intelligence."** (13.5–31.4 s). Part 4 is one ask, on the face, in the last 10 % of the script.

4. No comment-bait — and the data says the ban is free

59 % of the analysed corpus (158 of 266) runs a comment-keyword gate. It multiplies comment rate $\times 23$, save $\times 2.8$ and share $\times 1.8$, and moves reach by nothing at all: view_lift 2.08 against 1.72 ($p=0.58$), robust_z **1.2475 against 1.2481** ($p=0.63$) [I-12]. RULES §1.2 already bans it on positioning grounds; the measurement adds that **no distribution is forfeited**. Two consequences follow. M2's own published share and save rates will be structurally lower than gated references' and must never be compared with them. And selection must hold cta_type constant, because cards.py currently ranks on resh_1k and save_1k — the two rates the gate inflates.

The move that replaces the gate is already visible in the corpus. The purest gated reel, DVtgbY8Av6A(<https://www.instagram.com/reel/DVtgbY8Av6A/>), buys **57.3 saves per 1 000** from 84 words by naming fifteen skills and withholding the files. **Invert it:** name the artefact and publish it in the caption. The save comes from there being something to collect, not from having to ask [I-29].

5. The 50–70 second one-take template — keep it, and stop citing it as evidence

Keep the template: one location and light, one banner, one caption system, the screen as a full-frame insert, one closing line, **50–70 seconds**, and a change of speaker as the only permitted cut. It suits one filming day and five publications a week, and **nothing in the data argues against it**. But it is a production convenience, not a measured reach lever, and two numbers currently used to justify it do not reproduce.

PRODUCTION.MD CLAIM	SOURCE OF THE ORIGINAL	WHAT REPRODUCES ON 3 211 CREATOR-NORMALISED REELS
"the winners' median is 55 seconds against 47 for the rest"	a 100-reel top slice of an earlier 2 352-reel export	46.8 s for the 375 HIGH outliers against 47.8 s for the other 2 836 — a difference of one second, in the opposite direction
"under twenty seconds is 5 % of winners and 13 % of the rest"	same slice	14 % of HIGH outliers against 12 % of the rest — the direction is reversed

Duration correlates with robust_z at **-0.069** across 3 211 reels ($p=9.5e-05$, significant only because n is large) and with view_lift at -0.0004 ; the band medians are flat (under 20 s **0.25**, 20–45 s **0.00**, 45–70 s **-0.01**, 70–90 s **-0.19**, over 90 s **+0.05**) [I-02]. A third published number needs the same correction: **RULES §7's "a share separates a winner 3.6× more reliably, a save 2.8×" becomes $\times 2.62$ and $\times 2.09$** on the 2 899 reels clearing the 100-play guard [I-28]. Neither original is wrong about direction; both are wrong about magnitude, and all three should be edited before anyone quotes them again.

On the one-take rule specifically: cut frequency shows no relation to anything ($\rho +0.056$ on robust_z, $p=0.36$) and the metric is a scene-score threshold count that cannot be quoted as an edit count at all. Keep the rule because it is cheap. **Never publish a claim about M2's cut rate.**

6. The ten decisions, in order

1. **Own screen in every reel, and in frame one** — the only RELIABLE production instruction (share $+0.203$; first-frame screen, creator-normalised **+0.285**).

2. **Face → screen → face**, CTA on the face (39 reels, 32 creators, share 0.0213 vs 0.0140).
3. **Never split screen** (22.3 % of the niche's frames, robust_z 1.03 vs 1.43).
4. **Hook: one complete promise, 5–10 s, 20–30 words, a statement, opening on the artefact** — never on the series, the audience or the speaker (setup-first 0.67 vs 1.36, p=0.002).
5. **Three distinct named destinations per script**, and repeat the names rather than varying them.
6. **Friction as a named act with minutes, 8–10 s, first quarter**.
7. **Mechanism capped at a third; solution statement ~12 s; payoff names what changes on Monday**.
8. **Proof at ~22 s, on screen, M2's own run, including the failure**.
9. **One ask, ungated, last 10 % of the script, ~5 s and 20 words**.
10. **Judge on shares and saves per 1 000, with cta_type and creator reliability recorded** — and expect small multipliers once M2's own account becomes consistent (CONSISTENT creators: median robust_z 0.60 against HIGH_VARIANCE 1.37, with identical script shape).

Confidence. Recommendations 1, 5 (as a save lever) and the question ban are **RELIABLE**.

Recommendations 2, 3, 4, 6, 7, 8 and 9 are **PROBABLE** — real, but resting on categorical medians or on cells of 15 to 42 reels. The "what viewers are looking for" table is **PROBABLE** on its three largest rows and **INSUFFICIENT** on status, certainty and novelty (n<20 each). The topic verdict on M2's own territory is **PROBABLE at best and INSUFFICIENT as a prediction**: eight reels across eight creators cannot establish that a subject fails, and this corpus is 8.3 % of the database, pre-selected for strength. **No recommendation above is a causal claim, and none of them promises reach.**

34. Architecture of the Script Writer

A card is the end of a nine-link chain, every link a real file or table: data → analysis → insight → hypothesis → ranking → hypothesis-specific references → transformation → script → review. Nothing after "insight" is invented at write time — it is assembled from JSON and SQLite rows that already exist before a script-writer prompt runs.

Data to analysis. The corpus starts as `reels / transcripts / frames` rows in `db.py`'s legacy schema, passes through `engine/local_pipeline.py`, then two LLM analysis passes: `engine/prompts/transcript-analysis.md` (schema `ta-v1`: beats, semantics, hook/pain/proof/cta typing) and `engine/prompts/frame-analysis.md` (schema `fa-v1`: per-frame labels and visual aggregates). `engine.ingest_analysis` loads both into `beats / frame_labels / scenes`; `engine.features.build()` consolidates them plus `engine.stats.py`'s performance columns into one `video_features` row per code (`engine/SPEC.md` §5–§6).

Analysis to insight to hypothesis. `reports/analysis/08-hypotheses.md`: 30 numeric insights (`data/analysis/insights.json`, I-01...I-30, each with `statement`, `limitations`, `confidence` ∈ RELIABLE/PROBABLE/INSUFFICIENT) fed 24 hypotheses — 9 M2 Radar, 9 M2 Builds, 6 M2 Teardown, written before scoring so format mix wasn't an artefact of ranking. Denominators are carried forward explicitly: 268 analysis-ready reels for script/visual claims, 266 with both a semantic read and performance data, 132 creators with a stats row (17 CONSISTENT). `engine.ingest_insights` loads the JSON into `insights / hypotheses / hypothesis_refs` and opens `jobs` rows at `entity_kind='hypothesis'`, stages `CONCEPT_GENERATION` and `REFERENCE_SELECTION` (`engine/ingest_insights.py:279-322`) — the only two jobs stages this whole content-generation chain writes before Phase 3.

Ranking. Fourteen weighted keys (evidence and positioning at 0.12 each, assets and copy_risk at 0.03), each scored 1–5, summed to a `total_score` on the same 1–5 scale (`reports/analysis/08-hypotheses.md` §2). The cut to ten landed exactly on the 4 Radar / 4 Builds / 2 Teardown quota with a 0.49-point gap between the lowest selected hypothesis (H-19, 4.16) and the highest rejected one (H-14, 3.67) — "a coincidence worth naming rather than a validation of the weights," per the report itself.

Hypothesis-specific references. `reports/analysis/09-reference-selection.md`: 41 references across the 10 hypotheses, 27 distinct reels, in `data/analysis/hypothesis_refs.json` and `hypothesis_refs`, each row carrying one named `function` (hook/pain/explanation/proof/cta/a_roll/b_roll/split/screen_proof/rhythm/ transition — `engine/SPEC.md` §2.8), real beat/scene ids, a `performance_json` read from `video_features` at generation time, and a `transformation` field. Seven rules govern selection: CONSISTENT creators may contribute several references, a HIGH_VARIANCE creator's multiplier is not comparable across reliability labels, gated CTA rates are corrected for comment-gate inflation (×2.8 save, ×1.8 share, reach unchanged), a reel serves two hypotheses only under two different functions, and no reel flagged unsafe on content grounds is used.

The input package the writer actually receives. `engine/prompts/script-writer.md` names it directly: the selected hypothesis and its `hypothesis_refs` rows (copied verbatim from `data/analysis/hypothesis_refs.json`), plus `POSITIONING.md`, `PRODUCTION.md` (the 50–70s one-location template), `RULES.md` §2–§4, and `design/README.md` + `design/tokens/tokens.json` for overlay/caption constraints. Every number the script states must come from `insights.json`, `reports/analysis/*.md`, a reference reel's real metrics, or the team's own runs (`reports/audit/*.md`, `reports/data/*.json`) — "we do not know yet" is the explicit fallback.

Hard rules, a violation rejects the card. The strategic filter is four literal, non-empty fields (`process` , `friction` , `ai_boundary` , `next_action`); every claim carries a state (OBSERVED/PLANNED/TO_MEASURE/MISSING); no comment-bait CTA (`comment_keyword` is banned, per the insight that it inflates comment rate 23× while reach moves not at all); originality means borrowing structure only, each borrowing recorded in `references[].transformed_how` and `traceability` ; timing runs 2.3–2.6 words/second within a 50–70s band; hooks ship 3 candidates.

The card-v2 contract. Output is one `cards/<card_id>.json` , schema `m2radar.card.v2` (`engine/SPEC.md` §8), validated by `engine/cards_v2.py` 's `validate()` against closed vocabularies (21-value `frame_type` , 13-value `hook_type` , 9-value `cta_type` , the 4-value `layout` set the Remotion EDL contract also enforces) and structural rules (contiguous storyboard scenes, `script.total_s` a hard error outside 20–120s, a warning outside 50–70s). `save()` refuses any card with a hard error.

Reviewer step and `script_versions`. On save, `cards_v2.save()` appends a `writer` row to `script_versions` at `script.version` , and — only when `review.revised` is `true` — a `reviewer` row at `script.version + 1` carrying `review.findings` ; both rows point at the same `script_json` , since the schema keeps only the final text, not the pre-review draft (a documented simplification, not a bug — `cards/README.md`).

Traceability, answered from the DB. For a given hypothesis and card:

```
SELECT hypothesis_id, title, total_score, supporting_insights_json, candidate_refs_json
FROM hypotheses WHERE hypothesis_id = 'H-11';
```

```
SELECT insight_id, statement, confidence FROM insights WHERE insight_id IN
  (SELECT value FROM json_each(
    (SELECT supporting_insights_json FROM hypotheses WHERE hypothesis_id = 'H-11')));
```

```
SELECT code, function, reason, performance_json, transformation
FROM hypothesis_refs WHERE hypothesis_id = 'H-11';
```

```
SELECT card_id, title, status, script_json FROM cards_v2 WHERE hypothesis_id = 'H-11';
```

```
SELECT version, kind, findings, created_at FROM script_versions
WHERE card_id = (SELECT card_id FROM cards_v2 WHERE hypothesis_id = 'H-11')
ORDER BY version;
```

```
SELECT stage, state, started_at, error FROM jobs
WHERE entity_kind = 'hypothesis' AND entity_id = 'H-11' ORDER BY started_at;
```

The last query returns only `CONCEPT_GENERATION` / `REFERENCE_SELECTION` rows today.

`CARD_GENERATION` , `SCRIPT_GENERATION` , `SCRIPT_REVIEW` and `FRAME_PLAN` are named in the canonical `jobs.stage` list (`engine/state.py:STAGES`) but no module — `engine/cards_v2.py` included — ever opens a `job()` context around them; the trace for those stages lives only in `script_versions.created_at` and `cards_v2.updated_at` , not in `jobs` .

What is not tracked, honestly. `jobs.cost_json` exists as a column in `engine/schema.py` , threaded through `engine/state.py` , but no call site in `engine/ingest_insights.py` , `engine/cards_v2.py` or `engine/production.py` ever populates it (verified by `grep`). The hypothesis-generation and script-writing passes are session-driven LLM prompts (`engine/prompts/*.md`) run outside any scheduled job; `ingest_insights.py` only loads already-written JSON, it never calls a model or meters tokens. There is no per-hypothesis or per-card token or dollar figure anywhere in this pipeline — a real gap, worth closing before this chain justifies its own cost.

35. Architecture of Frame Generation

Frame generation in this engine means one thing today: `engine/storyboard_render.py` turns each storyboard scene into a labelled 1080×1920 PNG with Pillow, drawing only from `design/tokens/tokens.json` — never a hardcoded palette, never a stock photo, never AI-generated imagery. There is no image model in the render path, and that is a deliberate design decision, not a missing feature.

Why templates, not AI images

Two rules make the choice explicit. First, `design/tokens/tokens.json` carries a `banned` list — no centred text, no rounded corners, no shadows, no gradients — and `storyboard_render.py` enforces it in every layout it draws. Second, and more fundamental: text is never generated inside an image in this system. A generative image model that burns a caption into pixels cannot be proofread, corrected, or re-timed the way a Pillow-drawn caption band from `script_text` can; the template renderer keeps every word an editable string until a human presenter actually says it on camera. Combined with `engine/SPEC.md §0.8` ("no paid API calls from any new module by default; anything that costs money must print the estimate and require `--yes`"), a card can move from hypothesis to a full, reviewable storyboard for zero dollars, with every visual still a placeholder honestly labelled as one.

`render_card(card, out_dir='cards/frames')` writes one PNG per scene plus a contact sheet at `cards/frames/<card_id>-sheet.png`, and mutates `card['storyboard']` in place: each scene's `asset_status` becomes `'template'` and `asset_path` points at the rendered file (`cards/README.md`). Four layouts exist, matching the same four values the Remotion EDL contract accepts (`studio/remotion/contract.mjs` 's `UNSUPPORTED_LAYOUT` check): `a_roll` (one placeholder presenter box + banner + caption band), `split_screen` (a proof panel on top, presenter below, split at 58% — the contract's default `split_ratio`), `demo` (a full-frame `[SCREEN: ...]` placeholder), and `motion_graphic` (a left-aligned statement card, no placeholder box; `storyboard_render` additionally accepts the rendering-time-only alias `"text"`, which `cards_v2.validate()` deliberately refuses in a saved card so nothing invalid ever reaches the EDL builder).

How the analysis actually informs the frame

This is not cosmetic. `fa-v1` (`engine/prompts/frame-analysis.md`) produces, per code, `first_frame_type`, `visual_sequence`, and share estimates for A-roll/B-roll/split/ screen states; `engine.features.build()` folds these into `video_features` 's visual block. `reports/analysis/01-general-conclusions.md §3` states the single most reliable production instruction the corpus yields: a screen on camera is the only visual feature that survives creator normalisation on two independent metrics, worth roughly half again the forwarding rate. `engine/prompts/script-writer.md` 's hard rule 7 turns that finding into an instruction — use a screen insert or split-screen proof within the first 10 seconds wherever the hypothesis has a screen to show — and the writer records which `layout / frame_type` each scene gets accordingly. `storyboard_render.py` then renders exactly that decision: a `demo` or `split_screen` layout appears because the semantic layer upstream said a screen belongs there, not because the renderer chose it.

asset_status semantics

`card_scenes.asset_status` has four defined values, and only one is produced by any module in this repository today:

VALUE	MEANING	WRITTEN BY
<code>missing</code>	nothing rendered yet	default state before render
<code>template</code>	this module's Pillow placeholder	<code>engine.storyboard_render.render_card</code> (the only writer that exists)
<code>ready</code>	real owner footage or graphic bound in	intended for Phase 3, not produced by any code in this repo yet
<code>generated</code>	an AI-generated visual	not used anywhere in this codebase — no paid image/video generation is called from any owner's files

How a future provider would plug in

`engine/providers.py` defines a `Provider` registry (`kind`, `role` $\in$ `DEFAULT|OPTIONAL_PROVIDER|FALLBACK`, `env_vars`, a `resolve()` reporting only presence/length of a credential, never its value). Two rows already exist for this future: `openai` (`kind='image_gen'`, `OPTIONAL_PROVIDER`, `OPENAI_API_KEY`) and `higgsfield` (`kind='video_gen'`, `OPTIONAL_PROVIDER`, `HIGGSFIELD_API_KEY`) — both `NOT_CONFIGURED` here, and both verified (`grep -rn` for their client imports) to have zero call sites anywhere in the code. `config/provider-catalog.json`, ported verbatim from `origin/Latest`, extends this into a real capability catalogue — models like `seedance2_5` and `veo3.1` with their admitted durations, aspect ratios and a `runtime_adapter` string (`m2_studio.providers.RunwayImageTransport`) that does not exist in this repository — dated `2026-09-05`, `recheck_after` a week later, and `"execution_enabled": false` at the top level. This is an escrow pattern, not a stub waiting to be finished: capabilities are priced and catalogued in advance, but nothing calls them until a human flips `execution_enabled` and a real credential lands in `.env`, at which point the existing `providers.py` registry — not a new integration — is what would report the provider `CONFIGURED` and let a future `render_card`-equivalent choose it over the Pillow template.

FRAME_GENERATION_READY and VIDEO_GENERATION_READY — one real, one not

Only one of these is an actual state this pipeline writes. `VIDEO_GENERATION_READY` is a real `jobs.stage` value, opened by `engine.production.ingest_takes` once real MP4 takes exist for a card (`engine/production.py:392`) — it belongs to Phase 3, supplied footage, not to frame generation. `FRAME_GENERATION` is named in the same canonical stage list (`engine/state.py:STAGES`) alongside `FRAME_PLAN`, but neither `engine.storyboard_render` nor any other module ever opens a `job()` context around it — verified by reading `storyboard_render.py` directly, which mutates the card and writes files but never touches `engine.state`. So there is no `FRAME_GENERATION_READY` flag anywhere in this schema; the only signal that a card's frames are done is `card_scenes.asset_status = 'template'` on every scene, checked by eye or by querying the table — a gap that matches the same under-instrumented pattern the render stages show later in the pipeline (chapter 36).

36. Video Production Architecture

Everything from an approved card to a reviewed MP4 runs through two modules, `engine/production.py` and `engine/storage.py`, on top of one file format: the Remotion EDL. `engine/edl.py` 's `card_to_edl()` turns a validated card into an `m2.remotion-edl.v1` document at `cards/edl/<card_id>.json`, and `studio/remotion/contract.mjs` is the single source of truth for its validity — both the Python builder and the Node renderer call into behavior that file defines.

PREVIS vs PRODUCTION

An EDL's `render_mode` is either `PREVIS` or `PRODUCTION`. `card_to_edl()` always writes `PREVIS`: every scene carries a `placeholder` layer standing in for real A-roll, an `image` layer for the rendered template PNG (`rights_approved: true`, `rights_receipt_id: "internal-template"`), and `pending_audio_assets: ["speech"]` — this module never invents a speech asset. `PRODUCTION` mode is gated by `contract.mjs` behind `review_state='APPROVED'` and no pending audio or placeholder layers remaining; nothing in this branch produces a `PRODUCTION` -mode EDL yet. Phase 3's `engine.production.edl_with_takes()` moves a scene from placeholder to real: for every scene with a selected take it swaps in a real `video` layer (verified sha256, timing from the probed file), leaves every other scene untouched, and re-validates the whole document through `contract.mjs` again.

The verified PREVIS render

This is not a theoretical contract — it was run. `docs/PRODUCTION_PIPELINE.md` records a real render of the example card (C-2026-09-11-EX) on 2026-09-12, after `npm ci` (179 packages, `tsc --noEmit clean`):

```
node studio/remotion/render.mjs --edl cards/edl/C-2026-09-11-EX.json --assets . \
  --output <scratch>/render-ex.mp4 --browser "/Applications/Google
  Chrome.app/Contents/MacOS/Google Chrome"
```

Result: 1,800 frames at 30fps (60s), 1080×1920 H.264, 3.4 MB. The receipt, `reports/evidence/remotion-previs-receipt-C-2026-09-11-EX.json`, carries `"status": "RENDERED_REVIEW_REQUIRED"`, `"render_mode": "PREVIS"`, `"remotion_version": "4.0.520"`, an EDL sha256 (`bfd8aa9a159981bb4fa3eebaf560b3fdb24910273b70e17c643a7d3a58ab0510`), an output sha256 (`0716c154c1d4b076a22b4475d9ad623726e9315e67b51ada1b9b22d93b72245f`), and per-asset hashes for all seven storyboard images — a mid-render frame is saved at `reports/evidence/remotion-previs-frame-12s.jpg`. The MP4 itself was not committed. This confirms what `reports/audit/01-branch-comparison.md` §7 had listed as explicitly unverified a day earlier — "whether Remotion actually renders" — and closes that gap with a real artefact.

The supplied-MP4 take contract

`schemas/supplied-take.schema.json` defines one take: `card_id`, `scene_id`, `script_segment`, `take`, `shot_type`, `roll (A|B)`, `duration_s`, `fps`, `width`, `height`, `orientation`, `has_audio`, `audio_quality`, `subject`, `in_s`, `out_s`, `sync_notes`, `quality_status (PENDING|OK|RETAKE)`, `retake_of`, `sha256`, `path`. Six fields (`duration_s`, `fps`, `width`, `height`, `has_audio`, `sha256`) stay

`null` until `probe_media(path)` reads the real file — via `ffprobe` if present (verified absent here: `imageio_ffmpeg` 's bundled binaries hold only `ffmpeg`, not `ffprobe`) or a regex fallback over `ffmpeg -i 's` `stderr`. Everything else is filled in by hand at shoot time. `orientation` must be `"portrait"` — every M2Radar card is a 9:16 reel, so anything else is a hard validation error, never a warning.

Take selection and `edl_with_takes`

`engine.production.ingest_takes(con, card_id)` reads `cards/takes/<card_id>.json`, probes every file that exists on disk, validates each take against the card's storyboard, and writes `cards/takes/<card_id>.validated.json`. It always records exactly one `jobs` row (`entity_kind='card'`, `stage='VIDEO_GENERATION_READY'`) and never claims success it did not observe — a missing card is `SKIPPED` / `FAILED`; no sidecar or zero real files on disk is `SKIPPED` / `PENDING`; at least one real, validated take is `DONE` / `DONE`. `select_takes()` then picks, per scene, the take marked `quality_status: "OK"` with no validation errors and the highest take number — a scene with no such take is simply absent, so its EDL slot stays a placeholder rather than being guessed at.

Render plan and QA

`engine.production.render_plan(card_id)` builds the exact `node studio/remotion/render.mjs --edl ... --assets ... --output ... --browser ...` command (read from `render.mjs` itself, never guessed) and returns four states: `NOT_CONFIGURED` (no EDL yet, `node_modules` missing, no Chrome/Chromium binary, or `node` missing from `PATH`), `READY` (command built, nothing run), `EXECUTED` (subprocess exited 0, receipt attached), `FAILED` (non-zero exit, or the output path already exists — `render.mjs` refuses to overwrite on purpose). QA is explicitly a human step: nothing in `engine/production.py` auto-approves a render; `docs/PRODUCTION_PIPELINE.md` §6 has the operator record `qa_status: 'OK'` (or `'FAILED'`, "never invent a pass") by hand before `record_render()` is called.

Storage: Supabase canonical, Cloudflare delivery-only

`engine/storage.py` is a fresh implementation — `git grep` across this repo and `origin/Latest` for Supabase/Cloudflare/R2 code turns up only documentation and false positives (a regression variable named `r2`, a test fixture reel named `"R2"`), confirmed by reading each hit. `LocalStorage` copies into `data/renderers/objects/` and is always configured; `SupabaseStorage` does `POST {SUPABASE_URL}/storage/v1/object/{bucket}/{key}` behind an injectable `transport()` so no test makes a real HTTP call, and raises `StorageNotConfigured` naming the exact missing variable when `SUPABASE_URL` / `SUPABASE_SERVICE_KEY` are absent — true here, verified. `CloudflareDelivery` implements no upload path at all by design: its only justified role is turning an already-Supabase-stored key into a public preview URL from `CF_PUBLIC_BASE_URL`, a variable `.env.example` does not yet define. `record_render()` writes `data/renderers/<run_id>-<card_id>.json` and upserts into a guarded `render_records` table this module creates itself, the same "create your own table" pattern `local_pipeline.py` uses.

What is not implemented or verified

No owner footage exists yet anywhere in this repository — every take example above is illustrative. Supabase and Cloudflare are both `NOT_CONFIGURED` (no credentials in this checkout's `.env`), so every real render today can only land in `LocalStorage`. There is no `cron.sh` step for Phase 3:

`engine.production` and `engine.storage` are operator-run, confirmed by reading `cron.sh` , which stops at the watchdog and features stages. `VIDEO_RENDER` and `QUALITY_REVIEW` are named in the canonical `jobs.stage` list but no module opens a job around either — a render is traced only if the operator manually wraps the render command in an `engine.state.job()` context (`docs/PRODUCTION_PIPELINE.md` §5). None of this is hidden: every gap above is named in `docs/M2RADAR_CONTENT_ENGINE_ARCHITECTURE.md` §16, not smoothed over as "coming soon."

37. Limitations

This chapter consolidates limitations named elsewhere in this project's own documents. None of it is new; gathering it here means no claim in this report, or in a card built from this engine, should be read without the caveat that applies to it.

Coverage is 268 of 3,211, and it is a selected sample. Every structural conclusion about scripts or visuals rests on the 268 analysis-ready reels (8.3%), not the full 3,211-reel corpus with metrics. That 268 is the top of `score.py`'s own ranking — median `robust_z` +1.26 against 0.00 across all ingested reels — so every contrast compares strong reels against other strong reels, never strong against average. `docs/M2RADAR_ANALYSIS_METHOD.md` §10: "That 8.3% is a selected sample, not a random one." Insight I-01 in `data/analysis/insights.json` names the only fix — "a random sample of the 2,914 unanalysed codes" — which does not exist today.

Nine-fixed-frame sampling means shares are estimates, not counts. The legacy corpus samples 9 frames per video at fixed offsets, not detected cuts (`frames_version= 'fixed9-v1'`). `engine/SPEC.md` §5: shares are estimates from samples, transitions between them are `unknown` unless two adjacent frames obviously differ, and `cut_metric_quality='ffmpeg_scene_0.35_count_only'` marks every legacy cut figure as a count, never a real cut. `reports/analysis/09-reference-selection.md` §5: "A scene listed as running 44 seconds may have been four shots. Use the scene ids to locate the moment, never to plan a cut."

Media is never retained; the legacy corpus cannot be re-extracted. Every mp4 is deleted after frame extraction (`docs/data-lifecycle.md`). `reports/audit/ 02-data-inventory.md` §15: whether `cuts=0` on 38 deep dives is a genuine single take or a failed detection cannot be settled without the original video, only by eye against the surviving contact sheets; no cut count can be re-measured with a better method because the source no longer exists.

ASR damage and provider gaps. The corpus's own transcript analysis records mis-transcriptions used as evidence in shipped hypotheses — "Kenwa" for Canva, "cloud code" for Claude Code, "Chachi BT" for ChatGPT (`reports/analysis/09-reference-selection.md` rule R7) — borrowing a reel's structure never means borrowing its wording. Separately, `engine/HANDOFF_NOTES.md`'s schema-owner section records 8 August-archive codes with `transcript_meta.provider='unknown'` , `asr_version=NULL` : "Do not 'fill in' the unknown ones."

Two music-only reels carry no usable transcript, by design, not failure.

`docs/M2RADAR_CONTENT_ENGINE_ARCHITECTURE.md` §16 names them directly; 7 codes in total have an empty `transcripts` row because Whisper with `vad_filter` heard no speech — music-only or text-on-screen — and get their own `TRANSCRIPT_UNUSABLE / EMPTY_NO_SPEECH` tier rather than being folded into "failed."

Max's corpus (1,062 transcripts, 19,808 frames) is not in this git repo. `origin/Latest`'s output exists only on his side. `reports/audit/01-branch-comparison.md` §7: "Whether Max's 1,062 transcripts overlap main's 273 — a stable-ID overlap audit is still required." §5.2 recommends exporting them via the `m2_signal` receipt format; that has not happened.

1,312–1,313 of the newest 1,535 collected reels are unprocessed, links expired. `reports/audit/02-data-inventory.md` §7 counts 1,313 codes with neither transcript nor frames in the newest snapshot; `reports/charts/corpus_coverage_funnel.json` : "reprocessing them costs another Hiker call." `engine/watchdog.py` never calls Hiker again — it only replays already-cached signed URLs — so these codes are stuck until a fresh, paid collection re-establishes a live link.

Notion attachments are unreadable via MCP, and owned measurement is empty. `reports/audit/05-notion-state.md` §6/§298: pre-existing file attachments on a Notion page (CSV/JSON/HTML dashboards) resolve only to unusable `attachment:<uuid>` references, not downloadable URLs — an exhibit already in Notion must be re-published somewhere fetchable. Separately, `reports/audit/02-data-inventory.md` §15: "`our_posts`, `our_metrics` and `followers` are empty" — every performance number here describes someone else's account, never M2 Lab's own.

Comment-gate contamination of the ranker. `reports/analysis/01-general-conclusions.md` §4, verbatim: "The comment gate contaminates our own selection. 59% of the analysed corpus runs one. It moves reach not at all and moves exactly the two rates the radar ranks by. Until `cta_type` is held constant in selection, the radar is partly ranking DM funnels." Not yet fixed in `cards.py` 's ranking — see chapter 38.

Two PRODUCTION.md numbers do not reproduce on the full corpus. `reports/analysis/01-general-conclusions.md` §6: "winners' median 55s vs 47s" and "under twenty seconds is 5% of winners against 13% of the rest" become 46.8s vs 47.8s and 14% vs 12% on 3,211 reels; `RULES.md` §7's "a share separates a winner 3.6× more reliably, a save 2.8×" becomes 2.62× and 2.09×. Both corrections are on record, not silently carried forward.

The Teardown evidence base is weak. `reports/analysis/08-hypotheses.md` §4: of the four RULES-permitted Teardown sources, only "our own processes" is available today. The six Teardown candidates scored lowest of the three formats (mean 3.37 against 3.77 for Radar, 3.93 for Builds); rejected H-17 is named as "precisely the defect the 8 September audit found in ten of Max's cards — designed fixtures, no owner footage, no measured number."

Every semantic and visual label is LLM output, not a human judgement.

`docs/M2RADAR_ANALYSIS_METHOD.md` §10: "they are normalised to closed vocabularies and the raw label is always kept, but they are judgements, not measurements," each carrying its own `ta-v1 / fa-v1 confidence` (RELIABLE/PROBABLE/INSUFFICIENT). `data/analysis/ ingest_warnings.md` records the concrete leakage this produces — beat roles like `thesis`, `demo_first`, `story_open` arriving outside the SPEC §4 vocabulary and coerced to `other`, and one `frame_type` value coerced to `UNKNOWN`.

No causal claim is licensed by anything in this pipeline. `docs/M2RADAR_ANALYSIS_METHOD.md` §10's final line summarises the above: every number here describes a corpus built one particular way, not evidence that any script choice will cause a specific outcome for M2 Lab's own account, which has not published enough to measure at all.

38. Next Steps

1. Run the Phase 2 watchdog on the server, after the next Monday collection. `engine/watchdog.py` is safe today — it never calls Hiker, only replays already-cached signed URLs — but 1,312 of the 1,535 newest-snapshot codes have no live cached link left (`reports/charts/corpus_coverage_funnel.json`), so a run today would mostly mark codes `EXPIRED`. Correct order: let `cron.sh` 's `run.py` step do its weekly collection first (re-establishing live URLs for whatever is still reachable), then run

```
timeout 3600 python -m engine.watchdog --limit 40 --yes
```

as `cron.sh` already schedules. Cost note: watchdog itself is free; a fresh Hiker fetch for a live link is billed at Hiker's own per-unit price — \$0.02/unit, "Start" tariff (`lib/hiker.py` 's `PRICE_RECEIPT`) — so re-fetching all 1,312 would run roughly \$26 at one unit per reel, worth confirming against the actual unit cost first.

2. Fix `cards.py` 's ranking contamination by holding `cta_type` constant. `01-general-conclusions.md` §4: the comment-gate CTA moves comment rate 23×, save 2.8× and share 1.8× while reach moves not at all, so "until `cta_type` is held constant in selection, the radar is partly ranking DM funnels." The fix is mechanical — group or filter candidates by `cta_type` before comparing `robust_z` /rate columns in the legacy `cards.py` ranking, matching `09-reference-selection.md` 's rule R3 already done by hand. Until it lands, any round reusing `cards.py` rather than the hand-curated `08-hypotheses.md` process inherits the same bias.

3. Re-tag topics across snapshots. `topics.py` / `tag_topics.py` apply the 27-label vocabulary at collection time; `docs/M2RADAR_ANALYSIS_METHOD.md` §3 records at least one creator identity already splitting across snapshots, and the finding that M2's territory is thin (`business_process` : n=8 at 0.65× lift) depends on topic labels staying comparable release to release. Re-running the tagger against the newest snapshot, and spot-checking `creator_stats.topic_dist_json` for the same creator across snapshots, is a cheap way to confirm the label set hasn't drifted.

4. Build decoupled owned measurement (`our_posts`). `our_posts` / `our_metrics` are empty (`reports/audit/02-data-inventory.md` §15) — nothing here has ever measured M2 Lab's own published performance; every insight describes someone else's account. Populating `our_posts` (even manually, one row per published reel, the schema's existing shape) is the only way a future report can compare "what the corpus says should work" against what actually worked — the causal question chapter 37 says nothing here can answer.

5. Import path for Max's export. `origin/Latest` holds ~1,062 transcripts and 19,808 frames not in this repo; `reports/audit/01-branch-comparison.md` §5.2 recommends exporting via the `m2_signal` receipt format rather than a raw dump, and its overlap check ("a stable-ID overlap audit is still required") is still open. Run that audit against this branch's `beats` / `scenes` by `code` before importing, so the merge doesn't duplicate codes already analysed independently.

6. Notion decision (a) vs (b). `docs/NOTION_DASHBOARD.md` : this build implemented option (b) — keep the operational Reels/Accounts DBs separate from the new engine databases, linking them into the page tree rather than migrating — "as a working assumption, not Misha's sign-off." Confirm or override before `engine.notion_sync --apply` ever runs for real; today it has made zero writes to Notion.

7. Phase 3 first shoot, using the MP4 take contract. With one of the ten selected hypotheses (`08-hypotheses.md` §5) as the source card: validate the card, render its storyboard, build its PREVIS EDL, shoot to `schemas/supplied-take.schema.json` 's fields, then follow `docs/PRODUCTION_PIPELINE.md`

steps 1–6 (`engine.production takes` → `select_takes` → `edl_with_takes` → `plan --execute` → human QA → `record_render`) — the first time Phase 3's tested-but-unused code runs against real footage, not a synthetic fixture.

8. A provider pilot approval process, if any `OPTIONAL_PROVIDER` is turned on. Every non-default provider (`loore` , `supabase` , `cloudflare` , `openai` , `higgsfield`) is `NOT_CONFIGURED / DISABLED` and stays that way until a credential is added to `.env` . `engine/SPEC.md` §0.8 already states the rule: print the cost estimate and require an explicit `--yes` before the first paid call. Before piloting a provider from `config/provider-catalog.json` , its `execution_enabled: false` flag and dated `recheck_after` should be revisited — the pricing may already be stale.

9. Monitor the RELIABLE insights as new data lands. Insights I-01 through I-30 each carry a `confidence` and a stated denominator; several RELIABLE ones — the screen-on-camera finding, the comment-gate inflation, the duration null-correlation — were computed on 266–268 analysis-ready reels. As watchdog and the Max import (item 5) grow that denominator, re-running `engine/stats.py` and `engine/ingest_insights.py` against the larger corpus and diffing against `insights.json` checks these findings hold rather than being an artefact of a small, selected sample.

10. Open decisions for Misha:

1. `FRAMES_ONLY` — ratify 29 codes (frames + transcript not DONE, incl. 7 empty-transcript codes) or the spec's literal 22 (frames + no `transcripts` row at all)? (`engine/HANDOFF_NOTES.md` , schema-owner §1.1)
2. The two disagreeing corpus-tier definitions (`video_state.corpus_tier` as processing state vs `engine.corpus.tiers()` as evidence tier) — separate names, or collapse to one? (`engine/HANDOFF_NOTES.md` , features-owner §1)
3. `check.py` 's one discrepancy — 8 frame rows with no file on disk in `DcxV37-CJ0C` — re-fetch the reel or delete the rows? (schema-owner §6)
4. Notion decision (a) vs (b) above — confirm (b), or migrate to (a)?
5. The two orphaned "📺 M2 Lab — Radar" stub Notion pages — delete, or keep linked under Legacy?
6. Approve a pilot budget for any `OPTIONAL_PROVIDER` (Loore, Supabase, Cloudflare, or an image/video model from `config/provider-catalog.json`) — and at what ceiling?
7. Priority: items 1–5 versus the first Phase 3 shoot (item 7) — wait on the ranking fix (2) and owned-measurement table (4), or run in parallel?